[STAGE 1.1 (PART 1)] Real-World Build Log: Stage 1.1 Heatsink & Push-Pin Lock on Raspberry Pi 5 [P01] (Photos 1–7)

Chronological Photos 1–7 (1 Step Per Row): Unboxing, peeling blue film, aligning pins & clicking push-pins in.

ARCTIC TRAIL CAM SYSTEM — STAGE 1.1 (PART 1): INSTALLING ACTIVE COOLER [P02] ON PI 5 [P01]

Chronological Benchtop Photos 1–7 (1 Step Per Row): Unboxing, Peeling Thermal Film, Pre-Click Alignment & Locking Push-Pins

PHOTO 1 — STEP 1.0: Unboxed Stage 1 Parts on the Benchtop Cloth ([P01], [P02] & [P07])

- Left: Raspberry Pi 5 (8GB) [P01] removed from red/white box and placed on white anti-static cloth.
- Center: Official Raspberry Pi 5 Active Cooler [P02] (6x7 bare aluminum fin grid + offset circular radial blower fan).
- Right: Raspberry Pi Pico 2 With Headers [P07] (SKU 828988, 40 metal legs pre-soldered) in ESD bag.

File: docs/images/real_build/stage1_01_unboxed_parts.jpg

VERIFIED — All 3 Stage 1 hardware items verified on benchtop (including Pico 2 With Headers [P07])

PHOTO 2 — STEP 1.1A: Peeling Clear Protective Film off the 3 Blue Thermal Pads on [P02]

- Flip the Official Active Cooler [P02] upside down to expose the 3 factory-applied blue thermal interface pads (for the broad BCM2712 CPU heatspreader, RP1 Southbridge chip, and PMIC power circuitry).
- Peel the clear plastic backing film completely off all 3 blue thermal pads before placing the cooler on [P01].

File: docs/images/real_build/stage1_02_peeling_thermal_film.jpg

VERIFIED — Clear protective backing sheet removed — sticky blue thermal pads exposed and ready

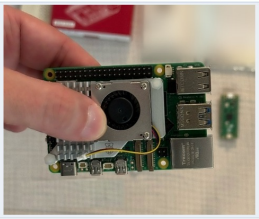


PHOTO 3 — STEP 1.1B: Placing Official Active Cooler [P02] on Top of Pi 5 [P01] (Before Clicking In)

- Turn the Active Cooler [P02] right-side up: 6x7 square aluminum fin grid on the left (above USB-C & HDMI ports) and round black radial blower fan on the right (next to the silver USB & Gigabit Ethernet ports).
- Rest the cooler lightly on the board so the 2 diagonal white spring push-pins sit over the 2 mounting holes.

File: docs/images/real_build/stage1_03_cooler_top_alignment.jpg

VERIFIED — 6x7 fin grid on left + circular blower fan on right positioned over Pi 5 mounting holes

PHOTO 4 — STEP 1.1C: Checking Bottom-Left Push-Pin & CAM/DISP 0 & 1 Clearance (Before Clicking In)

- Before pressing the push-pins down, check the bottom-left white spring push-pin between USB-C and micro-HDMI.
- Notice the two vertical 22-pin camera/display ribbon slots (CAM/DISP 0 & 1) to the right of the micro-HDMI ports — completely unobstructed and ready for your Arducam IMX708 ribbon cable [P06] in Stage 2!

File: docs/images/real_build/stage1_04_hdmi_camdisp_clearance.jpg

VERIFIED — Bottom-left push-pin aligned above hole; both CAM/DISP 0 & 1 ribbon slots 100% clear

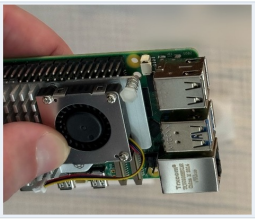


PHOTO 5 — STEP 1.1D: Checking Top-Right Push-Pin & J11 FAN Socket Location (Before Clicking In)

- Still before pressing the push-pins down, inspect the top-right corner beside the top USB 2.0 port:
- The top-right white spring-loaded push-pin sits right below its round gold-ringed PCB mounting hole.
- Directly between that mounting hole and the top USB 2.0 port is the tiny cream-white 4-pin FAN socket (J11).

File: docs/images/real_build/stage1_07_fan_socket_j11_closeup.jpg

VERIFIED — Top-right push-pin aligned with upper PCB hole; J11 4-pin FAN socket located beside USB 2.0

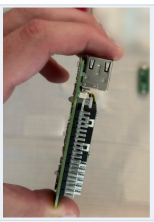


PHOTO 6 — STEP 1.1E: Press Both Push-Pins Down (CLICK!) — Side Profile Confirming Barbs Locked

- Press both white spring-loaded push-pins straight down until they CLICK through the board!
- Look horizontally under the 40-pin GPIO header edge of the Raspberry Pi 5 [P01]:
  Both white plastic locking barbs now protrude all the way through the bottom of the PCB & pads sit flat.

File: docs/images/real_build/stage1_05_gpio_side_profile_pins_locked.jpg

VERIFIED — Both white push-pin barbs clicked through underside of PCB; blue thermal pads compressed flat

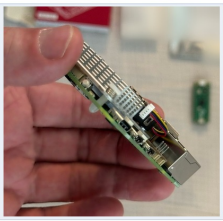


PHOTO 7 — STEP 1.1F: Port-Edge Profile Check Along USB-C, Micro-HDMI & Ethernet (Pins Locked)

- Inspect the opposite edge along USB-C PWR IN, dual micro-HDMI ports, CAM/DISP slots, and Gigabit Ethernet.
- Confirm the bottom-left white push-pin barb is locked through the underside and the heatsink base sits level.

File: docs/images/real_build/stage1_06_ports_edge_profile.jpg

VERIFIED — Bottom-left push-pin barb locked through PCB; heatsink base sits level across all ports

[STAGE 1.1 (PART 2)] Real-World Build Log: Stage 1.1 Removing J11 Dust Cap & Seating 4-Wire Fan Plug (Photos 8–11)

Chronological Photos 8–11 (1 Step Per Row): Locating J11 socket, popping out factory dust cap & plugging fan in.

ARCTIC TRAIL CAM SYSTEM — STAGE 1.1 (PART 2): J11 DUST CAP REMOVAL & 4-WIRE FAN PLUG

Chronological Benchtop Photos 8–11 (1 Step Per Row): Identifying J11 Dust Cap, Popping Out Cap & Seating 4-Wire Fan Plug

PHOTO 8 — STEP 1.1G: Identifying the Factory Plastic Dust Cap Inside the J11 FAN Socket

- With both white push-pins locked in, point directly at the top-right J11 4-pin FAN socket:
- Notice that a brand-new Raspberry Pi 5 [P01] ships from the factory with a tiny removable cream-white plastic dust cap (pick-and-place nozzle cap) sitting inside the J11 socket! You must pop this cap out first.

File: docs/images/real_build/stage1_07b_pointing_at_j11_dust_cap.jpg

VERIFIED — Factory pick-and-place plastic dust cap identified inside the white J11 FAN socket

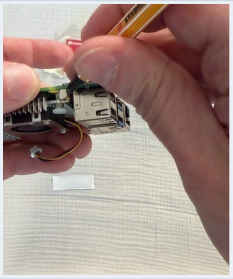


PHOTO 9 — STEP 1.1H: Gently Lifting & Popping Out the Factory Dust Cap from J11

- Use a fine tip, fingernails, or tweezers [T01] to gently lever the tiny cream-white plastic cap straight UP out of the J11 socket (making sure you only lift the inner cap, not the outer socket housing soldered to the PCB).
- The tiny plastic dust cap pops right out with gentle upward pressure.

File: docs/images/real_build/stage1_07c_prying_j11_dust_cap.jpg

VERIFIED — Lifting the removable factory dust cap straight up out of the J11 FAN socket

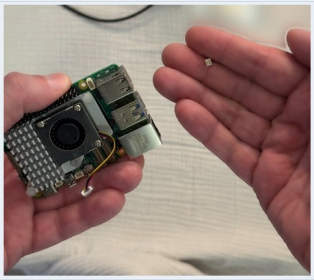


PHOTO 10 — STEP 1.1I: Factory Dust Cap Removed — Open 4-Pin J11 Socket Ready for Fan Plug

- RIGHT HAND: Holds the tiny square cream-white plastic dust cap that was just popped out of J11.
- LEFT HAND: Shows the Raspberry Pi 5 [P01] with the J11 4-pin FAN socket now completely open (4 tiny vertical metal pins visible inside the white housing) and the 4-wire fan plug ready to connect.

File: docs/images/real_build/stage1_07d_j11_cap_removed_socket_open.jpg

VERIFIED — Factory dust cap removed (right hand); 4 metal pins inside J11 socket now open (left hand)

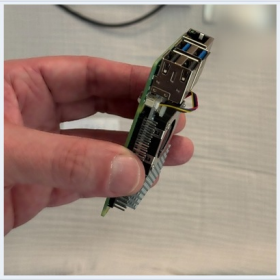


PHOTO 11 — STEP 1.1J: 4-Wire Fan Plug Seated All the Way into J11 — Cooler [P02] 100% Complete!

- Line up the white 4-pin plug on the cooler's 4-wire cable (black/red/yellow/blue) with the open J11 socket (raised key ribs facing the outer board edge, blue wire toward USB ports / black wire toward GPIO header).
- Push straight down until the white plug seats snugly into J11 — Step 1.1 is now 100% finished!

File: docs/images/real_build/stage1_07e_fan_cable_plugged_into_j11.jpg

VERIFIED — COMPLETE: 4-wire fan cable seated in J11; Official Active Cooler [P02] 100% installed!

ARCTIC TRAIL CAM SYSTEM — STAGE 1.4: FLASHING PICO 2 [P07] ON MACBOOK PRO [T02]

Chronological Benchtop Photos 12–17 (1 Step Per Row): USB-C Adapter, Unplugged Trapezoid Alignment, Holding BOOTSEL & Flashing UF2

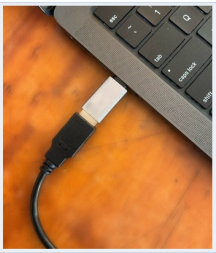


PHOTO 12 — STEP 1.4A: Plug USB-A to USB-C Adapter into Left Side of MacBook Pro [T02]

- Plug the rectangular USB-A data cable into a silver USB-A to USB-C adapter.
- Insert the USB-C plug into the left-side Thunderbolt port of your MacBook Pro (beside esc / tab / caps lock).
- Leave the small Micro-USB end unplugged in your hand until you hold down BOOTSEL in Step 1.4D.

File: docs/images/real_build/stage1_13_usb_c_adapter_in_macbook_pro.jpg

VERIFIED — Data-capable 4-wire USB cable connected to MacBook Pro [T02] (Micro-USB end still unplugged)

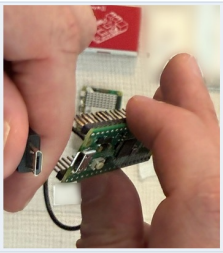


PHOTO 13 — STEP 1.4B: Hold Pico 2 With Headers [P07] & Align Trapezoid Micro-USB Plug (Unplugged)

- Confirm you are holding Pico 2 With Headers [P07] (SKU 828988, 40 pre-soldered metal legs pointing down).
- Before plugging in, match the trapezoid shape of the Micro-USB plug to the Pico 2 silver socket:
- Narrower FLAT face points UP toward the top PCB (BOOTSEL side); wider BEVELED face points DOWN toward the 40 legs.

File: docs/images/real_build/stage1_08_pico2_micro_usb_alignment.jpg

VERIFIED — Pico 2 With Headers [P07] verified & trapezoidal Micro-USB key aligned (still unplugged)

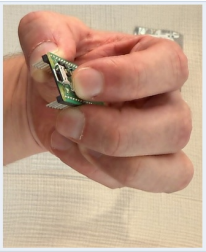


PHOTO 14 — STEP 1.4C: Press & Hold Down the White BOOTSEL Button First (While Still Unplugged)

- With the Micro-USB cable still unplugged, place your index fingertip on the tiny white BOOTSEL button.
- Feel the crisp tactile click as you press and HOLD BOOTSEL down prior to inserting the Micro-USB plug.
- Holding BOOTSEL during USB power-up tells the RP2350 chip to enter USB UF2 bootloader mode.

File: docs/images/real_build/stage1_14_pressing_bootsel_button.jpg

VERIFIED — White BOOTSEL button held down firmly with index fingertip before connecting USB

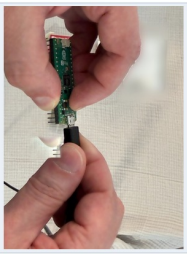


PHOTO 15 — STEP 1.4D: Push Micro-USB Plug Into Pico 2 [P07] While Still Holding BOOTSEL Down

- Keep holding the white BOOTSEL button down firmly with your top hand.
- With your bottom hand, push the aligned Micro-USB plug straight into the Pico 2 silver socket.
- Wait 1 full second after the plug seats, then release your fingertip from the white BOOTSEL button.

File: docs/images/real_build/stage1_15_plugging_cord_while_holding_bootsel.jpg

VERIFIED — Micro-USB connected once while holding BOOTSEL — Pico 2 enters RP2350 BOOTSEL mode

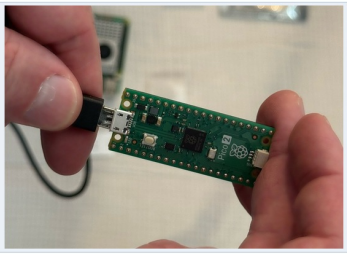


PHOTO 16 — STEP 1.4E: Release BOOTSEL & Check Insertion Depth (Push Past Halfway Catch!)

IMPORTANT: New Micro-USB sockets have a tight metal friction catch halfway in (power-only depth, as shown here).

- After releasing BOOTSEL, push the black Micro-USB plug firmly inward another 2 mm until the 2 tiny silver barbs click in.
- Only ~1 mm of silver metal should remain visible between the black plug housing and the socket lip.

File: docs/images/real_build/stage1_09_pico2_top_bootsel_view.jpg

VERIFIED — BOOTSEL released & Micro-USB pushed past halfway catch for full data-pin engagement

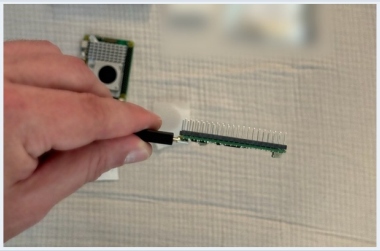


PHOTO 17 — STEP 1.4F: Connected Side Profile & Flash All-in-One UF2 (MicroPython + main.py)

COMMAND: picotool load -v -x pi-diode-trailcam/pico_gatekeeper/arctic_trailcam_pico2_complete.uf2

- Flashes Official Stable MicroPython v1.25.0 (0x10000000) + LittleFS v2 filesystem (0x10100000) in one step.
- Unplug the Micro-USB cord after 'Verifying Flash: 100% OK' before pressing Pico 2 into Breadboard [P09] in Stage 2.

File: docs/images/real_build/stage1_11_pico2_side_profile_over_stage1_bench.jpg

VERIFIED — Verifying Flash: 100% OK — MicroPython v1.25.0 + main.py flashed in a single step!

[STAGE 2.1 REAL BUILD] Real-World Build Log: Stage 2.1 Seating Pico 2 [P07] in Breadboard [P09] (Photos 18–23)

Chronological Photos 18–23 (1 Step Per Row): Pico 2 seated in Rows 1–20 (Cols b & h) beside Raspberry Pi 5 [P01].

ARCTIC TRAIL CAM SYSTEM — STAGE 2.1: SEATING PICO 2 [P07] IN BREADBOARD [P09]

Chronological Benchtop Photos 18–23 (1 Step Per Row): Pressing Pico 2 into Rows 1–20 (Cols b & h) Beside Raspberry Pi 5 [P01]

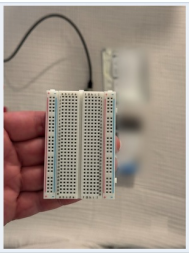


PHOTO 18 — STEP 2.1A: Orient the 30-Row Half-Size Breadboard [P09] Vertically

- Hold the white Inland 400-Hole Half-Size Breadboard [P09] with Row 1 at the TOP and Row 30 at BOTTOM.
- Left terminal strips are Columns a, b, c, d, e; Right terminal strips are Columns f, g, h, i, j.
- The deep vertical groove down the middle is the center ravine (isolates left pins from right pins).

File: docs/images/real_build/stage2_01_empty_breadboard_30rows.jpg

VERIFIED 30-Row Half-Size Breadboard [P09] oriented vertically with Row 1 at top

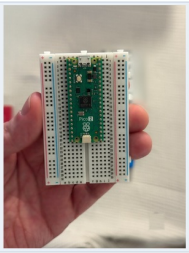


PHOTO 19 — STEP 2.1B: Align Pico 2 [P07] Across the Center Ravine (Rows 1 to 20)

- Point the silver Micro-USB port of Raspberry Pi Pico 2 [P07] UP overhanging Row 1.
- Line up the 20 LEFT metal legs over Column b, Rows 1–20 (leaves Column a: a1–a20 open on the left).
- Line up the 20 RIGHT metal legs over Column h, Rows 1–20 (leaves Columns i & j open on the right).

File: docs/images/real_build/stage2_02_pico2_top_alignment_rows1_to_20.jpg

VERIFIED Pin 1 aligned with Hole b1 and Pin 40 aligned with Hole h1 across center ravine

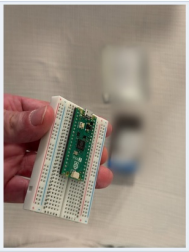


PHOTO 20 — STEP 2.1C: Press Pico 2 [P07] Straight Down Firmly into Rows 1–20

- Press evenly with both thumbs along the top of Pico 2 [P07] so all 40 legs slide straight down.
- Verify open jumper-wire holes remain visible on both outer sides (Col a on left; Cols i & j on right).
- Rows 21–30 at the bottom of the breadboard remain completely open for the QMC5883L sensor [P08].

File: docs/images/real_build/stage2_03_pico2_angled_seated_in_breadboard.jpg

VERIFIED All 40 header legs pressed cleanly into Rows 1–20 (Columns b & h)

PHOTO 21 — STEP 2.1D: Side-Profile Check — Black Header Strip 100% Flush on Breadboard

- Inspect the breadboard from eye level along the side edge.
- The black plastic pin insulator strip underneath Pico 2 [P07] sits 100% flat against the white board.
- Zero exposed silver pin shafts between the black plastic strip and the breadboard surface.

File: docs/images/real_build/stage2_04_pico2_side_profile_flush_on_breadboard.jpg

VERIFIED Black header insulator strip seated 100% flush against white breadboard face

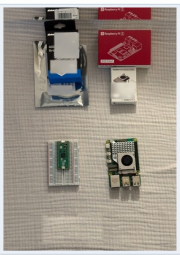


PHOTO 22 — STEP 2.1E: Full Stage 2 Benchtop Inventory & Sub-Assemblies Check

- Top row: Component boxes, SanDisk 64GB MicroSD [P04], and QMC5883L magnetometer [P08] in ESD bag.
- Bottom-left: Breadboard [P09] with Raspberry Pi Pico 2 [P07] seated in Rows 1–20.
- Bottom-right: Raspberry Pi 5 (8GB) [P01] with Official Active Cooler [P02] locked & plugged into J11.

File: docs/images/real_build/stage2_05_bench_overview_boxes_and_boards.jpg

VERIFIED Both core hardware sub-assemblies completed and staged side-by-side on bench

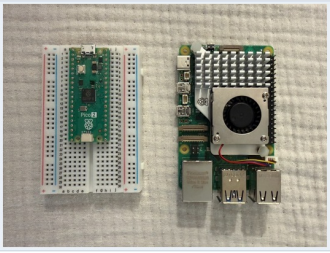


PHOTO 23 — STEP 2.1F: Place Breadboard + Pico 2 on Left, Raspberry Pi 5 on Right (1:1 Match)

- Position Breadboard [P09] + Pico 2 [P07] on the LEFT and Raspberry Pi 5 [P01] on the RIGHT.
- Matches the Stage 2 Breadboard & GPIO Pin Schedule (BENCHTOP_QUICKSTART_GUIDE.md & WIRING_PINOUT.md) 1:1.
- Keeps the short jumper-wire bridge (W5 Purple, W6 Orange, W7 Green) direct and tangle-free.

File: docs/images/real_build/stage2_06_topdown_breadboard_pico2_next_to_pi5.jpg

VERIFIED Stage 2.1 side-by-side board layout complete — ready for jumper wiring!

ARCTIC TRAIL CAM SYSTEM — STAGE 2.2: PRE-STAGING & PLUGGING 7 RAINBOW JUMPER WIRES [P10]

Chronological Benchtop Photos 31–37 (1 Step Per Row): Separating 4 M-M + 3 M-F Wires, Straightening Pins & Wiring Pico 2

PHOTO 31 — STEP 2.2A: Stage ELEGOO Jumper Wire Box [P10], Breadboard [P09] & Pi 5 [P01]

- Unplug all USB power cords before wiring.
- Place the ELEGOO 3-ribbon jumper wire kit [P10] above Breadboard + Pico 2 [P07] and Raspberry Pi 5 [P01].
- Keep the GY-271 (QMC5883L) magnetometer pouch [P08] staged at the top right.

File: docs/images/real_build/stage2_14_elegoo_jumpers_flatlay.jpg

VERIFIED — All Stage 2.2 wiring components staged unpowered on the benchtop cloth

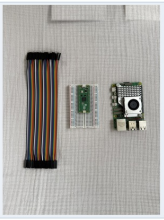


PHOTO 32 — STEP 2.2B: Unpack the 20 cm Color-Coded 40-Wire Dupont Jumper Ribbons [P10]

- Your ELEGOO box contains 3 separate 40-wire ribbons: Male-to-Male (pin-to-pin), Male-to-Female (pin-to-socket), and Female-to-Female (socket-to-socket).
- We only use 4 wires from the Male-to-Male ribbon and 3 wires from the Male-to-Female ribbon (7 wires total).

File: docs/images/real_build/stage2_15_unpacked_40wire_jumper_ribbon.jpg

VERIFIED — 40-wire rainbow jumper ribbon unpacked beside Breadboard [P09] and Pi 5 [P01]

PHOTO 33 — STEP 2.2C: Peel & Pre-Stage Two Separate Wire Groups (4 M-M + 3 M-F = 7 Unique Colors)

- GROUP A (Left — 4 Male-to-Male / Pin-to-Pin wires): Red (W1), Black (W2), Yellow (W4), Blue (W3).
- GROUP B (Right — 3 Male-to-Female / Pin-to-Socket wires): Purple (W5), Green (W7), Orange (W6).
- Every single wire has a unique color — zero duplicate colors anywhere on the board!

File: docs/images/real_build/stage2_16_4_male_male_and_3_male_female_wires_selected.jpg

VERIFIED — 4 Male-to-Male wires (Red/Black/Yellow/Blue) + 3 Male-to-Female wires (Purple/Green/Orange) staged

PHOTO 34 — STEP 2.2D: Unbox HiLetgo GY-271 (QMC5883L) [P08] & 5-Pin Male Header Strip

- Remove the blue GY-271 (HW-246 / QMC5883L) magnetometer breakout board and its 5-pin black male header strip from the pink anti-static bag.
- Notice the 5 labeled through-holes along the left edge: VCC, GND, SCL, SDA, and DRDY (5th bottom pin unused).

File: docs/images/real_build/stage2_17_unboxed_gy271_compass_and_5pin_header.jpg

VERIFIED — Blue GY-271 (QMC5883L) PCB and 5-pin male header strip unboxed and verified

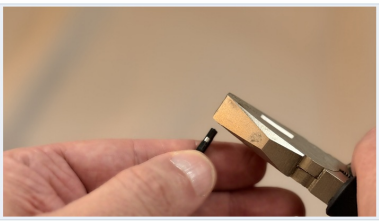


PHOTO 35 — STEP 2.2E: Pinch Black Plastic Housing 2mm Above Tip & Keep Pins 90° Straight

IMPORTANT: Brand-new breadboard spring clips are stiff! Never push a jumper wire while holding the soft wire.

- Always pinch the square black plastic housing 2 mm above the metal tip and push straight down vertically (90°).
- If a pin bends, squeeze it flat between the smooth jaws of your flat-nose pliers as shown here.

File: docs/images/real_build/stage2_18_straightening_jumper_pin_with_pliers.jpg

VERIFIED — Male jumper pin straightened to 90° with smooth flat-nose pliers before vertical insertion

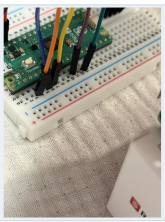


PHOTO 36 — STEP 2.2F: Plug All 7 Jumper Wires into Pico 2 Rows (Col a on Left, Col j on Right)

- Right side of Pico 2 (Column j): Red (W1) -> Hole j5 (3V3_OUT); Black (W2) -> Hole j3 (GND).
- Left side of Pico 2 (Column a): Orange (W6) -> a4 (GP2); Blue (W3) -> a6 (GP4 SDA); Yellow (W4) -> a7 (GP5 SCL); Purple (W5) -> a11 (GP8 TX); Green (W7) -> a13 (GND).

File: docs/images/real_build/stage2_19_w1_to_w4_wired_on_breadboard_top_view.jpg

VERIFIED — All 7 color-coded jumper wires seated in exact Pico 2 rows (j3, j5, a4, a6, a7, a11, a13)

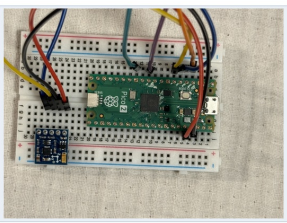


PHOTO 37 — STEP 2.2G: Angled Inspection of All 7 Jumper Wires Seated on Breadboard [P09]

- Verify every black pin housing sits upright and snug in its assigned row alongside Pico 2 [P07].
- Lower ends of Red (W1), Black (W2), Yellow (W4), and Blue (W3) route cleanly down to Column f (Rows 24–27).
- Female socket ends of Purple (W5), Green (W7), and Orange (W6) route to Raspberry Pi 5 [P01] on the right.

File: docs/images/real_build/stage2_20_all_7_wires_on_breadboard_rows_visible.jpg

VERIFIED — Step 2.2 breadboard jumper harness complete and verified across all 7 rows

[STAGE 2.3A REAL BUILD] Real-World Build Log: Stage 2.3A GY-271 5-Pin Header Twist & Col j Seating (Photos 38–42)

Chronological Photos 38–42 (1 Step Per Row): 4-pin pliers press-fit twist, W1–W4 in Col f, and GY-271 in Col j facing right.

ARCTIC TRAIL CAM SYSTEM — STAGE 2.3 (PART A): GY-271 COMPASS [P08] PRESS-FIT & COL J SEATING

Chronological Benchtop Photos 38–42 (1 Step Per Row): 5-Pin Header Orientation, Pliers Twist Lock & Column j Placement

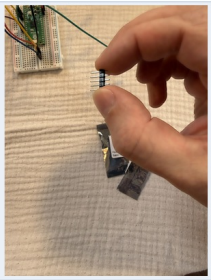


PHOTO 38 — STEP 2.3A: Hold 5-Pin Header Strip With SHORT GOLD PINS UP & LONG GOLD PINS DOWN

- Inspect the black 5-pin male header strip from the side before sliding the blue GY-271 board onto it:
- The SHORT gold pins (~3.0 mm) must point UP (to slide through the 5 holes of the blue GY-271 PCB).
- The LONG gold pins (~6.0 mm) must point DOWN (to plug deep into the breadboard spring clips in Column j).

File: docs/images/real_build/stage2_21_holding_5pin_header_short_up_long_down.jpg

VERIFIED **5-pin header oriented correctly: SHORT gold pins UP, LONG gold pins DOWN**

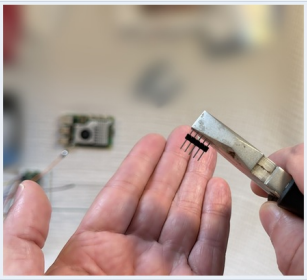


PHOTO 39 — STEP 2.3B: Twist the 4 Active Short Pins (15°–30°) With Pliers for Unsoldered Press-Fit

IMPORTANT: Unsoldered round through-holes (0.90mm) are wider than flat 0.64mm pins — straight pins wobble!

- Grip each of the 4 active SHORT gold pins (VCC, GND, SCL, SDA) with flat-nose pliers and twist axially ~15°–30°.
- Twisting makes the 0.905mm square pin diagonal bite snugly inside the gold plated holes like a press-fit pin!

File: docs/images/real_build/stage2_22_pliers_zigzag_bend_for_snug_unsoldered_fit.jpg

VERIFIED **4 active short gold pins twisted ~15°–30° axially for solid unsoldered press-fit contact**

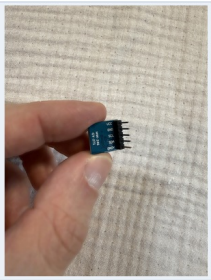


PHOTO 40 — STEP 2.3C: Press Blue GY-271 Compass Board Snugly Down Onto the Short Gold Pins

- Hold the blue GY-271 board with the black square sensor chip and printed white labels (VCC..DRDY) facing UP.
- Slide all 5 holes down over the short gold pins until the blue PCB sits flush against the black plastic strip.
- Orient the blue board so it extends out to the RIGHT away from the breadboard center ravine.

File: docs/images/real_build/stage2_23_gy271_slid_onto_5pin_header.jpg

VERIFIED **GY-271 PCB (chip & labels facing UP) locked snugly onto twisted short header pins**

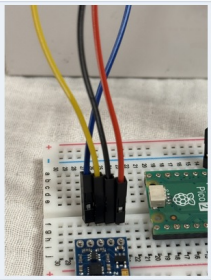


PHOTO 41 — STEP 2.3D: Plug W1–W4 Into Column f (f24–f27) & 5-Pin Header Into Column j (j24–j28)

IMPORTANT: Count EMPTY HOLES from the ends (do not trust printed row numbers 24–28 on the plastic!):

- Column f (4 Wires: Red, Black, Yellow, Blue): Leave 3 empty holes below Pico 2 AND 3 empty holes at bottom!
- Column j (5-Pin Strip): Leave 3 empty holes below Pico 2 AND ONLY 2 EMPTY HOLES at the bottom (j29, j30)!

File: docs/images/real_build/stage2_24_w1_to_w4_in_col_f_and_5pin_strip_in_col_j.jpg

VERIFIED **W1–W4 in Column f (3 empty bottom holes) aligned 1:1 with 5-pin strip in Column j (2 empty bottom holes)**

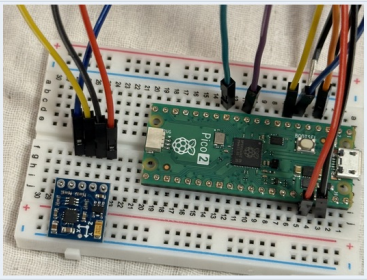


PHOTO 42 — STEP 2.3E: Completed Breadboard With Blue GY-271 Seated in Column j Facing RIGHT

- With the 5 long pins pressed into Column j (j24=VCC, j25=GND, j26=SCL, j27=SDA, j28=DRDY), the blue GY-271 board hangs out cleanly to the RIGHT over the +/- rails with 3 open holes (Cols g, h, i) between it and Col f!
- Every wire color matches its compass pin 1:1 across the row: Red->VCC, Black->GND, Yellow->SCL, Blue->SDA.

File: docs/images/real_build/stage2_25_completed_breadboard_with_gy271_in_col_j_facing_right.jpg

VERIFIED **COMPLETE: GY-271 seated in Column j facing right with 100% row alignment to W1–W4**

ARCTIC TRAIL CAM SYSTEM — STAGE 2.3 (PART B): PI 5 [P01] OUTER-RIGHT GPIO COLUMN VERIFICATION

Chronological Benchtop Photos 43–46 (1 Step Per Row): Side, Angled & Top Verification of Pins 10 (Purple), 14 (Green) & 16 (Orange)

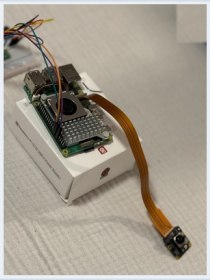


PHOTO 43 — STEP 2.3F: Side-Profile Check of Raspberry Pi 5 [P01] Outer-Right GPIO Column

- Look along the outer-right edge of the Raspberry Pi 5 [P01] 40-pin GPIO header:
- All 20 INNER-LEFT pins (closest to the aluminum heatsink fins) remain 100% BARE / EMPTY!
- Counting down the OUTER-RIGHT column from top-right: #1–#4 bare, #5=Purple, #6 bare, #7=Green, #8=Orange.

File: docs/images/real_build/stage2_26_pi5_gpio_outer_column_verified_side_view.jpg

VERIFIED **Side profile confirms Purple (#5), skip 1 (#6 bare), Green (#7), Orange (#8) on outer column**

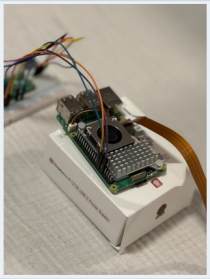


PHOTO 44 — STEP 2.3G: Angled Check Confirming All 20 Inner GPIO Pins Are 100% Bare

- Inspect from an angled view above the Active Cooler [P02] and 40-pin GPIO header:
- Verify that Purple (Pin 10 / GPIO15 RX), Green (Pin 14 / GND), and Orange (Pin 16 / GPIO23 DONE) sit strictly on the outer-right row of gold pins with zero sockets accidentally pushed onto the inner row.

File: docs/images/real_build/stage2_27_pi5_gpio_outer_column_verified_angled_view.jpg

VERIFIED **Angled inspection confirms zero inner-row pins touched; outer-right pins 10, 14, 16 locked**

PHOTO 45 — STEP 2.3H: Top-Down View of Verified GPIO Sockets & J4 (CAM/DISP 0) Ribbon Slot

- Top-down view showing the 3 female sockets (Purple, Green, Orange) on the outer-right GPIO header.
- Also shows the two vertical 22-pin camera/display slots between the micro-HDMI ports and Gigabit Ethernet: the upper slot is J4 (CAM/DISP 0), ready for the narrow 22-pin end of your orange camera cable in Step 2.4!

File: docs/images/real_build/stage2_28_pi5_gpio_and_cam_disp0_top_view.jpg

VERIFIED **Outer-right GPIO wiring verified; upper J4 (CAM/DISP 0) slot located for Step 2.4**

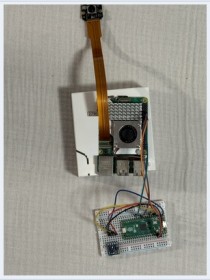


PHOTO 46 — STEP 2.3I: Full Golden-Path Overview of Completed Stage 2.3 Benchtop Wiring

- Left: 30-Row Breadboard [P09] with Pico 2 [P07], 7-color jumper harness [P10], and GY-271 [P08] in Column j.
- Right: Raspberry Pi 5 [P01] with Active Cooler [P02], SanDisk 64GB MicroSD [P04] in J9, and W5–W7 on Pins 10/14/16.
- Top right: Camera Module 3 NoIR Wide [P05] box staged for Step 2.4.

File: docs/images/real_build/stage2_29_completed_stage2_3_full_assembly_golden.jpg

VERIFIED **COMPLETE: Stage 2.3 7-wire breadboard & Pi 5 GPIO assembly 100% verified!**

ARCTIC TRAIL CAM SYSTEM — STAGE 2.4: CONNECTING 200MM ORANGE CSI CAMERA RIBBON [P06]

Chronological Benchtop Photos 47–53 (1 Step Per Row): Wide vs. Narrow Ends, Gold Teeth Orientation, Pi 5 J4 & Camera Latch Lock

PHOTO 47 — STEP 2.4A-1: Stage the 3 White Boxes for Steps 2.4–2.6 ([P05], [P06] & 27W USB-C PSU)

- Left: Official Raspberry Pi 27W USB-C Power Supply (for powering Raspberry Pi 5 [P01] in Step 2.6).
- Center: Official Raspberry Pi 5 Camera Cable (Standard - Mini | 200mm) [P06] (orange flexible ribbon).
- Right: Raspberry Pi Camera Module 3 NoIR Wide [P05] (12MP Sony IMX708 wide-angle night-vision sensor).

File: docs/images/real_build/stage2_30a_three_white_boxes_golden.jpg

VERIFIED 27W USB-C PSU, 200mm Pi 5 Camera Ribbon [P06], and Camera Module 3 NoIR Wide [P05] staged

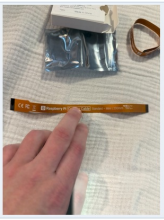


PHOTO 48 — STEP 2.4A-2: Inspect Orange Ribbon SIDE A (Black Stiffeners & White Printed Text Side)

- Lay the 200mm orange ribbon [P06] flat with the black stiffener strips and white text facing UP (Side A):
- LEFT END (WIDE 15-Pin Standard, 16.0 mm wide, labeled 'CE FC'): Plugs into Camera Module 3 [P05].
- RIGHT END (NARROW 22-Pin Mini, 11.5 mm wide, labeled 'Mini | 200mm'): Plugs into Pi 5 J4 (CAM/DISP 0).

File: docs/images/real_build/stage2_32_flat_camera_ribbon_wide_vs_narrow_ends.jpg

VERIFIED Side A verified: Wide 15-pin ('CE FC') for Camera on left; Narrow 22-pin ('Mini | 200mm') for Pi 5 on right

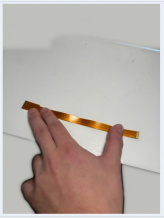


PHOTO 49 — STEP 2.4A-3: Flip Ribbon Over to Inspect SIDE B (Shiny Gold Contact Teeth on Both Ends)

- Flip the orange ribbon over so the plain orange side faces UP (Side B):
- Notice that BOTH sets of shiny gold metal contact teeth (15 wide teeth on left, 22 fine teeth on right) are on this exact same side (Side B), while the black stiffeners back them on Side A.

File: docs/images/real_build/stage2_33_flat_camera_ribbon_gold_teeth_side.jpg

VERIFIED Side B verified: Shiny gold contact teeth exposed on both the 15-pin and 22-pin ends

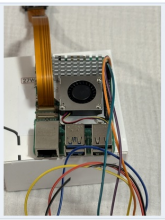


PHOTO 50 — STEP 2.4A-4: Lock NARROW 22-Pin End ('Mini | 200mm') Into Pi 5 Upper Slot J4 (CAM/DISP 0)

- Lift the brown plastic latch on upper slot J4 (CAM/DISP 0) straight UP by ONLY 1.5 mm with two fingernails.
- Slide the NARROW 22-pin end 2 mm deep into J4: GOLD TEETH face RIGHT (toward Ethernet), BLACK STIFFENER faces LEFT (toward Micro-HDMI). Support Pi 5 on the white PSU box and press the brown latch straight down flush!

File: docs/images/real_build/stage2_34_pi5_j4_ribbon_seated_and_locked.jpg

VERIFIED Narrow 22-pin end seated 2mm deep in J4 (gold teeth toward Ethernet) & brown latch locked flush

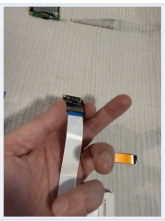


PHOTO 51 — STEP 2.4B-1: Inspect Factory White Ribbon Orientation on Camera Module 3 NoIR Wide [P05]

- Remove Camera Module 3 NoIR Wide [P05] from its ESD bag and look at the back of the black camera PCB:
- Notice how the factory white Pi 4 cable is installed: its BLUE stiffener tape faces UP (on the brown latch side, away from the black PCB), while its metal teeth face DOWN against the 15 silver solder pins on the black PCB.

File: docs/images/real_build/stage2_35_camera_module3_factory_white_ribbon.jpg

VERIFIED Factory white ribbon orientation noted: stiffener UP on latch side, metal teeth DOWN against PCB

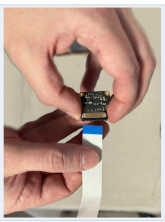


PHOTO 52 — STEP 2.4B-2: Slide Camera Brown Latch Out 1.5 mm & Pull Out Factory White Ribbon

- Use two thumbnails on the left and right ears of the camera's brown horizontal latch to slide it out 1.5 mm.
- Do NOT pull the brown latch completely off the connector housing.
- Slide the factory white ribbon straight out and store it in the box.

File: docs/images/real_build/stage2_36_camera_module3_white_ribbon_removed.jpg

VERIFIED Camera brown horizontal slide-latch opened 1.5mm and factory white ribbon removed

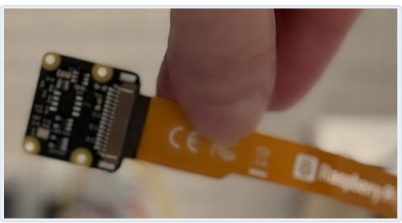


PHOTO 53 — STEP 2.4B-3: Insert WIDE 15-Pin End ('CE FC') Into Camera & Lock Brown Latch Flush

- Slide the WIDE 15-pin end ('CE FC') of your orange ribbon 2 mm deep into the camera slot:
- SHINY GOLD TEETH face DOWN against the black camera PCB; BLACK STIFFENER + WHITE 'CE FC' LETTERS face UP (visible on the brown-latch side, zero gold teeth showing outside). Push both ears of the brown latch closed!

File: docs/images/real_build/stage2_37_camera_module3_orange_ribbon_seated_and_locked.jpg

VERIFIED COMPLETE: Wide 15-pin orange ribbon seated 2mm deep (gold teeth down, 'CE FC' up) & latch locked!

ARCTIC TRAIL CAM SYSTEM — STAGE 2.4C–2.6: PIN CHECK, HOLE-COUNT VERIFICATION & POWERED GOLDEN BUILD

Chronological Benchtop Photos 54–58 (1 Step Per Row): Fixing Hooked Pin Tips, Verifying 2 Bottom Holes in Col j & Full Powered System

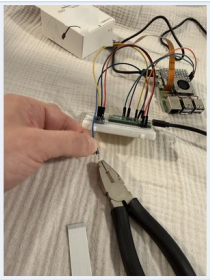


PHOTO 54 — STEP 2.4C-1: Inspect Any Jumper Pin That Resists Seating for a Bent 'J-Hook' Tip

IMPORTANT: If a male jumper pin feels spongy or refuses to seat flush in a stiff breadboard hole, pull it out!

- As shown on this Blue SDA (W3) pin, a tip bent into a J-hook either rides on top of the metal clip (open circuit) or hooks sideways inside the slot where it can touch an adjacent row.

File: docs/images/real_build/stage2_38_blue_wire_hook_pin_risk.jpg

VERIFIED **Bent J-hook tip on Blue W3 jumper pin identified before re-insertion**

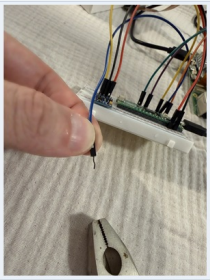


PHOTO 55 — STEP 2.4C-2: Squeeze Hooked Pin Tip Dead-Flat (90°) With Smooth Flat-Nose Pliers

- Clamp the metal pin shaft between the smooth flat jaws of your pliers and squeeze firmly until 100% straight.
- Re-insert by pinching the black plastic collar 2 mm above the pin and pushing straight down at 90°.
- Always verify solid vertical seating before applying the peelable Permatex silicone damper bead in Stage 5.

File: docs/images/real_build/stage2_39_straightening_hook_pin_with_pliers.jpg

VERIFIED **Hooked jumper pin straightened 100% flat with smooth pliers and re-seated vertically**

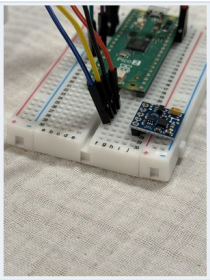


PHOTO 56 — STEP 2.4D-1: Close-Up Verification of the Bottom Empty-Hole Rule (Col f = 3, Col j = 2)

IMPORTANT: Because the Compass header has 5 pins (VCC, GND, SCL, SDA, DRDY) while there are 4 wires:

- Column f (4 Wires: Red, Black, Yellow, Blue) has 3 EMPTY HOLES below Pico 2 and 3 EMPTY HOLES at bottom.
- Column j (5-Pin Compass Strip) has 3 EMPTY HOLES below Pico 2 and ONLY 2 EMPTY HOLES at bottom (j29, j30)!

File: docs/images/real_build/stage2_40_compass_shifted_down_2_empty_bottom_holes_closeup.jpg

VERIFIED **Verified: 3 empty bottom holes under 4 wires in Col f; 2 empty bottom holes under 5-pin strip in Col j**

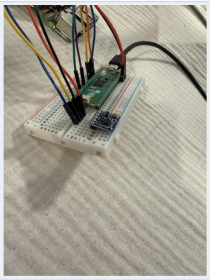


PHOTO 57 — STEP 2.4D-2: Overhead Golden-Path Verification of Completed Breadboard [P09]

- Straight-down verification of the finished breadboard assembly:
- Pico 2 [P07] in Rows 1–20, 3 empty rows (21–23), W1–W4 in f24–f27 (3 empty bottom holes), and GY-271 [P08] in j24–j28 (2 empty bottom holes) facing right.

File: docs/images/real_build/stage2_41_completed_breadboard_topdown_golden.jpg

VERIFIED **Overhead golden-path breadboard wiring & GY-271 alignment 100% verified**

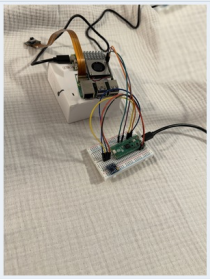


PHOTO 58 — STEP 2.5/2.6: Complete Stage 2 Golden-Path Benchtop System Powered & Ready!

- CORD 1 (Black Micro-USB): Plugged firmly into Pico 2 [P07] on Breadboard [P09] -> MacBook Pro [T02] left USB-C.
- CORD 2 (White 27W USB-C Power Supply): Plugged into Raspberry Pi 5 [P01] USB-C PWR IN -> wall outlet.
- Camera Module 3 NoIR Wide [P05] connected via 200mm orange ribbon [P06] — Stage 2 hardware 100% complete!

File: docs/images/real_build/stage2_42_complete_benchtop_system_powered_golden.jpg

VERIFIED **COMPLETE: Full Stage 2 benchtop hardware assembled, powered, and verified!**

ARCTIC TRAIL CAM SYSTEM — STAGE 2.5: MICRO-USB CORD, MAC SD STAGING & PI 5 SLOT

Chronological Benchtop Photos 24–30 (1 Step Per Row): Connecting CORD 1 to Pico 2, Staging Code on Mac & Seating MicroSD in Pi 5

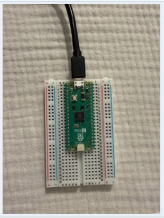


PHOTO 24 — STEP 2.5A: Plug Black Micro-USB Data Cord (CORD 1) Firmly into Breadboard Pico 2

- Do NOT hold down the white BOOTSEL button (MicroPython v1.25.0 + main.py is already flashed!).
- Push the black Micro-USB plug firmly all the way into Pico 2 [P07] past the 2 tiny silver barbs.
- Pico 2 boots straight into main.py and enumerates on MacBook Pro as /dev/cu.usbmodem*.

File: docs/images/real_build/stage2_07_micro_usb_cord_connected_to_breadboard_pico2.jpg

VERIFIED — CORD 1 seated firmly in Pico 2 [P07] on Breadboard [P09] (normal application boot)

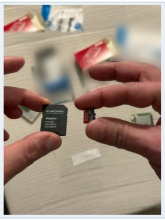


PHOTO 25 — STEP 2.5B: Take SanDisk 64GB MicroSD Card [P04] & Full-Size SanDisk Adapter

- Left hand: Full-size black SanDisk microSD-to-SD Adapter sleeve (check left Lock switch is UP).
- Right hand: Red/grey SanDisk Ultra PLUS 64GB MicroSD Card [P04].
- Used whenever flashing Raspberry Pi OS or pre-staging updated sender code from MacBook Pro [T02].

File: docs/images/real_build/stage2_08_sandisk_64gb_microsd_and_adapter.jpg

VERIFIED — SanDisk Ultra PLUS 64GB MicroSD Card [P04] and SanDisk Adapter ready in hand

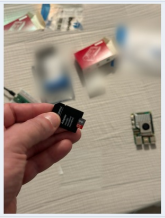


PHOTO 26 — STEP 2.5C: Align & Slide MicroSD Card [P04] into Bottom of SanDisk Adapter

- Hold both the black SanDisk Adapter and the 64GB MicroSD Card [P04] with printed labels facing UP.
- Slide the narrow top edge of the MicroSD card straight into the bottom slot of the adapter.
- Never flip the MicroSD upside down relative to the adapter label.

File: docs/images/real_build/stage2_09_sliding_microsd_into_adapter.jpg

VERIFIED — MicroSD [P04] aligned label-up and sliding into bottom slot of SanDisk Adapter

PHOTO 27 — STEP 2.5D: Push MicroSD Card [P04] Flush Inside the SanDisk Adapter

- Push the bottom edge of the 64GB MicroSD card with your thumb until it seats completely flush.
- Verify the small white write-protect switch on the left edge of the adapter is slid UP (unlocked).
- The adapter can now plug directly into the MacBook Pro SDXC reader slot.

File: docs/images/real_build/stage2_10_microsd_seated_flush_in_adapter.jpg

VERIFIED — MicroSD Card [P04] seated 100% flush inside full-size SanDisk Adapter

PHOTO 28 — STEP 2.5E: Insert Adapter into MacBook Pro [T02] & Stage Refactored Code Cleanly

- Push the SanDisk Adapter into the right-side SDXC slot of MacBook Pro [T02] (beside return / shift).
- Automated stage_sd_card_without_macos_metadata() disables Spotlight (mdutil -i off /Volumes/bootfs).
- Copies refactored pi5_sender_node/ with COPYFILE_DISABLE=1 (0 ._* or .DS_Store files) + UART config.

File: docs/images/real_build/stage2_11_sd_adapter_in_macbook_pro_slot.jpg

VERIFIED — /Volumes/bootfs updated with refactored code (zero macOS junk) & cleanly ejected

PHOTO 29 — STEP 2.5F: Flip Raspberry Pi 5 [P01] Upside Down & Align MicroSD Card in Slot J9

- Pull the 64GB MicroSD Card [P04] out of the adapter and flip Raspberry Pi 5 [P01] upside down.
- Hold the MicroSD card so its SHINY GOLD CONTACT TEETH face UP toward the green Pi 5 PCB.
- The red/grey printed SanDisk label faces AWAY from the green board (visible to your eyes).

File: docs/images/real_build/stage2_12_sliding_microsd_into_pi5_underside_slot.jpg

VERIFIED — Gold contact teeth facing UP toward green Pi 5 PCB; label facing away in slot J9

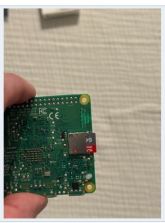


PHOTO 30 — STEP 2.5G: Push SanDisk 64GB MicroSD [P04] All the Way Flush into Slot J9

- Slide the MicroSD card straight into the silver metal J9 slot until it stops firmly (~12 mm deep).
- Raspberry Pi 5 [P01] uses a friction-fit MicroSD slot (no spring click) — card lip sits flush with PCB edge.
- Flip Raspberry Pi 5 [P01] back right-side up beside Breadboard [P09].

File: docs/images/real_build/stage2_13_microsd_seated_flush_in_pi5_underside_slot.jpg

VERIFIED — COMPLETE: SanDisk 64GB MicroSD [P04] seated flush in J9 — Pi 5 ready for boot!

ARCTIC TRAIL CAM SYSTEM — STAGE 3 (PART 1): UNBOXING MEDIA CONVERTERS & ETHERNET CABLING

Chronological Benchtop Photos 59–64 (1 Step Per Row): StarTech MCMGBSCSM10 [P13], 5V DC Bricks & Cat6 Ethernet to Pi 5 / MacBook

PHOTO 59 — STEP 3.0A: Stage the 4 Optical Fiber & USB-Barrel Power Packages ([P14]–[P16], [P19])

- Top-Left: StarTech USB-to-Type-M 5V Barrel Power Cables [P19] (for 5V USB field/bench power).
- Top-Right: Duttek 1x2 Single-Mode SC/UPC Optical Fiber Splitter [P14] (1 White Input -> 1 Blue + 1 Orange Output).
- Bottom-Left: Tripp Lite 2m Duplex Yellow OS2 Single-Mode Patch Cable [P16]; Bottom-Right: Blue SC Couplers [P15].

File: docs/images/real_build/stage3_01_four_fiber_and_power_packages_staged.jpg

VERIFIED — All 4 optical passive & power cable packages ([P14], [P15], [P16], [P19]) staged on bench

PHOTO 60 — STEP 3.0B: Stage Two Identical StarTech MCMGBSCSM10 Gigabit Fiber Converters [P13]

- Left Box (Media Converter #1 — Field / Pi 5 Side): Converts Pi 5 Gigabit Ethernet into 1310nm single-mode laser.
- Right Box (Media Converter #2 — Collector / MacBook Side): Receives 1310nm laser over a single RX fiber strand.
- Both units use 1000BASE-LX single-mode SC transceivers (up to 10 km reach over 9/125 µm OS2 glass).

File: docs/images/real_build/stage3_02_two_startech_media_converter_boxes.jpg

VERIFIED — Two identical StarTech MCMGBSCSM10 1000BASE-LX SC media converters [P13] verified

PHOTO 61 — STEP 3.0C: Unbox Both Black StarTech Media Converters & +5.0V DC Wall Adapters

CAUTION — 5V ONLY RULE: Each StarTech MCMGBSCSM10 operates on strictly +5.0V DC (center-positive).

- On the indoor bench, power both boxes using their included 5V DC wall bricks (or 5V USB-to-Type-M cables [P19]).
- NEVER connect 12V battery power directly to the 5V DC barrel jack on the back of the media converter!

File: docs/images/real_build/stage3_03_unboxed_startech_converters_with_5v_power_bricks.jpg

VERIFIED — Both StarTech MCMGBSCSM10 chassis and +5.0V DC center-positive power bricks unboxed

PHOTO 62 — STEP 3.0D: Inspect Front Panel Ports (TX, RX, 4 LEDs, LFP Switch, RJ45) & Rear 5V Jack

- Front left: Duplex SC optical transceiver covered by a black rubber factory dust cap (TX on left, RX on right).
- Front center: 4 status LEDs (Fiber LNK/ACT, TP LNK/ACT, PWR, 1000) and the LFP (Link Fault Pass-Through) switch.
- Front right: Shielded Gigabit RJ45 copper Ethernet jack; Rear: +5V DC barrel socket + grounding screw.

File: docs/images/real_build/stage3_04_startech_front_ports_and_5v_dc_jack_closeup.jpg

VERIFIED — Front optical/copper ports, 4 LEDs, LFP slide switch, and rear +5V DC barrel jack inspected

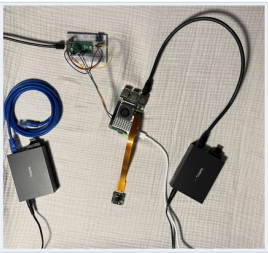


PHOTO 63 — STEP 3.1A: Connect Black Cat6 Cable (Pi 5 -> Box #1) & Blue Cat6 Cable (Box #2 -> Mac)

- Field Box #1 (Right): Plug Black Cat6 Ethernet cable [P23] between Raspberry Pi 5 [P01] and Box #1 RJ45.
- Collector Box #2 (Left): Plug Blue Cat6 Ethernet cable [P23] into Box #2 RJ45.
- Pull the black rubber factory dust caps out of the SC optical ports on both Media Converter #1 and #2.

File: docs/images/real_build/stage3_05_both_startech_boxes_ethernet_cables_connected.jpg

VERIFIED — Black Cat6 connected between Pi 5 & Box #1; Blue Cat6 connected to Box #2

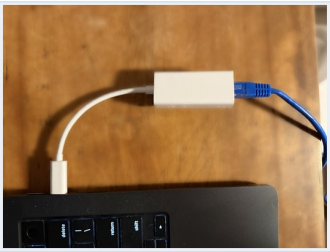


PHOTO 64 — STEP 3.1B: Plug Blue Cat6 Ethernet Cable from Box #2 Into MacBook USB-C Adapter [P24]

- Plug the other end of the Blue Cat6 Ethernet cable [P23] into your USB-C Gigabit Ethernet Adapter [P24] on MacBook Pro.
- Configure static IP 10.10.0.100/24 on the USB Ethernet interface (e.g., sudo ifconfig en10 10.10.0.100/24 up).
- The copper Ethernet path from Box #2 to your MacBook Pro Collector Hub is now physically connected.

File: docs/images/real_build/stage3_06_blue_ethernet_in_macbook_usbc_adapter.jpg

VERIFIED — Blue Cat6 Ethernet cable seated in USB-C Gigabit Ethernet adapter [P24] on MacBook Pro

ARCTIC TRAIL CAM SYSTEM — STAGE 3 (PART 2): SPLITTER, COUPLER & UNZIPPING SINGLE-STRAND FIBER

Chronological Benchtop Photos 65–71 (1 Step Per Row): Duttek 1x2 Splitter [P14], Blue Coupler [P15] & Removing Duplex Clip / Label [P16]

PHOTO 65 — STEP 3.1C: Pull Clear Dust Caps Off the Blue & Orange Output Legs of 1x2 Splitter [P14]

- Unbag the Duttek 1x2 SC/UPC Single-Mode Optical Splitter [P14] and locate its 2 output legs (Blue & Orange cords).
- Pull the clear plastic protective dust caps straight off the blue square SC/UPC plugs on both output legs.
- Never touch the white 2.5mm ceramic ferrule tip at the end of each plug (9 µm laser core at center!).

File: docs/images/real_build/stage3_07_duttek_blue_and_orange_plugs_caps_removed.jpg

VERIFIED — Clear dust caps removed from Blue (Leg 1) and Orange (Leg 2) output plugs of 1x2 splitter

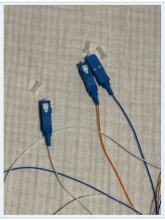


PHOTO 66 — STEP 3.1D: Identify All 3 Color-Coded Cords on the 1x2 Optical Splitter [P14]

- WHITE Cord (Left — 1x Input): Plugs into Media Converter #1 'TX' (receives 100% of Pi 5's outbound 1310nm laser).
- BLUE Cord (Top-Right — Output Leg 1): Loops back into Media Converter #1 'RX' (satisfies local link check).
- ORANGE Cord (Bottom-Right — Output Leg 2): Sends identical 50% laser copy out to Blue Coupler [P15] -> Box #2 'RX'.

File: docs/images/real_build/stage3_08_duttek_splitter_all_three_cords_white_blue_orange.jpg

VERIFIED — All 3 splitter legs uncapped and identified: White = TX Input, Blue = Box #1 RX, Orange = Outbound

PHOTO 67 — STEP 3.1E: Pull Both Black Dust Caps Off the Blue SC/UPC Female-to-Female Coupler [P15]

- Take one Blue Cable Matters SC/UPC Simplex Coupler [P15] and pull off both black plastic dust caps.
- Inside the blue coupler is a precision ceramic alignment sleeve that mates two SC/UPC ferrules tip-to-tip.
- Notice the rectangular U-slot keyway on one flat side of the coupler (matches the raised key tab on each SC plug).

File: docs/images/real_build/stage3_09_blue_sc_coupler_black_dust_caps_removed.jpg

VERIFIED — Blue SC/UPC coupler [P15] uncapped and keyway slot inspected

PHOTO 68 — STEP 3.1F: Unbag the 2-Meter Yellow OS2 Single-Mode Duplex Patch Cable [P16]

- Remove the yellow Tripp Lite N356-02M (2m, OS2 9/125 µm SC/SC) duplex fiber cable [P16] from its bag.
- By default, a duplex patch cable has TWO yellow fiber strands joined side-by-side by plastic clips at both ends.
- To create a true physical 1-way data diode (1 strand total, 0 return strand), we separate 1 single yellow strand!

File: docs/images/real_build/stage3_10_yellow_duplex_fiber_cable_unbagged.jpg

VERIFIED — 2m yellow OS2 single-mode duplex fiber patch cord [P16] unbagged on bench

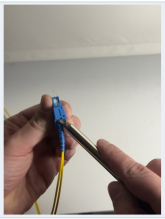


PHOTO 69 — STEP 3.1G: Pop Off the Blue Plastic Duplex Joining Clips at Both Ends of Yellow Fiber

- Slip the tip of a small screwdriver or tweezers [T01] under the center tab of the outer duplex clip holding the two blue SC plugs together, and gently pop the clip off.
- Do NOT remove the main rectangular blue sliding housing of the SC connector itself — only the outer bridge clip!

File: docs/images/real_build/stage3_11_prying_open_duplex_clip_on_yellow_fiber.jpg

VERIFIED — Outer plastic duplex bridge clip popped off cleanly without disturbing SC connector shells

PHOTO 70 — STEP 3.1H: Verify Both Blue SC Plugs Are Freed From the Outer Duplex Clip

- With the outer plastic duplex clip removed (shown at bottom-left), the two blue SC/UPC plugs separate cleanly.
- Repeat on the other end of the 2-meter yellow cable so both ends have independent SC/UPC plugs.
- Next, check along the yellow cord jacket for the white adhesive identification wrap label before unzipping.

File: docs/images/real_build/stage3_12_duplex_clip_removed_two_yellow_fiber_plugs_separated.jpg

VERIFIED — Outer duplex clip removed; individual blue SC/UPC connectors freed

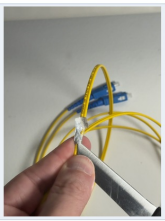


PHOTO 71 — STEP 3.1I: Peel Off the White Wrap-Around Label Binding the Two Yellow Jackets

- Near one end of the yellow zip-cord is a white adhesive wrap-around label binding the two strands together.
- Use tweezers [T01] or your fingernail to peel or snip through the white label between the two yellow cords (never cut into the yellow PVC jacket itself!).

File: docs/images/real_build/stage3_13b_peeling_white_wrap_label_to_separate_yellow_strands.jpg

VERIFIED — White adhesive wrap label peeled off so the two yellow fiber strands can unzip end-to-end

ARCTIC TRAIL CAM SYSTEM — STAGE 3 (PART 3): SINGLE FIBER STRAND, COUPLER & BOX #1 4 GREEN LEDS

Chronological Benchtop Photos 72–77 (1 Step Per Row): Unzipping 1 Yellow Strand, Seating Coupler [P15] & Lighting Box #1 (4/4 LEDs)

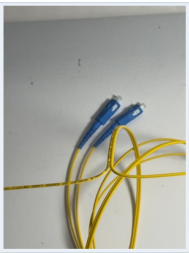


PHOTO 72 — STEP 3.1J: Gently Pull the Two Yellow Fiber Strands Apart Along the Center Web

- Hold one blue SC plug in each hand and pull the two yellow cords gently apart down the thin center zip-cord web for the full 2-meter length.
- Store the 2nd yellow strand in your kit as a 100% ready spare single-mode fiber!

File: docs/images/real_build/stage3_14_unzipping_duplex_yellow_fiber_into_single_strand.jpg

VERIFIED — Duplex yellow OS2 zip-cord unzipped cleanly into two independent single-strand fibers

PHOTO 73 — STEP 3.1K: Coil Your Single Yellow OS2 Fiber Strand (Dia >= 60mm) & Remove Dust Caps

- You now have a single 2-meter yellow OS2 fiber strand with one blue SC/UPC plug on each end.
- Keep any loose loops wider than a soda can (>= 60 mm diameter) so light never leaks at tight bends.
- Pull the clear plastic dust caps off both blue SC/UPC plugs right before clicking them in.

File: docs/images/real_build/stage3_15_single_yellow_fiber_strand_ready_caps_removed.jpg

VERIFIED — Single yellow OS2 fiber strand coiled loosely (>= 60mm diameter) with dust caps removed

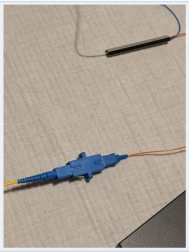


PHOTO 74 — STEP 3.1L: Click Orange Splitter Leg & Yellow Fiber Deep Into Blue Coupler [P15]

CRITICAL DEPTH CHECK: Match the raised plastic key tab on both blue SC plugs to the U-slot on the Blue Coupler.

- Push both the Orange splitter plug (top) and Yellow fiber plug (bottom) ALL THE WAY IN until both metal clips LOCK with a click (~1 mm gap to center flange) so the two white ceramic tips touch inside!

File: docs/images/real_build/stage3_16_blue_coupler_joining_orange_splitter_and_yellow_fiber.jpg

VERIFIED — Orange splitter leg (top) and Yellow OS2 fiber (bottom) mated inside Blue SC/UPC Coupler [P15]

PHOTO 75 — STEP 3.1M: Plug WHITE into Box #1 'TX' & BLUE into Box #1 'RX' -> ALL 4 GREEN LEDS ON!

- On Field Media Converter #1 (Pi 5 side), align the raised key tabs facing UP:
- Push the WHITE splitter input plug firmly into 'TX' (left port) past the internal spring until it clicks.
- Push the BLUE splitter output leg firmly into 'RX' (right port) -> ALL 4 GREEN LEDS light up solid on Box #1!

File: docs/images/real_build/stage3_17_pi_startech_box1_white_tx_blue_rx_all_4_green_leds.jpg

VERIFIED — Field Box #1: White in TX, Blue loopback in RX, Black Cat6 in RJ45 -> ALL 4 GREEN LEDS SOLID ON!

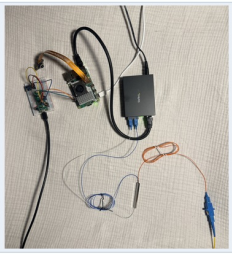


PHOTO 76 — STEP 3.1N: Top-Down Verification of Field Side (Pi 5 + Breadboard + Box #1 + Splitter)

- Top-Left: Breadboard [P09] + Pico 2 [P07] + GY-271 [P08] connected to Raspberry Pi 5 [P01] + Camera [P05].
- Top-Right: Field StarTech Box #1 [P13] connected to Pi 5 via Black Cat6 Ethernet cable.
- Bottom: Silver 1x2 Splitter tube [P14] splitting White TX into Blue (Box #1 RX) + Orange (to Blue Coupler [P15]).

File: docs/images/real_build/stage3_18_field_side_topdown_pi5_breadboard_box1_splitter_coupler.jpg

VERIFIED — Top-down Field Side architecture verified from Breadboard & Pi 5 through Box #1 & 1x2 Splitter

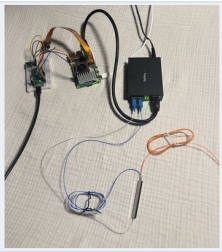


PHOTO 77 — STEP 3.1O: Angled Verification Showing All 4 Green LEDs Glowing on Field Box #1

- Angled benchtop check confirming all 4 green LEDs (Fiber LNK/ACT, TP LNK/ACT, PWR, 1000) are solid green on Box #1.
- Why the Blue loopback works: Box #1 sees its own 1310nm laser returning on RX, keeping its Gigabit link permanently active while broadcasting an identical laser stream out the Orange leg toward Collector Box #2!

File: docs/images/real_build/stage3_19_field_side_angled_box1_4_green_leds_and_splitter.jpg

VERIFIED — Field Side Box #1 100% operational with all 4 green LEDs lit and outbound laser active on Orange leg

ARCTIC TRAIL CAM SYSTEM — STAGE 3 (PART 4): LFP = OFF, HANDSHAKE SYNC & ALL 8 GREEN LEDS LIT!

Chronological Benchtop Photos 78–83 (1 Step Per Row): Sliding LFP Switch to OFF, Power-Cycle Reset & Verifying 4 + 4 Green LEDs

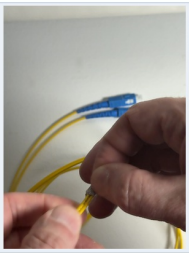


PHOTO 78 — STEP 3.2A: Pre-Check Yellow Fiber Plug Key Tab Up & White Label Clearance

- Before plugging the far end of the Yellow OS2 fiber into Collector Box #2, verify the raised key tab faces UP.
- Ensure the yellow strand runs smoothly without sharp kinks from the Blue Coupler [P15] to Collector Box #2.
- Plug the blue SC connector of the Yellow fiber firmly into the 'RX' port (right optical port) of Box #2.

File: docs/images/real_build/stage3_13a_locating_white_wrap_label_on_yellow_duplex_cord.jpg

VERIFIED — Yellow OS2 single-mode fiber routed cleanly and plugged key-tab-up into Box #2 RX port

PHOTO 79 — STEP 3.2B: Why Box #2 Shows Only 1 LED ('PWR') When Factory LFP Switch Is Left 'ON'

CRITICAL LESSON LEARNED: StarTech MCMGBSCSM10 ships with its front LFP (Link Fault Pass-Through) switch in 'ON' (UP).

- In a 1-way data diode, Box #2 'TX' is intentionally empty (unplugged air-gap).
- When LFP = ON at power-up, Box #2 detects no return fiber link and disables its RJ45 Ethernet port (only PWR lights)!

File: docs/images/real_build/stage3_20_macbook_box2_lfp_on_only_pwr_led_before_switch_off.jpg

VERIFIED — Diagnostic landmark: LFP switch in factory ON position holds Box #2 in fault mode (only 1 PWR LED)

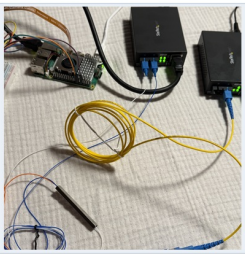


PHOTO 80 — STEP 3.2C: Slide LFP Switch DOWN to 'OFF' & Unplug/Re-Plug 5V Power -> 3 LEDs Light Up!

- Use a fingernail or tweezers tip to slide the tiny black LFP switch DOWN to 'OFF' on BOTH Box #1 and Box #2.
- IMPORTANT: The converter chip only reads the LFP switch at power-on reset! Unplug the rear 5V DC barrel plug for 3 seconds and plug it back in -> TP LNK/ACT and 1000 immediately light up green (3 green LEDs on Box #2)!

File: docs/images/real_build/stage3_21_lfp_switched_off_power_cycled_3_green_leds_ethernet_up.jpg

VERIFIED — LFP slid DOWN to OFF + 5V DC power-cycled -> Box #2 RJ45 Ethernet comes up (3 green LEDs lit)

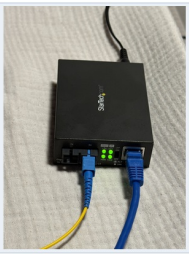


PHOTO 81 — STEP 3.2D: Unplug/Re-Plug WHITE ('Box #1 TX') for 2s -> ALL 4 GREEN LEDS SOLID ON BOX #2!

- Why the 4th LED (top-left 'Fiber LNK/ACT') needs a 2-second White TX re-plug if Box #1 booted first:
- Unplugging and re-plugging WHITE (Box #1 TX) forces Box #1 to restart its 1000BASE-X optical auto-negotiation burst (/C1/->/C2/->IDLE) so BOTH Box #1 RX and Box #2 RX lock onto the laser simultaneously -> ALL 4 LEDS SOLID!

File: docs/images/real_build/stage3_22_macbook_startech_box2_all_4_green_leds_rx_verified_closeup.jpg

VERIFIED — Collector Box #2 macro verification: LFP = OFF, Yellow in RX, TX empty air-gap & ALL 4 GREEN LEDS ON!

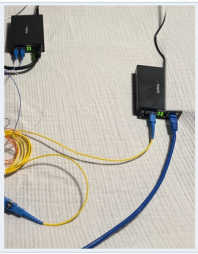


PHOTO 82 — STEP 3.2E: Angled Benchtop Check — All 8 Green LEDs Glowing (4 on Box #1 + 4 on Box #2)

- Left (Field Box #1 — Pi 5 side): All 4 green LEDs (Fiber LNK/ACT, TP LNK/ACT, PWR, 1000) glowing solid green.
- Right (Collector Box #2 — MacBook side): All 4 green LEDs glowing solid green with TX port 100% empty!
- Bottom: 1x2 Optical Splitter + Blue Coupler + Yellow OS2 fiber carrying 1-way UDP traffic at 1000 Mbps.

File: docs/images/real_build/stage3_23_full_benchtop_angled_both_boxes_all_8_green_leds.jpg

VERIFIED — Angled full-bench verification: 4 green LEDs on Box #1 + 4 green LEDs on Box #2 = 8/8 solid green!

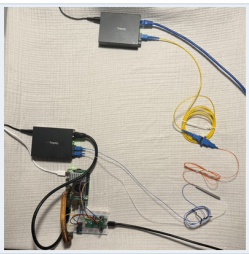


PHOTO 83 — STEP 3.2F: Golden-Path Top-Down Overview With Cleanly Separated Coils & Sub-Assemblies

- Top: Collector Box #2 (all 4 LEDs lit) with Yellow OS2 fiber in RX, Empty TX air-gap, and Blue Cat6 to MacBook.
- Left: Field Box #1 (all 4 LEDs lit) + Raspberry Pi 5 [P01] + Camera [P05] + Breadboard [P09] + Pico 2 [P07] + GY-271.
- Right: Neatly separated 1x2 Splitter [P14] coils (White/Blue/Orange) and Blue Coupler [P15] + Yellow OS2 loop [P16]!

File: docs/images/real_build/stage3_24_full_benchtop_clean_separated_bundles_golden_overview.jpg

VERIFIED — COMPLETE: Stage 3 1-Way Optical Fiber Data Diode 100% built, separated, and verified (8/8 LEDs)!

[STAGE 4.1 REAL BUILD] Real-World Build Log: Stage 4.1 Drilling the 20 mm Viewport & 12 mm Umbilical Hole

Stepped cone bit [T06], 20 mm circular front optical aperture, steel ruler verification [T10] & 12 mm bottom cable exit hole.

STAGE 4.1 REAL-WORLD BUILD: DRILLING THE 20 MM VIEWPORT & 12 MM UMBILICAL HOLE

Hammond 1591BSBK ABS Pod: 20 mm Front Camera Viewport (Hole 1) & 12 mm Bottom Umbilical Entry (Hole 2)

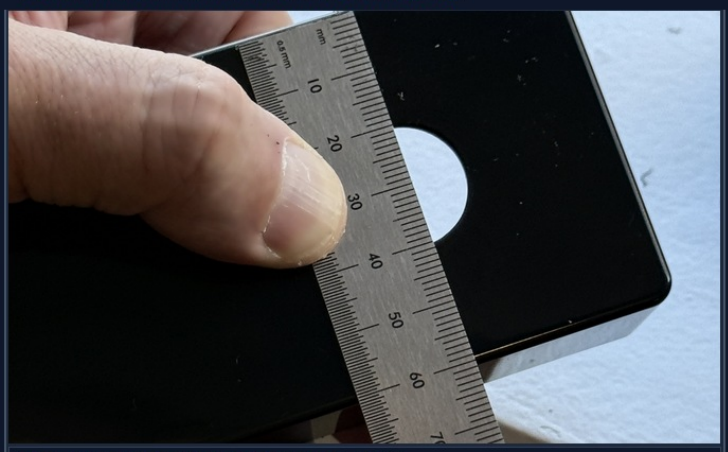
A. Titanium Stepped Cone Drill Bits ([T06])

Smooth Cone Step Bit ([T06]) — Zero Snagging Saw Teeth
Shaves a clean 20 mm viewport & 12 mm bottom cable hole in 2 mm ABS without cracking

B. Drilled 20 mm Circular Front Viewport (Hole 1)

20 mm Viewport Centered in Upper Half of Hammond 1591BSBK Lid
Leaves a wide, flat 5 mm perimeter plastic ledge all around (360 deg) for the 30 mm glass disc

C. Steel Ruler Verification of 20 mm Aperture ([T10])



Verifying 20 mm Diameter & 31 mm Side-to-Side Centering
Lower half of lid stays 100% solid (24 GHz radar sees straight through 2 mm ABS)

D. Drilling the 12 mm Bottom Umbilical Hole (Hole 2)

12 mm Bottom Exit Hole Centered in Narrow End-Wall
Allows the 2 m camera HDMI umbilical and 4-wire radar harness to exit cleanly at the bottom

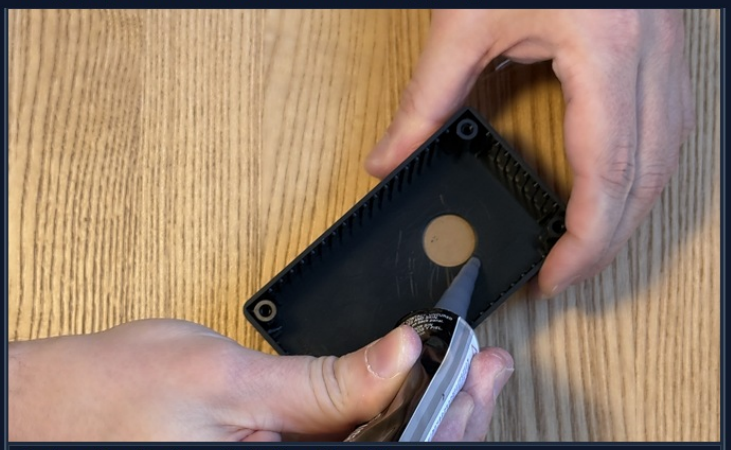
[STAGE 4.2 REAL BUILD] Real-World Build Log: Stage 4.2 Bonding the 30 mm Mineral Glass Viewport Window

Applying Permatex 81158 black RTV silicone [T11], seating the 30 mm glass disc [P20], 360-deg inside seal & cured exterior window.

STAGE 4.2 REAL-WORLD BUILD: BONDING THE 30 MM MINERAL GLASS VIEWPORT WINDOW

360-Degree Waterproof Permatex 81158 Black RTV Silicone Gasket Over the 20 mm Viewport Aperture

A. Applying Permatex 81158 Black Silicone Ring ([T11])



Continuous 3 mm Bead Around Inside Rim of 20 mm Viewport Hole
Applied to the flat inside ABS surface surrounding the hole for a 360-degree waterproof gasket

B. Pressing the 30 mm Mineral Glass Disc ([P20]) Into Place

Seating the 30 mm Optical Mineral Glass Disc Over the 20 mm Hole
5 mm overlap on all sides compresses the black RTV silicone into a continuous watertight ring

C. Inside Inspection of 360-Degree Waterproof Silicone Seal

Full-Perimeter Black RTV Silicone Gasket Verified Inside Lid
Corner screw bosses remain completely clear so the rear cover closes flush

D. Cured Crystal-Clear Exterior Optical Window

Exterior View After Curing: Flush, Watertight & Optically Clear
Ready for Camera Module 3 NoIR Wide (IMX708) and 2 m CSI-to-HDMI umbilical installation

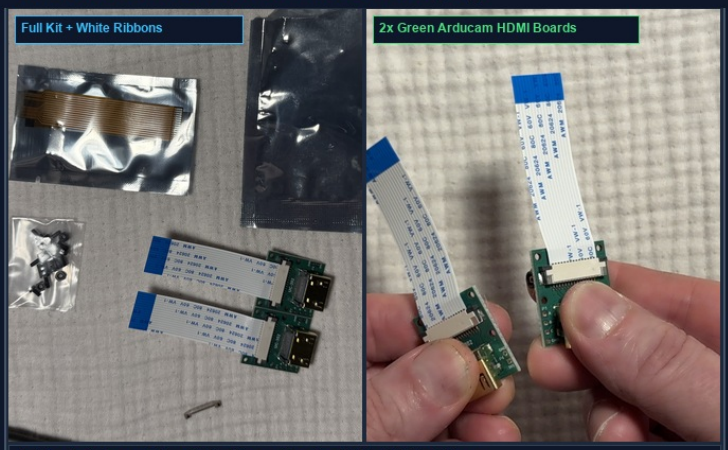
[STAGE 4.3A REAL BUILD] Real-World Build Log: Stage 4.3A Assembling the 2-Meter Arducam CSI-to-HDMI Extension [P06]

2x green CSI-to-HDMI boards, 200 mm orange ribbon on Pi-side adapter, white ribbon on Camera Module 3 & 2 m 8K HDMI cable.

STAGE 4.3A REAL-WORLD BUILD: ASSEMBLING THE 2-METER ARDUCAM CSI-TO-HDMI EXTENSION

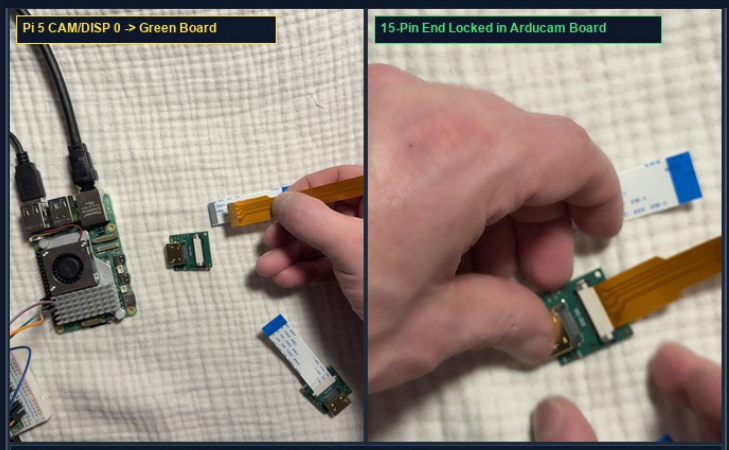
Extending Camera Module 3 NoIR Wide from Pi 5 (Zone 2 Vault) to External Sensor Pod (Zone 1) Over 2m 8K HDMI

A. Unboxed Arducam CSI-to-HDMI Extension Kit ([P06])



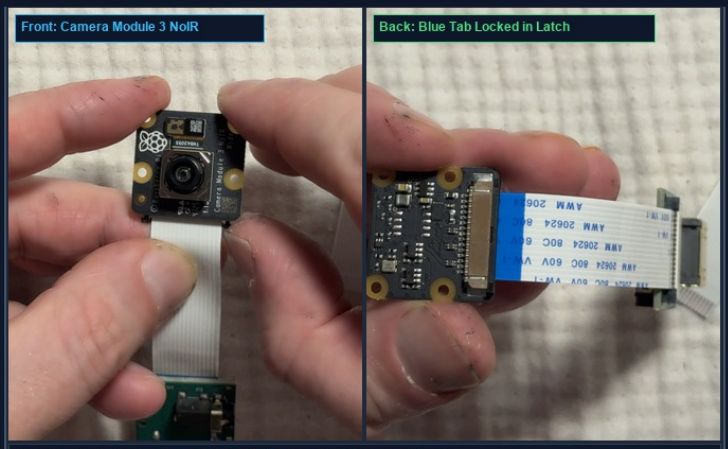
2x Identical Green CSI-to-HDMI Adapter Boards + White 15-Pin Ribbons
Converts high-speed MIPI CSI-2 camera signals to differential HDMI over a 2 m shielded cable

B. Connecting 200 mm Orange Pi 5 Ribbon to Pi-Side Adapter



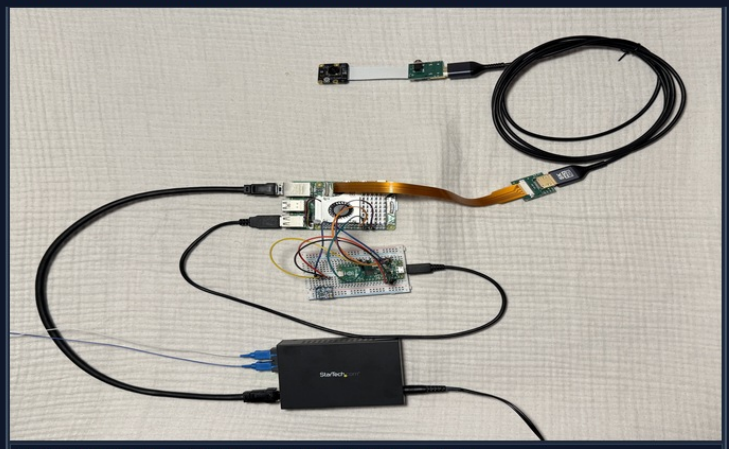
200 mm Orange 22-to-15-Pin Ribbon -> Pi-Side Arducam Adapter
Lock narrow 22-pin end in Pi 5 CAM/DISP 0 and wide 15-pin end in the Pi-side green HDMI board

C. Connecting Short White Ribbon to Camera Module 3 NoIR



Camera Module 3 NoIR Wide -> Pod-Side Arducam Adapter Board
Short white 15-pin ribbon links the camera board to the second green Arducam HDMI board

D. Complete 2-Meter 8K HDMI Camera Umbilical on Bench



2 m Black Shielded HDMI Cable Linking Zone 2 Pi 5 to Zone 1 Camera
Verified live IMX708 frame capture across the full 2-meter trunk umbilical extension

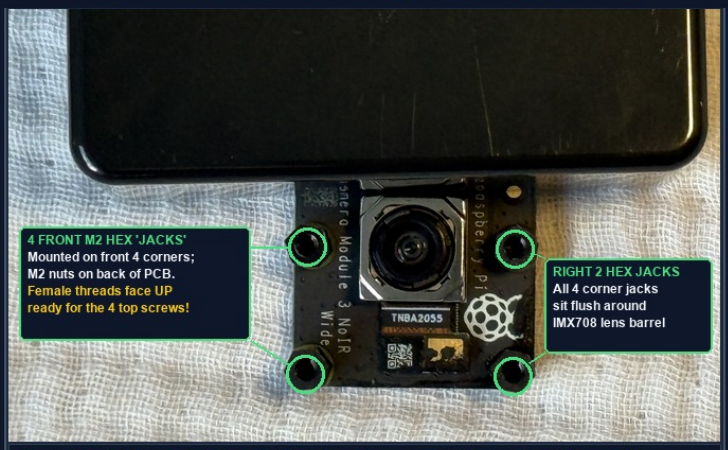
[STAGE 4.3B REAL BUILD] Real-World Build Log: Stage 4.3B Camera Front M2 Hex Standoff 'Jacks' & Top Focus-Clearance Sc

4 black M2 hex jacks on front of Camera Module 3 NoIR + 4 top screws left partially unthreaded (~1.5-2 mm gap = ~1.3 mm lens clearance).

STAGE 4.3B REAL-WORLD BUILD: CAMERA FRONT STANDOFF 'JACKS' & TOP FOCUS-CLEARANCE SCREWS

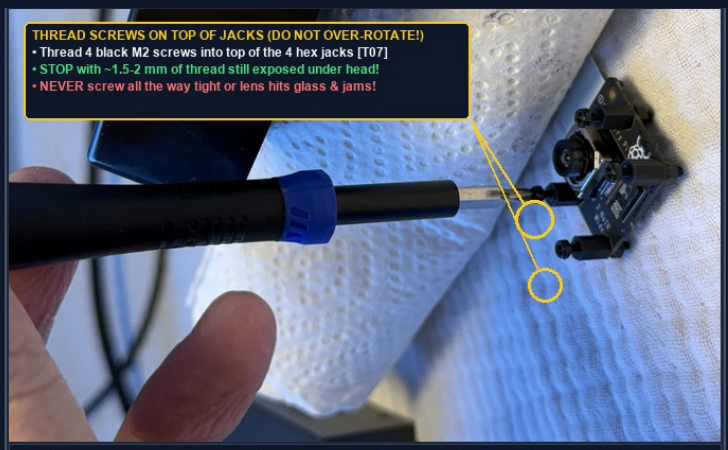
CRITICAL: Attach 4 M2 Hex Jacks + 4 Top Screws on Camera Module 3 NoIR & DO NOT Over-Rotate (Or Lens Jams Against Glass & Cannot Focus!)

A. Attach 4 Black M2 Hex Standoff 'Jacks' on Front of Camera (IMG_7394)



Mount 4 Black M2 Nylon Hex Standoffs ('Jacks', from Arducam [P06] bag) on Front 4 Corner Holes
Secure with 4 M2 nylon hex nuts on the back of Camera Module 3 NoIR Wide [P05] around IMX708 lens

B. Thread 4 M2 Screws Into Top of Jacks — DO NOT Over-Rotate! (IMG_7463)

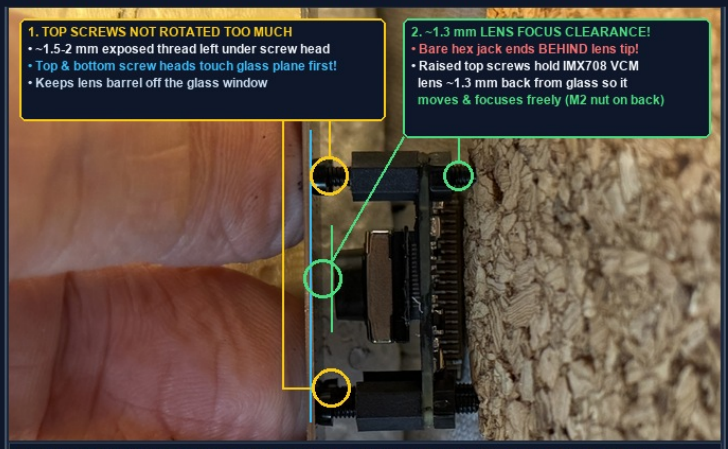


Use Precision Screwdriver [T07] to Thread 4 M2 Nylon Screws Into the Top of the 4 Hex Jacks
STOP after ~2-3 turns (leave ~1.5-2 mm thread showing) — DO NOT tighten flush!

C. 3/4 View: All 4 Front Jacks + Raised Top Screws Installed (IMG_7459)

All 4 Corner Standoff Jacks + Partially Unthreaded Top Screws Standing Proud of Lens Barrel
Notice the visible ~1.5-2 mm screw shaft gap under each screw head (not rotated all the way in)

D. Side-Profile Straightedge Check: Lens Clearance Gap Verified (IMG_7465)



CRITICAL FOCUS CHECK: 4 Raised Screw Heads Hold Glass Plane ~1.3 mm Ahead of Lens Barrel!
If screws are rotated too far in, the IMX708 lens jams against the glass and CANNOT MOVE / FOCUS!

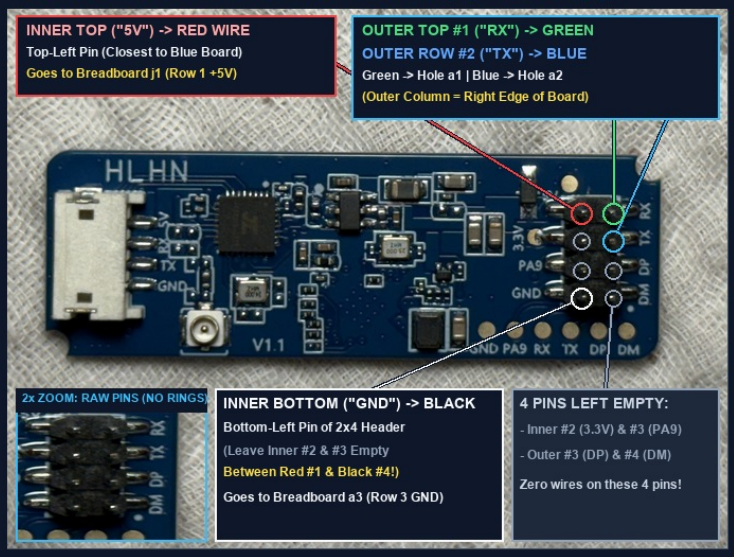
[STAGE 4.3C REAL BUILD] Real-World Build Log: Stage 4.3C Blue Radar Board (HLHN V1.1 / LD2450) 4-Wire Jumper Wiring [P

2x4 header pin map (Inner: 5V=Red->j1, 3.3V/PA9=Empty, GND=Black->a3; Outer: RX=Green->a1, TX=Blue->a2, DP/DM=Empty).

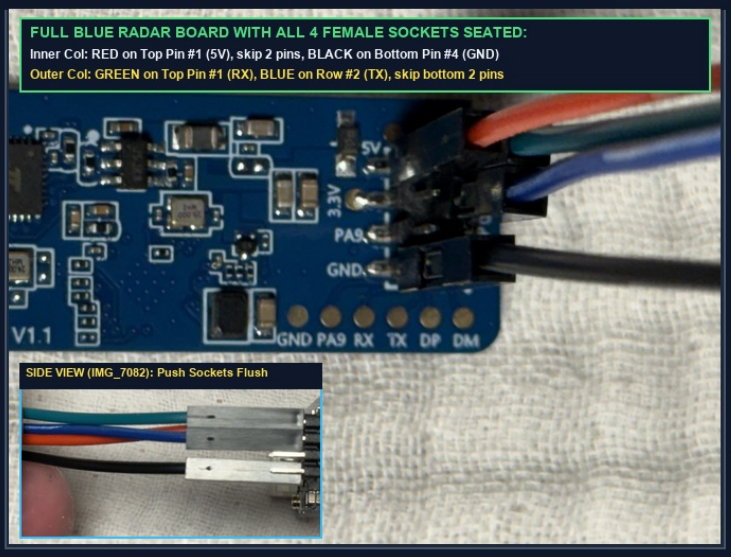
STAGE 4.3B REAL-WORLD GUIDE: BLUE RADAR BOARD (HLHN V1.1 / LD2450) 4-WIRE JUMPER WIRING

Full Uncut Blue HLHN V1.1 Radar Board Pinout (4 Active Jumpers, 4 Empty Pins) & Exact Breadboard Holes (a1, a2, a3 & j1)

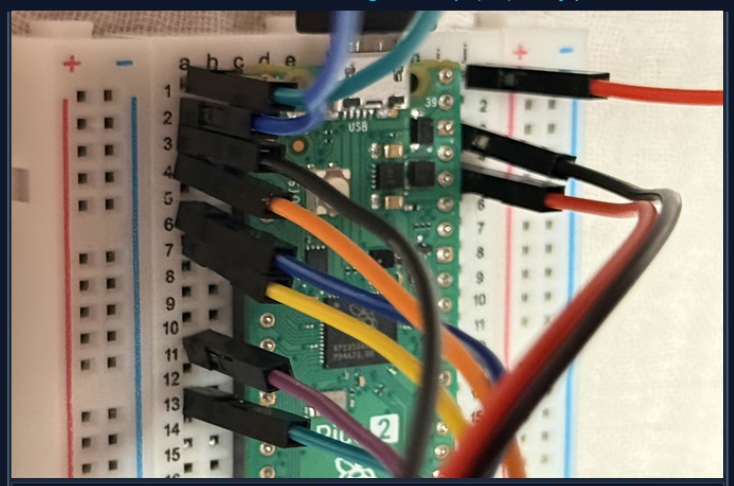
A. BARE BLUE BOARD 2x4 HEADER: Exact Pin Map (4 Used, 4 Empty)



B. ALL 4 JUMPERS PLUGGED INTO BLUE BOARD (TOP & SIDE VIEW)

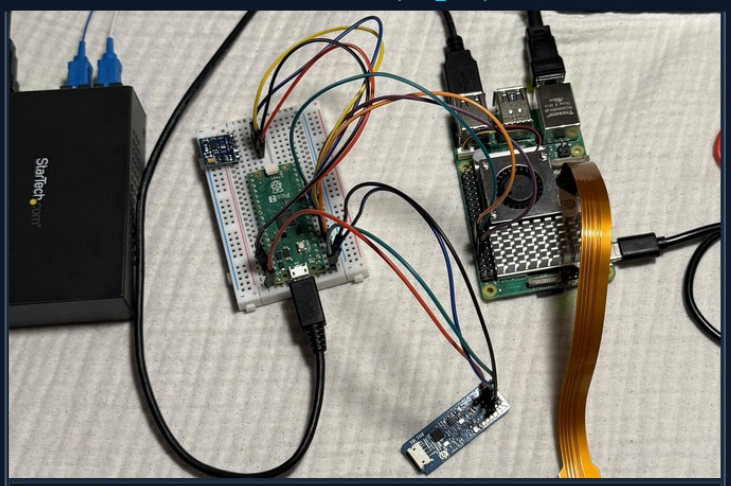


C. PICO 2 BREADBOARD END: Exact Target Holes (a1, a2, a3 & j1)



1. LEFT (Col a, Rows 1-3): a1 <- GREEN (Radar RX), a2 <- BLUE (Radar TX), a3 <- BLACK (GND).
2. RIGHT (Col j, Row 1): j1 <- RED (Radar +5V from Pico VBUS Pin 40).

D. COMPLETE CONNECTED BENCH CHECK (IMG_7086)



1. Blue HLK-LD2450 radar board wired to Pico 2 breadboard (a1, a2, a3, j1).
2. Ready to route alongside the 2 m HDMI camera umbilical into the Zone 1 Pod.

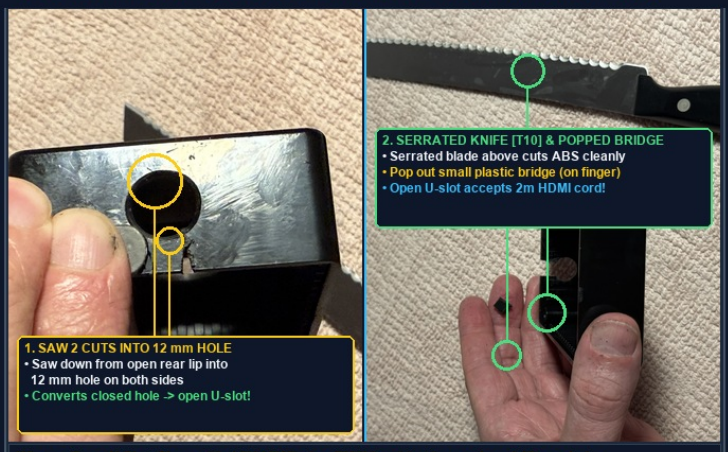
[STAGE 4.3D REAL BUILD] Real-World Build Log: Stage 4.3D Open U-Slot, Loctite Standoff Tack-Glue & Teaspoon RTV Smooth

Serrated-knife 12 mm open U-slot, dropping 2 m HDMI + Arducam in, Loctite Super Glue on 4 screw heads & teaspoon bowl RTV trowel.

STAGE 4.3C REAL-WORLD BUILD: SERRATED-KNIFE U-SLOT, 3M TAPE STRAP, LOCTITE MOUNT & SPOON RTV TROWEL

Drop Pre-Assembled 2m HDMI + Arducam Into Open U-Slot, Tack-Glue 4 Raised Standoff Screws w/ Loctite & Smooth RTV w/ Teaspoon Bowl

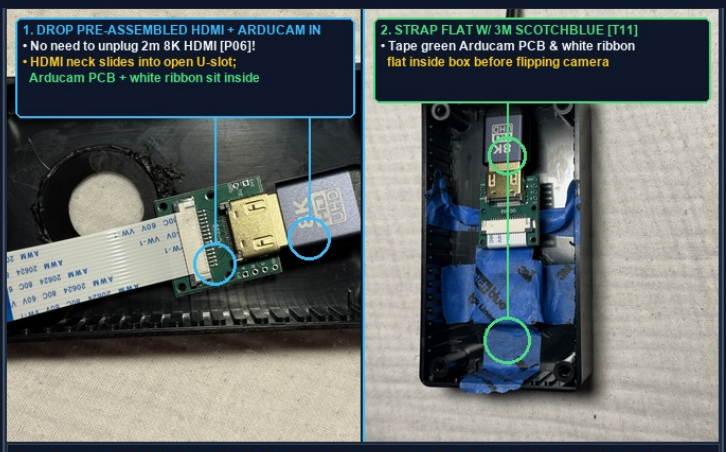
A. Saw 12 mm Bottom Hole Into Open U-Slot w/ Serrated Knife (IMG_7097 / 7099)



Why an Open U-Slot: The molded 8K HDMI plug [P06] is wider than a closed 12 mm hole

Saw 2 vertical cuts from the rear open lip into the 12 mm hole w/ Serrated Knife [T10] & pop bridge out

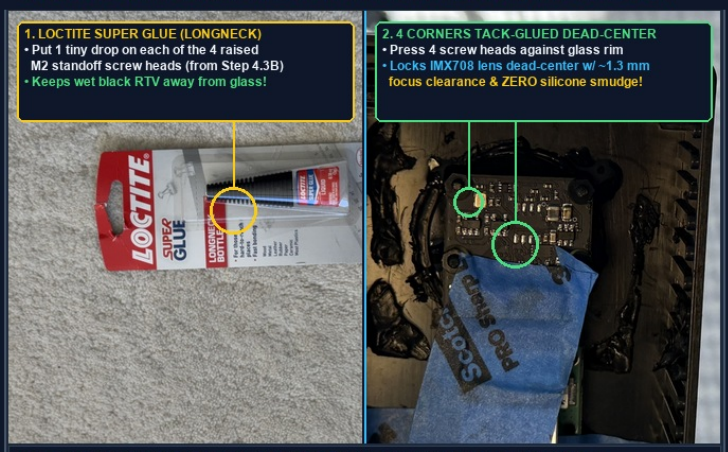
B. Drop Pre-Assembled 2m HDMI + Arducam In & Strap w/ 3M Tape (IMG_7100 / 7116)



Drop Pre-Assembled Camera + White Ribbon + Green Arducam Board + 2m HDMI Cord Straight In

Strap green Arducam CSI-to-HDMI board & white ribbon flat inside lower bay with 3M ScotchBlue [T11]

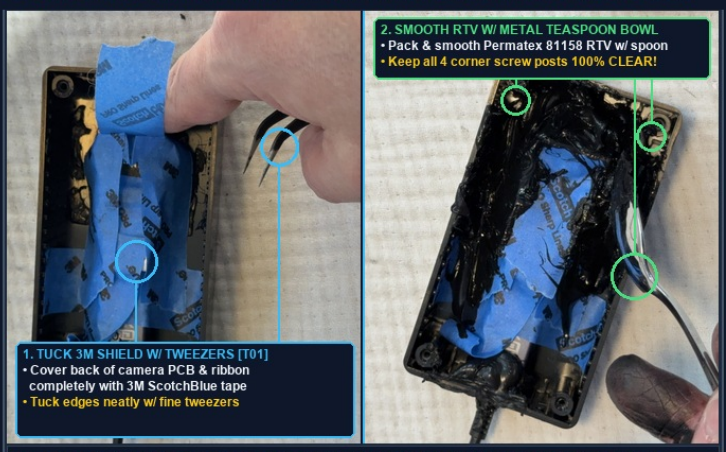
C. Tack-Glue 4 Raised Standoff Screw Heads to Glass w/ Loctite (IMG_7516 / 7534)



Put 1 Tiny Drop of Loctite Super Glue on Each of the 4 Raised M2 Standoff Screw Heads

Press camera centered onto inside perimeter of 30 mm glass (~1.3 mm focus gap, ZERO RTV near lens!)

D. Tuck 3M Tape Shield w/ Tweezers & Smooth RTV w/ Spoon Bowl (IMG_7573 / 7638)



Tuck 3M ScotchBlue Shield Over Camera Back w/ Tweezers [T01] Before Applying Any Black RTV

Use a Metal Teaspoon Bowl Like a Mini Trowel to Smooth Permatex 81158 RTV Around Perimeter & U-Slot

[STAGE 4.4 REAL BUILD] Real-World Build Log: Stage 4.4 3M Tape Shielding, Permatex Silicone Potting & Live Tripod Test

Centering Camera Module 3 behind glass window, 3M ScotchBlue protective barrier, Permatex RTV potting & live tripod verification.

STAGE 4.4 REAL-WORLD BUILD: 3M TAPE SHIELDING, SILICONE POTTING & LIVE TRIPOD TEST

Centering Camera Behind 30 mm Glass Window, 3M ScotchBlue Barrier, RTV Potting & Upright Tripod Test

A. Centering Camera Module 3 NoIR Behind Glass Window

Front-Look Alignment Through the 30 mm Mineral Glass Disc
Square 11.5 x 11.5 mm IMX708 lens barrel centered directly behind the clear optical aperture

B. Applying 3M ScotchBlue Protective Tape Barrier ([T11])

3M Painter Tape Isolates Camera, White Ribbon & Arducam PCB
Protects delicate electronics from direct RTV contact and allows clean future servicing

C. Applying Permatex 81158 Black RTV Potting & Strain Relief

Perimeter Waterproofing + Bottom HDMI Umbilical Strain-Relief Boot
Locks assembly rigidly in place while keeping all 4 corner screw posts 100% unobstructed

D. Live Tripod Optical & Radar Verification Test (IMG_7680)

Assembled Zone 1 Pod Mounted Upright on Tripod Connected to Pi 5
Verified crystal-clear IMX708 capture and zero-loss 2 m umbilical operation before closure

[STAGE 5.2 REAL BUILD] Real-World Build Log: Stage 5.2 12V Battery, Stripped/Twisted Splices in 3M Tape, 5V Buck & USB2T

Stage 5 power kit [P17]–[P19], stripping/twisting 3 splices in 3M ScotchBlue tape, multimeter +5.0V check & Pi 5 USB-A -> Box #1.

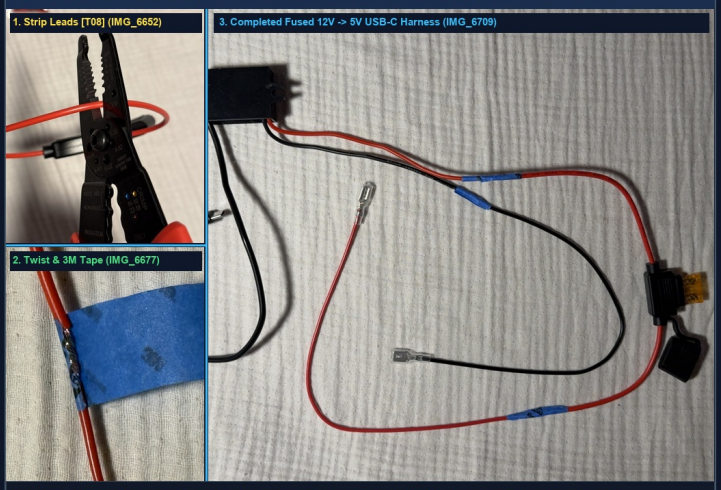
STAGE 5.2 REAL-WORLD BUILD: 12V BATTERY, FUSE, 5V BUCK & STARTECH USB-TO-BARREL CABLE

A. Stage 5 Power Kit & StarTech USB2TYPEM Cable ([P17]-[P19])

12V 12Ah Battery, Littelfuse Holder, 12V->5V 5A Buck & USB2TYPEM Cable

Yellow StarTech bag ([P19]) is the USB-A to 5V Barrel cable that powers Media Converter #1 from Pi 5 USB

B. Strip [T08], Twist Bare Copper & Wrap Splices in 3M Tape (IMG_6652 / 6677 / 6709)



Strip Leads [T08], Finger-Twist Bare Copper Strands & Wrap All 3 Splices in 3M ScotchBlue [T11]

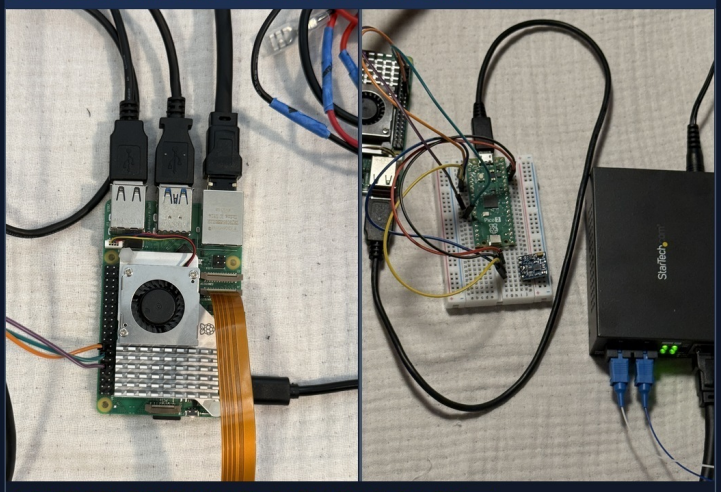
Links F2 Battery Spades + Littelfuse 5A Blade Fuse [P18] + 12V-to-5V 5A USB-C Buck Converter

C. Mandatory Digital Multimeter DC Voltage Verification ([T09])

Verifying +12.6V Battery Rail & Regulated +5.0V Buck Output

Always confirm +5.0V DC with multimeter on the bench BEFORE connecting Pi 5 USB-C power

D. Pi 5 USB-A Powers Both Pico 2 & StarTech Box #1 (IMG_7727 / 6764)



Left (IMG_7727): Pi 5 USB-A #1 -> Pico 2, USB-A #2 -> StarTech 5V Barrel

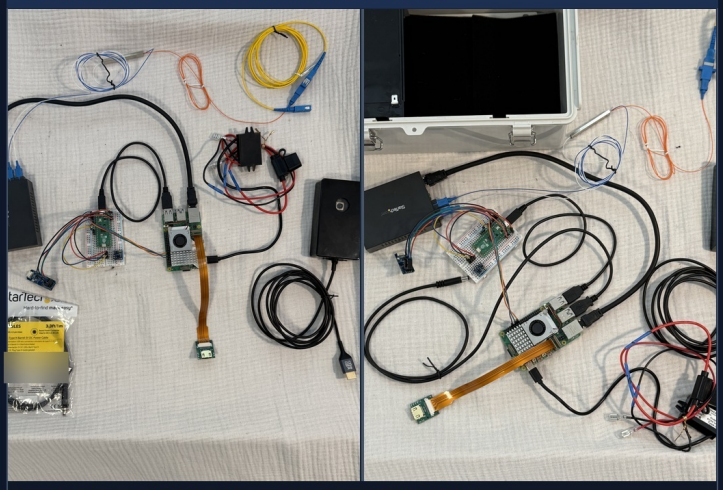
Right (IMG_6764): Entire field node (Pi 5 + Pico 2 + StarTech Box #1) running 100% off 12V battery

[STAGE 5.3A REAL BUILD] Real-World Build Log: Stage 5.3A Seating 12V Battery in Vault, Radar 3M Tape & Breadboard Lock

Pre-packing bench check, seating 12V 12Ah battery & Media Converter #1 on foam floor, 3M radar header wrap & breadboard straps.

STAGE 5.3A REAL-WORLD BUILD: SEATING BATTERY IN VAULT, RADAR 3M TAPE & BREADBOARD LOCK

A. Pre-Packing Bench Check Next to White IP67 Vault ([P21])



Complete Connected Field Node Alongside the Open White IP67 Enclosure

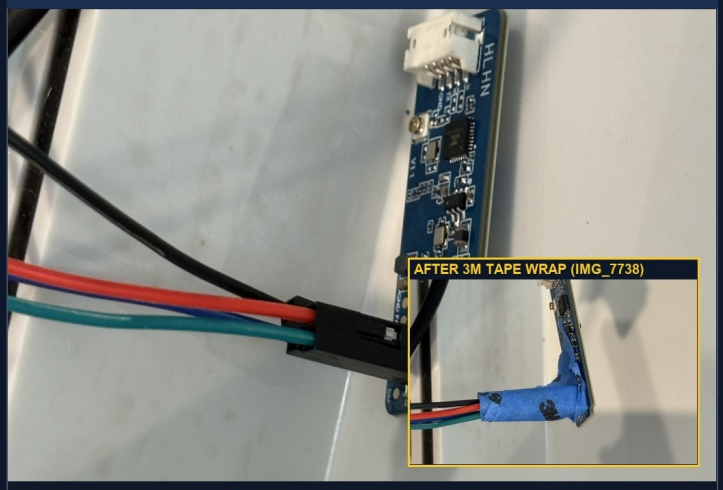
Left: Full connected bench assembly (IMG_7723) | Right: Positioned beside open White IP67 Vault (IMG_7734)

B. Seating 12V 12Ah Battery ([P17]) & Media Converter #1 ([P13])

Heavy 12V 12Ah Battery & StarTech Media Converter #1 on Foam Floor

Closed-cell foam bed cushions the 3.5 kg battery and optical converter at the base of the enclosure

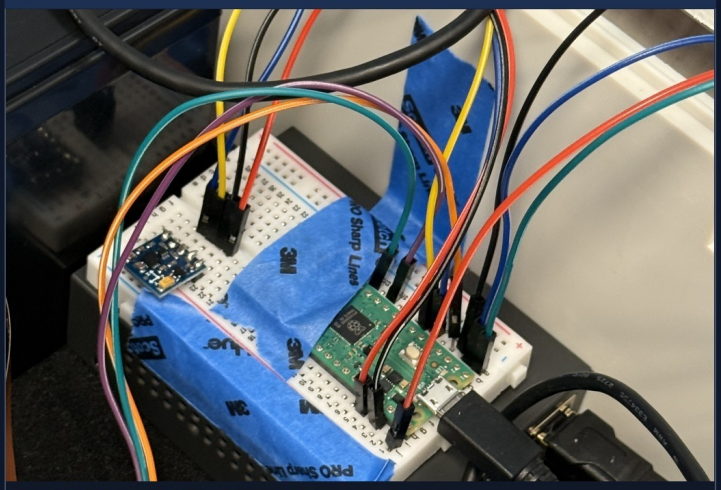
C. Wrapping Blue Radar 4-Wire Header in 3M Tape ([P11] / [T11])



3M ScotchBlue Strain-Relief & Electrical Insulation on HLK-LD2450 Header

Locks the 4 female sockets (Red, Black, Green, Blue) onto the 2x4 pins so they never pull loose

D. Securing Pico 2 + GY-271 Breadboard ([P09]) Inside Vault



3M ScotchBlue Retention Straps Over 30-Row Breadboard & Jumpers

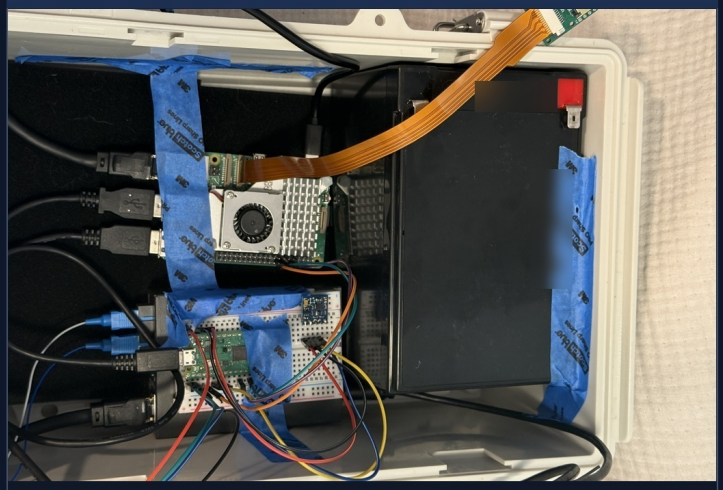
Anchors the Pico 2 gatekeeper and GY-271 magnetometer firmly inside the Zone 3 enclosure

[STAGE 5.3B REAL BUILD] Real-World Build Log: Stage 5.3B Zone 3 Vault Packing, Fiber Lid Spool & Black Battery Clip LAST

Strapping down Pi 5 & 12V battery, internal power/HDMI harness, optical fiber lid spool & connecting Black (-) battery clip LAST.

STAGE 5.3B REAL-WORLD BUILD: ZONE 3 VAULT PACKING, FIBER LID SPOOL & BATTERY CLIP LAST

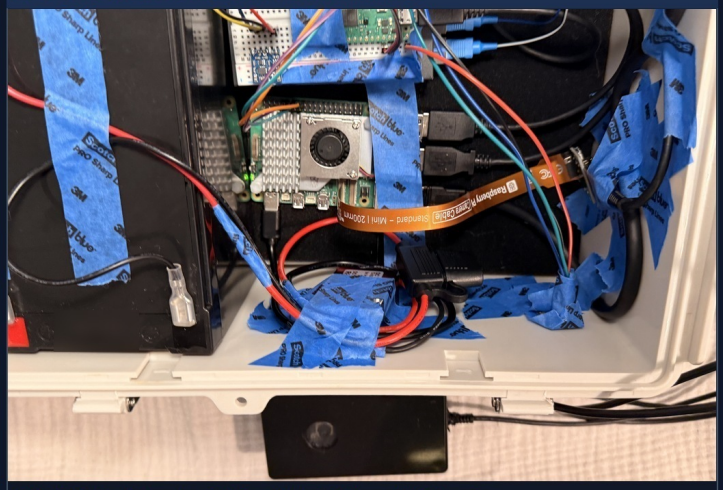
A. Strapping Down Pi 5 ([P01]), Cooler ([P02]) & Battery ([P17])



3M ScotchBlue Cross-Straps Secure Pi 5, Breadboard & 12V 12Ah Battery

200 mm orange ribbon connects Pi 5 to the Arducam CSI-to-HDMI adapter board

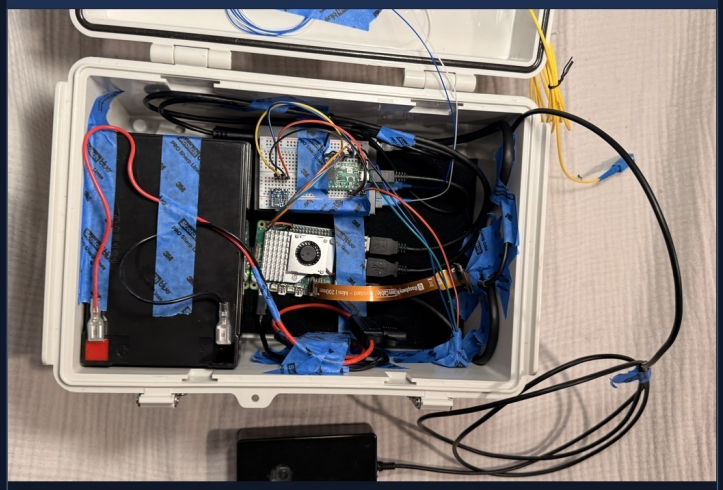
B. Close-Up of Packed Zone 3 Power & Umbilical Harness



12V->5V 5A Buck Converter, Littelfuse Holder & Arducam HDMI Coupling

All internal power, sensor, and HDMI umbilical cables neatly dressed and retention-taped

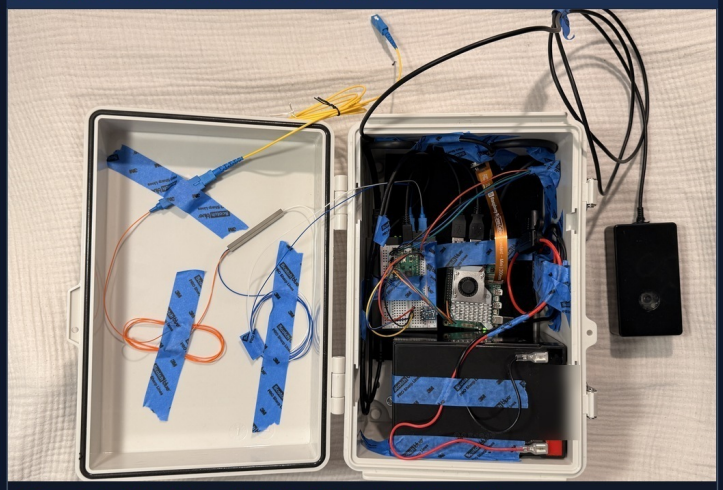
C. Packed Zone 3 Vault + Zone 1 Pod (Connect Black Battery Clip LAST)



Power Box #2 FIRST, Then Plug Black (-) Battery Clip Onto 12V Battery LAST

Unplugging & re-plugging Black (-) battery lead LAST locks all 4 Green LEDs on Collector Box #2

D. Complete Field-Ready System with Optical Fiber Lid Spool



0.25 mm Tactical Fiber & 900 um Buffer Strain-Relieved Inside Hinged Lid

Complete Arctic Trail Cam field node ready for Freezer Soak Test and field deployment

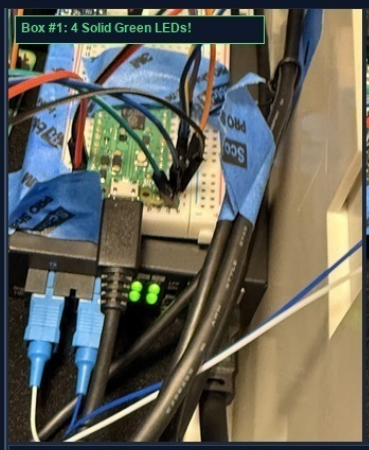
[STAGE 5.4 REAL BUILD] Real-World Build Log: Stage 5.4 All 8 Green LEDs on 12V Battery & Bottom Freezer 'Fast Freeze' Soak

4 green LEDs on Field Box #1 + 4 green LEDs on Collector Box #2, placed on frozen blue Igloo ice bricks in bottom freezer with Fast Freeze ON.

STAGE 5.4: 8 GREEN LED LINK LOCK & BOTTOM FREEZER 'FAST FREEZE' SOAK TEST

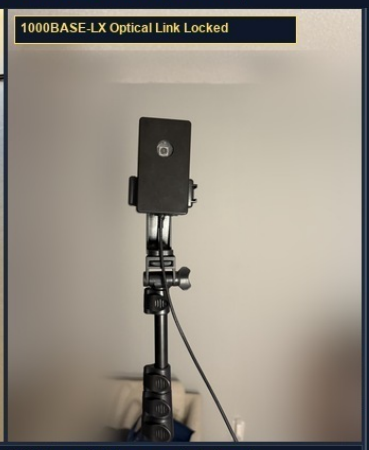
Verified 4+4 Green LEDs on Both StarTech Transceivers -> Seated on Blue Frozen Igloo Bricks in Fast Freeze Mode

A. StarTech Box #1 (Inside Vault): 4 Solid Green LEDs Verified



1. Connect 12V battery lead & LC/SC fiber patch cord to StarTech Box #1 inside the vault.
2. Confirm all 4 green LEDs illuminate on Box #1 inside the white QILIPSU IP67 vault.

B. StarTech Box #2 (Outside Receiver): 4 Solid Green LEDs (8/8 Total)



1. Verify all 4 green LEDs illuminate on Box #2 outside (8/8 green LEDs across both boxes).
2. Zero copper wires cross the vault boundary — only the yellow single-mode optical fiber.

C. Bottom Freezer Compartment: Placed on Frozen Blue Igloo Bricks

1. Line the bottom pull-out freezer compartment with pre-frozen blue Igloo ice bricks.
2. Place both the white IP67 vault and black Zone 1 Pod directly on top of the frozen bricks.

D. Activating 'Fast Freeze' Mode & Routing Fiber Past Door Gasket

1. Press 'Fast Freeze' (snowflake icon red LED lit + max 6/6 coldness bars active).
2. Close bottom freezer drawer over the yellow fiber & log continuous sub-zero soak telemetry.

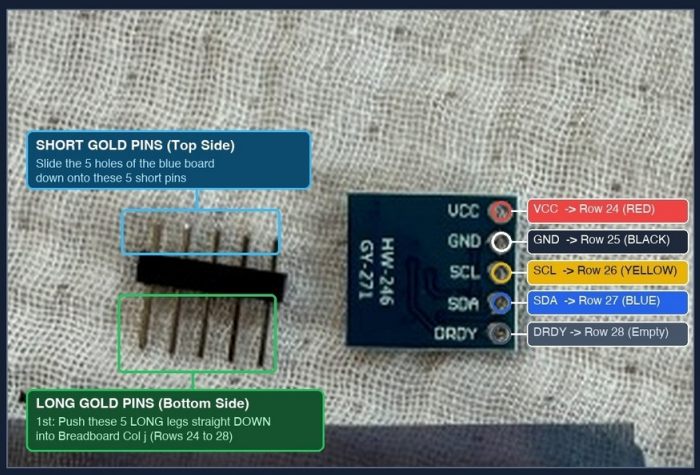
[COMPASS ZOOM GUIDE] 2-Panel Real-Photo Guides: GY-271 5-Pin Header & 3 vs 2 Empty Bottom Holes Landmark

Top: 5-pin header & Col f / Col j hole map. Bottom: 3 empty bottom holes in Col f vs 2 empty bottom holes in Col j.

HOW TO INSTALL YOUR COMPASS SENSOR (HiLetgo GY-271) INTO THE BREADBOARD

Step 1: Push the 5 LONG legs into Breadboard Col j (Rows 24-28) -> Step 2: Place the blue board over the 5 SHORT top pins -> Step 3: Plug in the 4 wires

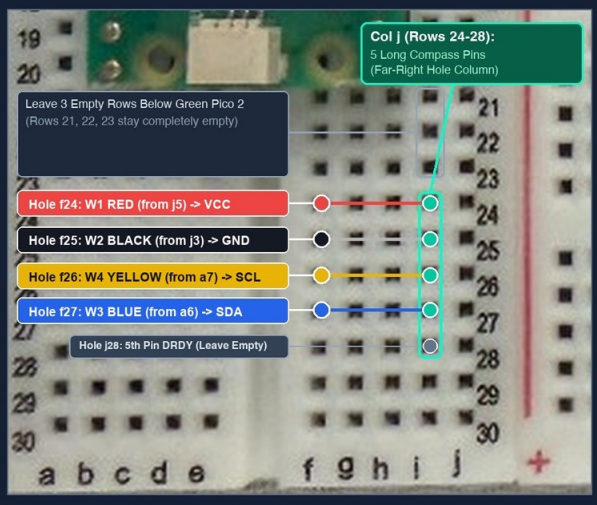
1. YOUR TWO COMPASS PIECES (CLOSE-UP OF YOUR PHOTO)



IMPORTANT TIP WHEN YOU PLACE THE BLUE BOARD ON THE 5 PINS:

- Keep the white letters (VCC, GND, SCL, SDA, DRDY) facing UP toward the ceiling so VCC is at the top (Row 24).
- For rock-solid unsoldered contact, grip the 4 active SHORT top pins (VCC, GND, SCL, SDA) with flat-nose pliers and twist each square pin slightly (~10-15 deg) in place so its diagonal corners bite snugly into the gold holes!
- Want it permanent later? Touch 1 tiny drop of solder to each of the pins with your soldering iron.

2. EXACT HOLES AT BOTTOM OF YOUR BREADBOARD



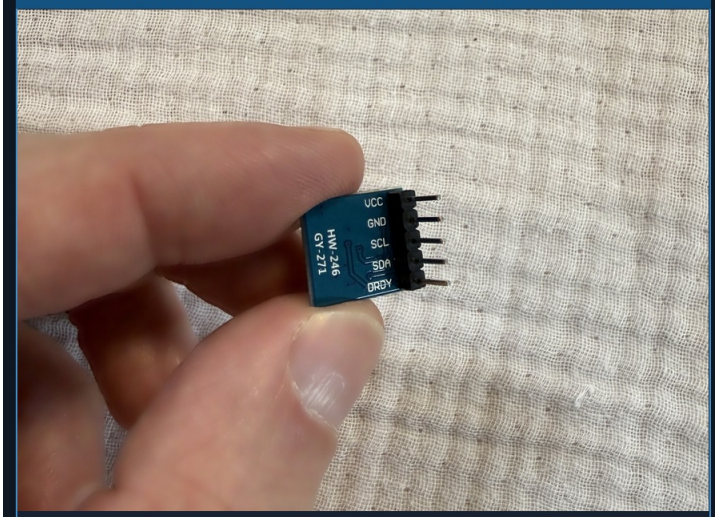
WHY THIS WORKS (ZERO ROOM FOR ERROR):

- In each horizontal row, the 5 holes (f, g, h, i, j) are connected together underneath by one metal strip.
- So plugging your 5 Compass pins into Column j (Rows 24-28) and your 4 wires (Red, Black, Yellow, Blue) into Column f on the EXACT SAME 4 ROWS (Rows 24-27) connects each wire straight to its Compass pin!

HOLE-COUNTING LANDMARK GUIDE: 4 WIRES IN COL f (3 EMPTY BOTTOM) vs. 5-PIN COMPASS IN COL j (2 EMPTY BOTTOM)

Because the GY-271 Compass header has 5 pins (vs. 4 wires), leave 3 empty bottom holes in Col f and 2 empty bottom holes in Col j!

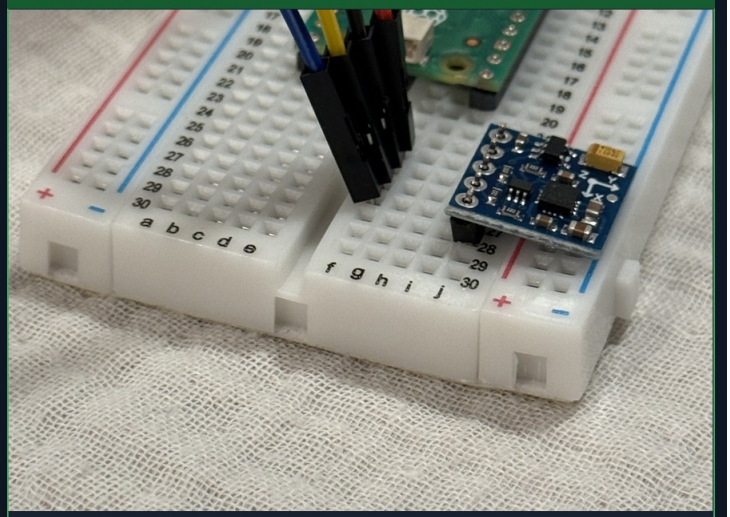
LEFT (PART IN HAND): 5-PIN GY-271 COMPASS MODULE (VCC AT TOP, DRDY AT BOTTOM)



WHY THE 5-PIN STRIP REACHES 1 HOLE LOWER THAN THE 4 WIRES:

- Top 4 Pins = VCC (Red Row 24), GND (Black Row 25), SCL (Yellow Row 26), SDA (Blue Row 27).
- 5th Bottom Pin = DRDY (Row 28, unused/bare). That 5th pin occupies 1 extra hole at the bottom!

RIGHT (GOLDEN-PATH PHOTO 56): 3 EMPTY HOLES IN COL f vs. 2 EMPTY IN COL j



EXACT HOLE-COUNTING LANDMARKS (ZERO RELIANCE ON PRINTED NUMBERS):

- TOP CHECK: Leave 3 EMPTY HOLES below Pico 2 (Rows 21, 22, 23) in BOTH Col f and Col j.
- BOTTOM CHECK: 3 empty holes at bottom of Col f (f28-f30) | 2 empty holes at bottom of Col j (j29-j30)!

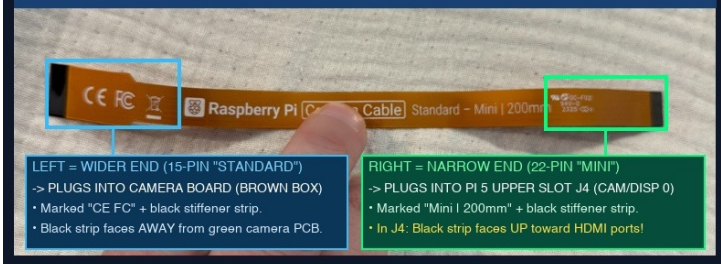
[RIBBON & LIVE TEST] 2-Panel Real-Photo Guides: 200mm Orange CSI Ribbon Orientation & Powered Bench System

Top: Side A/B ribbon check & Pi 5 J4 slot. Bottom: Camera Module 3 latch lock & complete powered Stage 2 system.

BOTH SIDES OF YOUR RIBBON CABLE VERIFIED: WHICH END & WHICH SIDE GO INTO PI 5

Left Top: Black-Tip / Printed Side (IMG_5775). Left Bottom: Gold-Teeth Side (IMG_5776). Right: Exact orientation in Upper Slot J4.

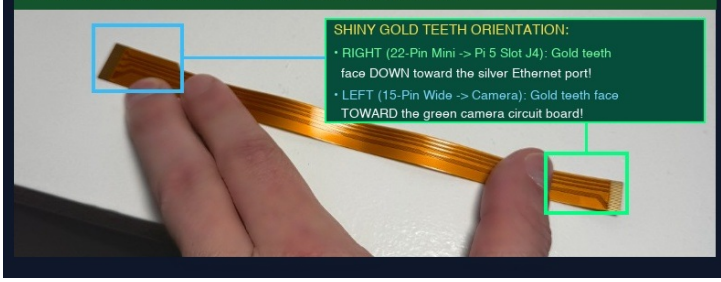
SIDE A (YOUR 1ST PHOTO): PRINTED TEXT & BLACK STIFFENER TIPS



LEFT = WIDER END (15-PIN "STANDARD")
-> PLUGS INTO CAMERA BOARD (BROWN BOX)
• Marked "CE FC" + black stiffener strip.
• Black strip faces AWAY from green camera PCB.

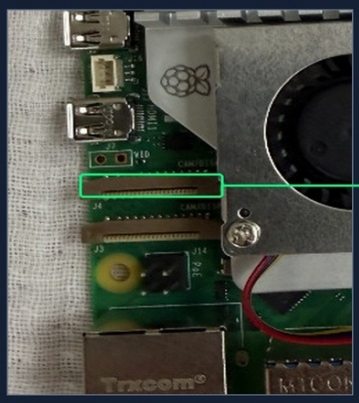
RIGHT = NARROW END (22-PIN "MINI")
-> PLUGS INTO PI 5 UPPER SLOT J4 (CAM/DISP 0)
• Marked "Mini I 200mm" + black stiffener strip.
• In J4: Black strip faces UP toward HDMI ports!

SIDE B (YOUR 2ND PHOTO): SHINY GOLD METAL CONTACT TEETH



SHINY GOLD TEETH ORIENTATION:
• RIGHT (22-Pin Mini -> Pi 5 Slot J4): Gold teeth face DOWN toward the silver Ethernet port!
• LEFT (15-Pin Wide -> Camera): Gold teeth face TOWARD the green camera circuit board!

1. EXACT SLOT ON YOUR PI 5: BEIGE CONNECTOR J4 (CAM/DISP 0)



BEIGE SLOT J4 (CAM/DISP 0):
• Boxed in green on the left: the UPPER horizontal beige-and-brown connector bar (right above the white J4 silkscreen label).

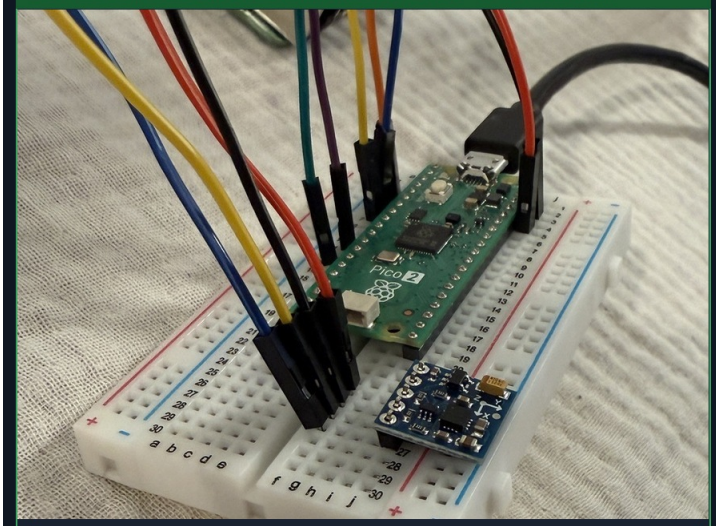
HOW TO PLUG RIBBON IN:
1. Gently lift the dark-brown latch ears up 2mm with fingernails.
2. Slide NARROW end of ribbon in (gold teeth facing DOWN toward the silver Ethernet port).

DO YOU NEED TO PLUG THE CAMERA IN RIGHT NOW TO TEST THE WIRES?
• NO! You can test your Pico 2 + Compass Sensor (W1-W4) first BEFORE plugging in the camera!
• When you DO plug in the camera ribbon (Step 2.4): NEVER plug or unplug the flat camera ribbon while the Raspberry Pi 5 power cord is plugged in. Always slide the ribbon in while Pi 5 is unplugged, and press the dark-brown plastic collar back down 2mm so it grips the flat ribbon snugly.

COMPLETED STAGE 2 GOLDEN-PATH BREADBOARD (PHOTO 57) & POWERED BENCH SYSTEM (PHOTO 58)

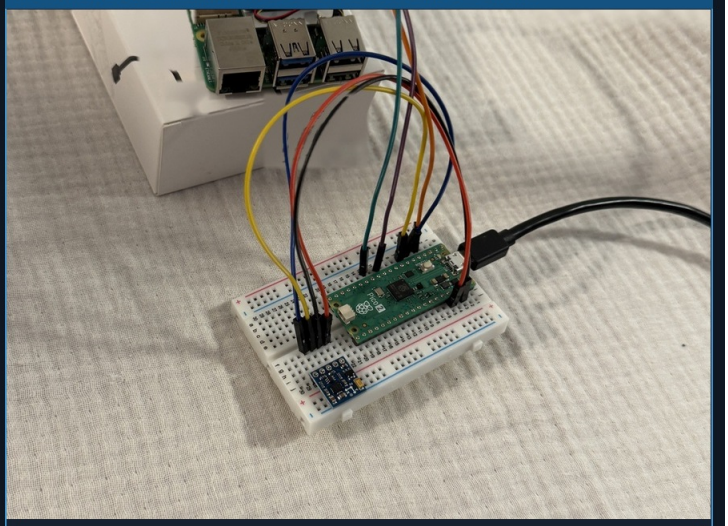
Left: Close-up of completed breadboard (Pico 2 + 7 wires + GY-271 + Micro-USB). Right: Full powered benchtop system (Photo 58).

LEFT (PHOTO 57): COMPLETED BREADBOARD CLOSE-UP (PICO 2 + W1–W7 + COMPASS)



GOLDEN-PATH BREADBOARD WIRING — 100% VERIFIED:
• Pico 2 in Rows 1–20; W1–W4 (Red/Black/Yellow/Blue) in Col f Rows 24–27 (3 empty bottom holes).
• GY-271 5-Pin Compass in Col j Rows 24–28 (2 empty bottom holes) + Micro-USB in Pico 2 Row 1!

RIGHT (PHOTO 58): COMPLETE POWERED BENCHTOP SYSTEM (GOLDEN PATH)



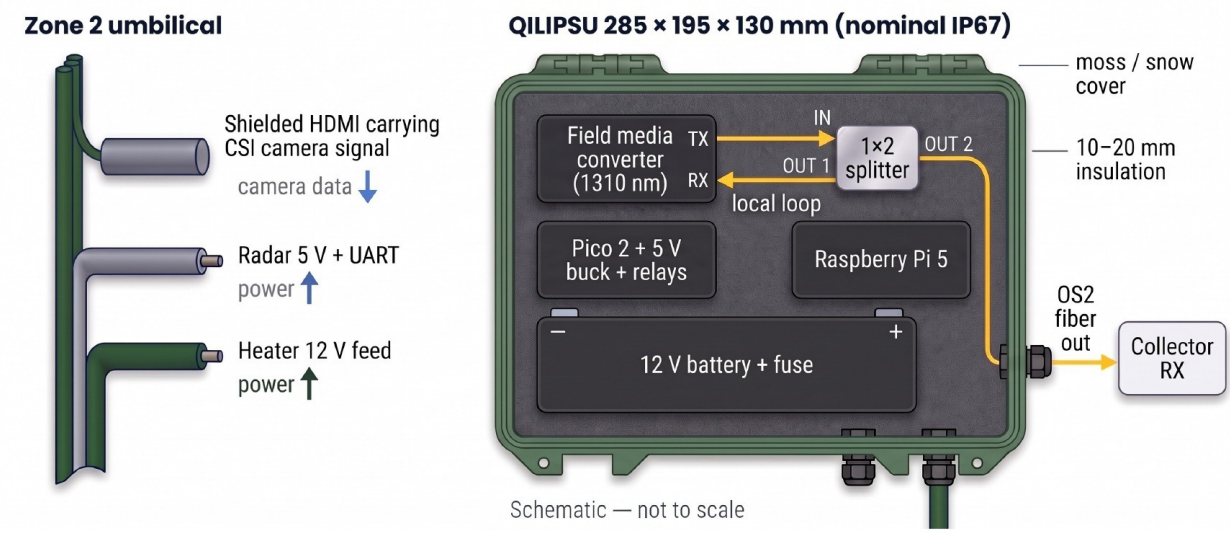
STEP 2.5 POWER & USB CONNECTIONS — 100% VERIFIED:
• Cable 1: Black Micro-USB in Pico 2 (Row 1) -> Laptop USB for live serial telemetry.
• Cable 2: Official 27W USB-C Power Supply in Pi 5 'PWR IN' + Camera Module 3 on 200mm ribbon!

[POD & VAULT] Inside Zone 1 Camera/Radar Pod & Zone 3 Base Vault (Wall-to-Wall 12V Battery Fit)

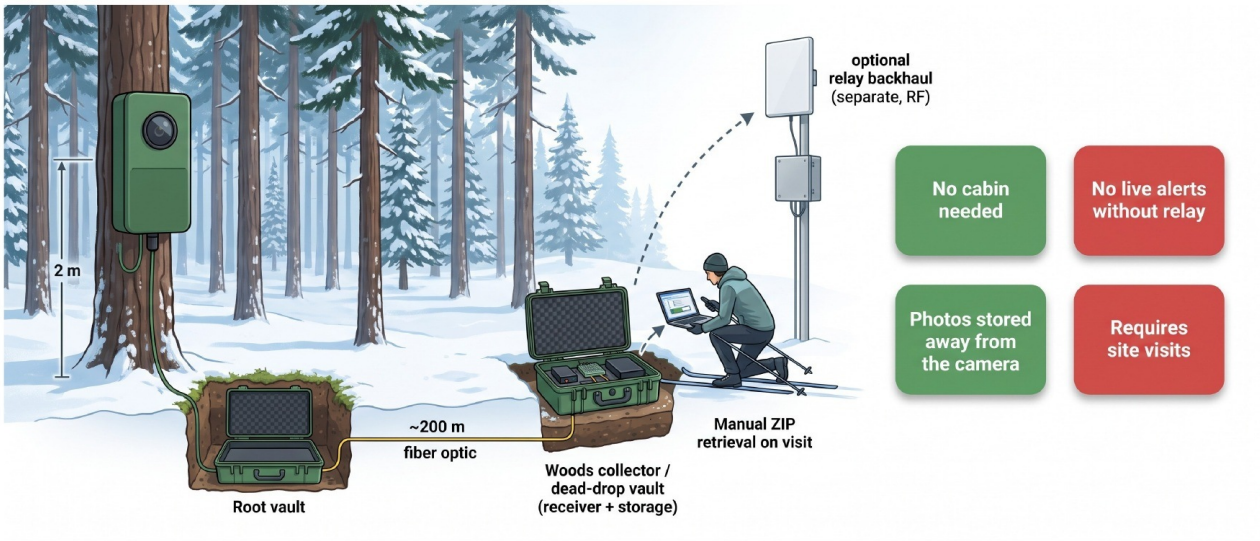
12V 12Ah AGM battery spans the bottom third; Pico 2, Relay & Pi 5 in middle row; Fiber Diode in top row.

Camera Pod Exploded View: One Window, One Camera, Hidden Radar

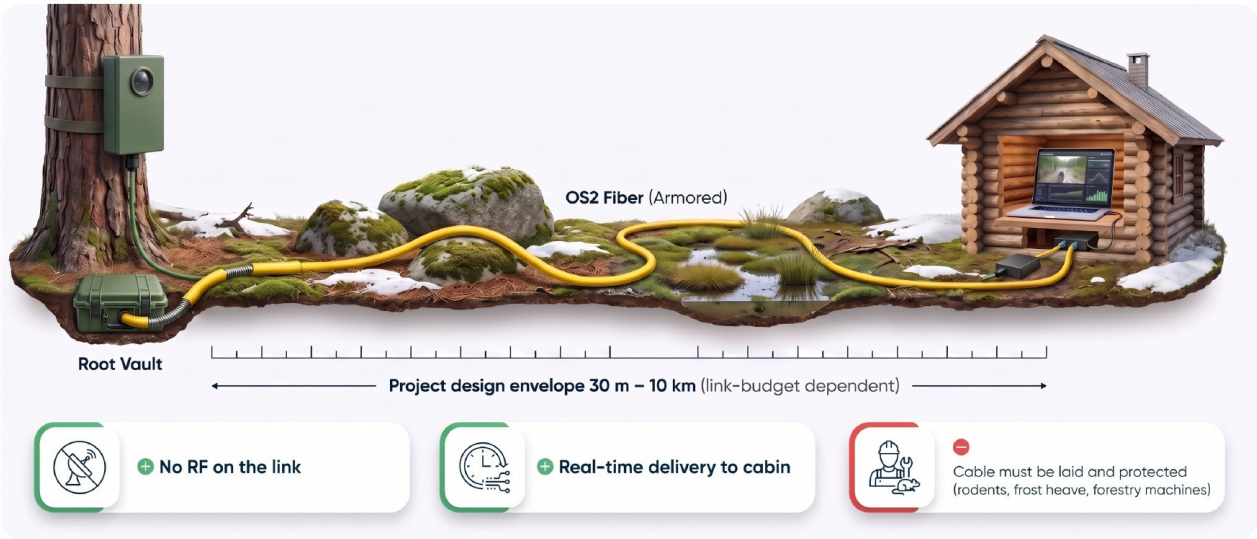
Umbilical and Root Vault: Power Goes Up, Camera Data Comes Down



Configuration 2 — Woods Dead-Drop: Images Survive Even If the Pod Does Not



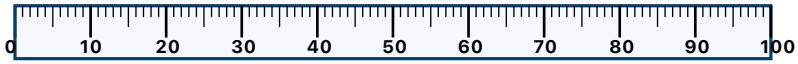
Configuration 1 — Direct Fiber: The Simplest Path When the Cabin Is Within Reach



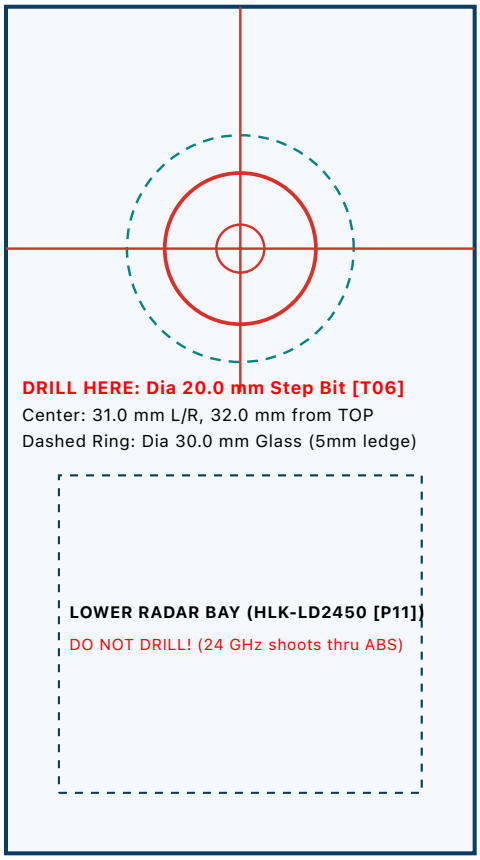
[TRUE 1:1 CAD TEMPLATE] Printable 1:1 Physical-Scale Vector Drilling & Foam Cutout Sheet (1.0 mm = 2.8346 pt)

PRINT AT 100% / ACTUAL SIZE (DO NOT 'FIT TO PAGE'). Verify 100.0 mm ruler below with Steel Ruler [T10] before drilling!

1:1 CALIBRATION RULER: 0 to 100.0 mm (283.46 pt) — MUST MEASURE EXACTLY 100.0 mm WITH [T10]!



TEMPLATE A — ZONE 1 LID [P20] (Hammond 1591BSBK: Exact 62.0 mm x 112.0 mm)

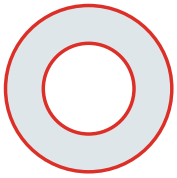


DRILL HERE: Dia 20.0 mm Step Bit [T06]
Center: 31.0 mm L/R, 32.0 mm from TOP
Dashed Ring: Dia 30.0 mm Glass (5mm ledge)

LOWER RADAR BAY (HLK-LD2450 [P11])
DO NOT DRILL! (24 GHz shoots thru ABS)

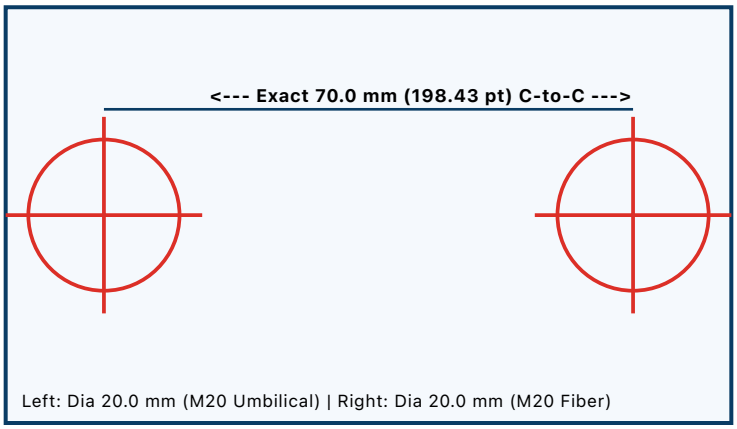
TEMPLATE C — 1:1 FOAM BAFFLE RING [P20]

Outer Circle: Exact Dia 22.0 mm | Inner Hole: Exact Dia 12.0 mm



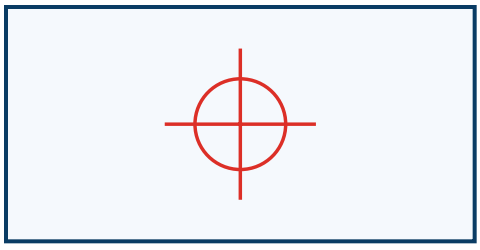
Fit snug over central IMX708 lens so side 940nm LEDs cannot glare!

TEMPLATE D — ZONE 3 VAULT GLAND SPACING [P21]



Left: Dia 20.0 mm (M20 Umbilical) | Right: Dia 20.0 mm (M20 Fiber)

TEMPLATE B — ZONE 1 BOTTOM WALL (62.0 x 31.0 mm)



Center: (31.0 mm, 15.5 mm) — Drill Dia 12.0 mm Step Bit [T06] for PG7/M12 Gland

Arctic Trail Cam System — IKEA-Style Manual Part 1: Indoor Benchtop Assembly & Bring-Up (Stages 1–3)

Zero Electronics Experience Required: This manual is written like an IKEA assembly guide. You do not need to know how to solder or read circuit schematics. Simply match each part in your hands to its [P##] Part Code, check the "Which Way Round Does It Fit?" box before plugging anything in, and follow each numbered step in order.

* Single pin reference: Wiring & Pinout Schedule

* Full hardware parts list: Hardware Configuration & BOM

* Next manual after bench bring-up: Part 2 — Enclosure Build, Camouflage & Tree Installation (Stages 4–6)

Step 0: Check Your Parts ([P01]–[P24]) & Identify Your 3 "Raspberry Pi" Boards

Full itemized part prices ([P01]–[P24]) and tool specifications ([T01]–[T12]) are maintained in HARDWARE_SPECS_AND_BOM.md.

* Stages 1–2 (Indoor Core Benchtop — $233.37): Use [P01], [P02], [P04]–[P10], [P23] (Cat6 Ethernet cables), [P24] (USB-C Gigabit Ethernet adapter), and household tools [T01]–[T03] (fingernails/tweezers, your laptop, and a small steel screwdriver/key).

* Stage 3 (Optical Fiber Data Diode — +$169.30 / $402.67 total): Adds [P13]–[P16] and [P19] (plus the 2nd Cat6 Ethernet cable in [P23]).

* Leave the 12V 12Ah Battery ([P17]) in its box during Stages 1–3! On the indoor table, power Raspberry Pi 5 [P01] with its included 27W USB-C wall adapter and Raspberry Pi Pico 2 [P07] via its Micro-USB data cable + USB-C adapter from your laptop ([T02]).

How to Identify Your 3 "Raspberry Pi" Boards at a Glance (DO NOT Mix Them Up!)

You have three different green circuit boards with "Raspberry Pi" in their name. Look at the physical ports on each board before starting so you never confuse them:

Board Code & Exact Name | Physical Size & Shape | How to Spot It Instantly in 3 Seconds | What Role It Plays in This Build

[P01] Raspberry Pi 5 (8GB) ($80.00) *(Primary Camera Computer — USE THIS BOARD!)* | Credit-card size (85 × 56 mm), flat underside with MicroSD slot | 1. Silver Gigabit Ethernet port is on the BOTTOM-RIGHT corner (below the blue USB 3.0 ports). ; 2. Has two narrow vertical beige/brown 22-pin slots labeled CAM/DISP 0 & CAM/DISP 1 between the micro-HDMI ports and Ethernet. ; 3. Has two round mounting holes beside the silver CPU for the [P02] Active Cooler + a white 4-pin FAN socket (J11) at top-right. | Main Field Camera & AI Computer (Stages 1–6). All [P02] Active Cooler, [P03] Hailo-8L, and [P06] 22-pin Camera Ribbon steps use this exact board (Raspberry Pi 5 8GB [P01]).

[P01-ALT] Raspberry Pi 4 Model B (8GB) ($75.00) *(Secondary Board — SET ASIDE)* | Credit-card size (85 × 56 mm), looks similar to Pi 5 at first glance! | 1. Silver Gigabit Ethernet port is on the TOP-RIGHT corner (above the blue USB 3.0 ports — the exact opposite of the Pi 5!). ; 2. Has only one wide (15-pin) white/black CAMERA slot. ; 3. Has NO mounting holes and NO 4-pin FAN socket for the Pi 5 Active Cooler ([P02]). | Do NOT use for the 22-pin camera cable [P06] or Active Cooler [P02]! Keep this Pi 4 aside as a spare or for a remote Collector Hub (Variant 3B).

[P07] Raspberry Pi Pico 2 (RP2350H) ($6.00) *(Tiny Gatekeeper Microcontroller)* | Stick-of-gum size (51 × 21 mm, 1/4 the size of Pi 5) | 1. Has 40 long metal legs soldered down both long edges (20 legs on the left, 20 legs on the right) so it presses into the white Solderless Breadboard ([P09]). ; 2. Has a tiny silver Micro-USB port at one end and a tiny white button labeled BOOTSEL. ; 3. Has NO MicroSD card slot and NO Ethernet/HDMI ports! | Always-On <4 mW Sensor Gatekeeper (Stages 1–6). Plugs into the white breadboard ([P09]), reads the QMC5883L magnetometer ([P08]) & radar ([P11]), and wakes the Pi 5 ([P01]).

STAGE 1: Prepare the Raspberry Pi 5 (8GB) ([P01]), Flash the MicroSD ([P04]) & Flash the Pico 2 ([P07])

Step 1.1 — Snap the Active Cooler ([P02]) onto the Raspberry Pi 5 (8GB) ([P01])

1. Unplug all power cables from the Raspberry Pi 5 (8GB) ([P01]) and set it flat on the table with the silver USB and Ethernet ports facing right. *(Confirm you are holding the Raspberry Pi 5 (8GB) [P01]—its silver Ethernet port is at the bottom-right corner, and you will see two empty round mounting holes on either side of the silver processor chip.)*

2. Pick up the Pi 5 Official Active Cooler ([P02]) and turn it upside down.

3. CRITICAL: Use your fingernail or tweezers ([T01]) to peel off the clear plastic protective sheet covering the pink/blue sticky thermal pads on the bottom of the cooler.

4. Hold the cooler right-side up over the Raspberry Pi 5 (8GB) ([P01]) so the two white plastic push-pins line up with the two round mounting holes on either side of the silver processor chip.

5. Press both white push-pins straight down with your thumbs until you hear two crisp CLICK sounds.

6. Pop out the factory plastic dust cap from J11 & plug in the 4-wire fan cable: Look at the tiny white 4-pin FAN socket (J11) at the top-right corner of the Raspberry Pi 5 (8GB) ([P01]) (between the black USB 2.0 ports and the 40 gold GPIO pins). A brand-new Raspberry Pi 5 ships with a tiny removable cream-white plastic pick-and-place dust cap sitting inside J11. Use a fine tip, fingernails, or tweezers ([T01]) to gently lift and pop that tiny plastic cap straight up out of J11, then push the white 4-pin connector on the cooler's 4-wire cable (black/red/yellow/blue) straight down into the open J11 socket. *Which way it fits:* The tiny raised plastic guides on one side of the plug slide into the matching notch on the socket—do not force it upside down.

* Documented Real-World Stage 1.1 Prototype Build Photos (11 Chronological Shots, Photos 1–11):

1. Photo 1 (stage1_01_unboxed_parts.jpg) — Step 1.0: Unboxed Raspberry Pi 5 [P01], Official Active Cooler [P02], and Raspberry Pi Pico 2 With Headers [P07] (SKU 828988, 40 pins pre-soldered) on the anti-static cloth.

2. Photo 2 (stage1_02_peeling_thermal_film.jpg) — Step 1.1A: Peeling the clear protective backing sheet off the 3 blue thermal pads on the underside of Official Active Cooler [P02].

3. Photo 3 (stage1_03_cooler_top_alignment.jpg) — Step 1.1B (Before Clicking Pins In): Top-down alignment of Active Cooler [P02] (6×7 bare aluminum fin grid on left, offset circular radial blower fan on right) positioned lightly over Raspberry Pi 5 [P01].

4. Photo 4 (stage1_04_hdmi_camdisp_clearance.jpg) — Step 1.1C (Before Clicking Pins In): Checking the bottom-left white spring push-pin by USB-C and clearance beside the two 22-pin CAM/DISP 0 & 1 ribbon slots prior to pressing the pins down.

5. Photo 5 (stage1_07_fan_socket_j11_closeup.jpg) — Step 1.1D (Before Clicking Pins In): Checking the top-right white spring push-pin aligned over its upper PCB mounting hole and locating the cream-white J11 4-pin FAN socket right beside the top USB 2.0 port.

6. Photo 6 (stage1_05_gpio_side_profile_pins_locked.jpg) — Step 1.1E (Push-Pins Clicked In): Press both white push-pins straight down (CLICK!). Side profile along the 40-pin GPIO header confirming both white push-pin barbs clicked completely through the underside of the Raspberry Pi 5 PCB with blue thermal pads compressed flat.

7. Photo 7 (stage1_06_ports_edge_profile.jpg) — Step 1.1F (Push-Pins Clicked In): Port-edge profile along USB-C, dual micro-HDMI, CAM/DISP, and Gigabit Ethernet confirming the bottom-left push-pin barb is locked through the PCB and the heatsink sits level.

8. Photo 8 (stage1_07b_pointing_at_j11_dust_cap.jpg) — Step 1.1G: With both push-pins locked in, pointing at the tiny cream-white factory plastic pick-and-place dust cap sitting inside the J11 4-pin FAN socket.

9. Photo 9 (stage1_07c_prying_j11_dust_cap.jpg) — Step 1.1H: Gently levering and popping the tiny cream-white factory dust cap straight up out of the J11 FAN socket.

10. Photo 10 (stage1_07d_j11_cap_removed_socket_open.jpg) — Step 1.1I: Removed factory dust cap held in the right hand, exposing the 4 vertical metal pins inside the open J11 FAN socket on Raspberry Pi 5 [P01] (left hand).

11. Photo 11 (stage1_07e_fan_cable_plugged_into_j11.jpg) — Step 1.1J: White 4-pin fan plug (black/red/yellow/blue) seated all the way down into J11 beside the top USB 2.0 port — Active Cooler [P02] installation 100% complete!

Step 1.2 — Flash the MicroSD Card ([P04]) for the Raspberry Pi 5 (8GB) ([P01]) on Your MacBook Pro ([T02])

1. Plug the SD card into your MacBook Pro ([T02]): Slide the SanDisk 64GB MicroSD card ([P04]) into its SD adapter sleeve and push it into the SDXC slot on the right side of your MacBook Pro (or plug your USB MicroSD card reader into a USB-C port on the side of your MacBook Pro).

2. Open Terminal on your MacBook Pro: Press ⌘ Cmd + Space on your MacBook Pro keyboard to open Spotlight Search, type Terminal, and press Enter (Return).

3. Launch Raspberry Pi Imager from Terminal: Type this exact command in Terminal and press Enter:

```
open -a "Raspberry Pi Imager"
```

(If Raspberry Pi Imager is not installed yet, type brew install --cask raspberry-pi-imager && open -a "Raspberry Pi Imager" and press Enter).

4. Follow the exact screens below in the Raspberry Pi Imager window (click each choice in order):

* Screen 1 — Choose Device: Click Raspberry Pi 5 (for your Raspberry Pi 5 (8GB) [P01]).

* Screen 2 — Choose OS: Click Raspberry Pi OS (other) -> select Raspberry Pi OS Lite (64-bit) (no desktop needed, boots fastest).

* Screen 3 — Choose Storage: Select your SanDisk 64GB MicroSD card ([P04]).

* Screen 4 — OS Customisation (Edit Settings):

* Hostname: Check Set hostname: and type trailcam-node

* Username & Password: Check Set username and password: -> type username pi and password trailcam (or your own password).

* Wireless LAN (Bench Setup Only): Check "Configure wireless LAN" and enter your home Wi-Fi name (SSID) and password so your MacBook Pro can talk to the Pi 5 over Wi-Fi during initial setup.

* Services Tab: Click the SERVICES tab at the top -> check "Enable SSH" and select "Use password authentication".

* Screen 5 — Write: Click SAVE -> click YES -> enter your MacBook Pro password when prompted, and wait ~2 minutes until Raspberry Pi Imager says "Write Successful — You can now remove the SD card from the reader".

Step 1.3 — Insert the MicroSD Card ([P04]) into the Underside of the Raspberry Pi 5 (8GB) ([P01])

* Which board & where the slot is:

* The MicroSD card ([P04]) goes into the Raspberry Pi 5 (8GB) ([P01])—*not* the tiny Pico 2 ([P07]), which has no SD slot.

* On the Raspberry Pi 5 (8GB) ([P01]), the metal MicroSD card slot is on the flat green UNDERSIDE (bottom face) of the board, under the left edge opposite the USB/Ethernet ports.

* Which way it fits (✓ CORRECT — See Panel 1 of the Isometric Orientation Guide):

1. Flip the Raspberry Pi 5 (8GB) ([P01]) upside down so you are looking at the flat green underside of the board.

2. Hold the SanDisk 64GB MicroSD card ([P04]) so its shiny gold metal contact teeth face UP toward the green Raspberry Pi 5 circuit board (and the printed red/grey SanDisk label faces DOWN toward the tabletop, away from the green board).

3. Slide the card straight into the silver metal slot until it stops flat (~12 mm deep). Flip the Raspberry Pi 5 (8GB) ([P01]) back right-side up.

Step 1.4 — Plug the Pico into Your MacBook Pro ([T02]) & Flash MicroPython from Terminal

1. Take a Micro-USB data cable (with a USB-C plug or USB-A to USB-C adapter on the computer end). Plug the USB-C end into a USB-C port on the left side of your MacBook Pro ([T02]) (right beside the esc / tab / caps lock keys)—leave the small Micro-USB end unplugged in your hand.

2. Pick up the stick-of-gum-sized Raspberry Pi Pico 2 with Headers ([P07], 40 metal legs underneath) *(or your bare Pico from the silver bag to test Stage 1)*. Find the tiny white rectangular button on top of the board labeled BOOTSEL (near the silver Micro-USB port).

3. Press and hold down the white BOOTSEL button with your fingertip, and while still holding BOOTSEL down, push the Micro-USB plug firmly all the way in (past the 2 tiny silver retention barbs on the metal tongue) into the Pico's silver Micro-USB port. Release your finger after 1 second.

4. Go to the Terminal window on your MacBook Pro, copy-paste this exact command (which flashes the all-in-one arctic_trailcam_pico2_complete.uf2 containing both official MicroPython v1.25.0 and pre-loaded main.py + test_qmc5883l.py at 0x10100000 with zero macOS metadata files), and press Enter:

```
   export COPYFILE_DISABLE=1; UF2="pi-diode-trailcam/pico_gatekeeper/
arctic_trailcam_pico2_complete.uf2"; xattr -c "$UF2" 2>/dev/null; if command -v picotool >/dev/null
2>&1; then picotool load -v -x "$UF2" && echo "SUCCESS: Flashed Pico 2 (RP2350) with MicroPython
v1.25.0 + main.py + test_qmc5883l.py!"; elif [ -d /Volumes/RP2350 ]; then mdutil -i off /Volumes/
RP2350 >/dev/null 2>&1; rm -rf /Volumes/RP2350/.Spotlight-V100 /Volumes/RP2350/.fseventsd /Volumes/
RP2350/._* 2>/dev/null; cp -X "$UF2" /Volumes/RP2350/ && echo "SUCCESS: Flashed Pico 2 (RP2350)!";
else echo "Pico drive not found — unplug USB-C, hold white BOOTSEL button, and plug back in!"; fi
```

5. What you should see: Terminal prints Verifying Flash: 100% OK and SUCCESS: Flashed Pico 2 (RP2350) with MicroPython v1.25.0 + main.py + test_qmc5883l.py! — installing both MicroPython and the complete Gatekeeper application in one single step!

* Documented Real-World Stage 1 Pico 2 [P07] & MacBook Pro [T02] USB Setup & BOOTSEL Flashing Photos (6 Chronological Shots Connecting Pico 2 Once With BOOTSEL Pressed, Photos 12–17):

12. Photo 12 (stage1_13_usb_c_adapter_in_macbook_pro.jpg) — Step 1.4A (Pico Unplugged): Plugging the rectangular USB-A cable + silver USB-A to USB-C adapter into the left-side USB-C (Thunderbolt) port of the MacBook Pro ([T02]) beside esc / tab / caps lock, leaving the small Micro-USB end unplugged in your hand.

13. Photo 13 (stage1_08_pico2_micro_usb_alignment.jpg) — Step 1.4B (Pico Unplugged): Holding Pico 2 With Headers [P07] (SKU 828988, 40 pre-soldered metal legs pointing down) and aligning the trapezoid key of the unplugged Micro-USB plug to the Pico 2 [P07] socket (narrow flat face toward top of PCB / BOOTSEL, wider beveled face toward the 40 long metal header pins).

14. Photo 14 (stage1_14_pressing_bootsel_button.jpg) — Step 1.4C (Pico Unplugged): Pressing and holding down the white BOOTSEL button on Pico 2 With Headers [P07] with the index fingertip before inserting the Micro-USB plug.

15. Photo 15 (stage1_15_plugging_cord_while_holding_bootsel.jpg) — Step 1.4D (Connecting With BOOTSEL Held): Pushing the Micro-USB plug straight into Pico 2 [P07] with the bottom hand while keeping the white BOOTSEL button pressed down with the top hand.

16. Photo 16 (stage1_09_pico2_top_bootsel_view.jpg) — Step 1.4E (Connected — Depth Check): Release BOOTSEL 1 second after plugging in and check insertion depth on Pico 2 [P07] — push firmly past the halfway friction catch so the 2 tiny silver retention barbs on the Micro-USB tongue slide inside the socket!

17. Photo 17 (stage1_11_pico2_side_profile_over_stage1_bench.jpg) — Step 1.4F (Connected — Flashing UF2): Pico 2 With Headers [P07] flashed in BOOTSEL mode via picotool load -v -x arctic_trailcam_pico2_complete.uf2 (Verifying Flash: 100% OK, bundling MicroPython v1.25.0 + main.py + test_qmc5883l.py at 0x10100000) over the Stage 1 work cloth.

STAGE 2: Exact Hole-by-Hole Breadboard ([P09]) & Raspberry Pi 5 (8GB) ([P01]) Wiring (7 Wires Total)

[CAUTION]

UNPLUG ALL USB POWER CABLES FIRST! Never plug or unplug breadboard wires while the Raspberry Pi 5 (8GB) ([P01]) or Raspberry Pi Pico 2 ([P07]) is plugged into power.

Look at the Real-World Golden-Path Top-Down Breadboard (Photo 57) & Complete Powered System (Photo 58) below. Every single connection is listed by its exact Row Number (1–30), Column Letter (a–j), and Empty-Hole Landmark Count.

How Your Half-Size Breadboard ([P09]) Works (In 20 Seconds)

* Turn the white breadboard ([P09]) so Row 1 is at the TOP and Row 30 is at the BOTTOM.

* Down the middle runs a deep vertical groove called the center canyon (ravine).

* On the left side of the canyon, in any single row (for example, Row 6), the 5 holes a6, b6, c6, d6, e6 are connected together underneath by one hidden metal clip.

* On the right side of the canyon, in any single row (for example, Row 5), the 5 holes f5, g5, h5, i5, j5 are connected together underneath by a separate metal clip.

* The left side (a–e) and right side (f–j) are not connected to each other across the center canyon.

Step 2.1 — Press the Raspberry Pi Pico 2 ([P07]) into Rows 1–20 (Columns b & h — DONE in Photos 18–23!)

1. Hold the Raspberry Pi Pico 2 With Headers ([P07]) over the top of the breadboard ([P09]) so its silver Micro-USB port points UP at Row 1.

2. Line up the left row of 20 metal legs over Column b, Rows 1 to 20 (Hole b1 at the top down to Hole b20).

3. Line up the right row of 20 metal legs over Column h, Rows 1 to 20 (Hole h1 at the top down to Hole h20).

4. Check that the green Pico 2 board bridges straight across the center canyon, leaving 1 open column on the left (Column a: a1–a20) and 2 open columns on the right (Columns i, j: i1–j20) in every row from 1 to 20.

5. Press straight down firmly and evenly with both thumbs until the black plastic pin strips sit completely flat against the white breadboard surface (verified in Photos 18–23 below!).

* Documented Real-World Stage 2.1 Breadboard [P09] + Pico 2 [P07] Seating Photos (6 Chronological Shots, Photos 18–23):

18. Photo 18 (stage2_01_empty_breadboard_30rows.jpg) — Step 2.1A: Holding the empty 30-Row Inland 400-Hole Half-Size Breadboard [P09] vertically (Row 1 at top, Row 30 at bottom, center ravine between Column e and Column f, plus outer red + and blue - power rails).

19. Photo 19 (stage2_02_pico2_top_alignment_rows1_to_20.jpg) — Step 2.1B: Top-down alignment of Raspberry Pi Pico 2 With Headers [P07] across the center ravine (Rows 1 to 20), with the silver Micro-USB port overhanging Row 1 at the top.

20. Photo 20 (stage2_03_pico2_angled_seated_in_breadboard.jpg) — Step 2.1C: Angled inspection after pressing Pico 2 [P07] straight down into Rows 1–20, leaving open holes beside the Pico on both left and right, with Rows 21–30 completely open below.

21. Photo 21 (stage2_04_pico2_side_profile_flush_on_breadboard.jpg) — Step 2.1D: Side-profile seating check confirming the black plastic header strip of Pico 2 [P07] is pressed 100% flush against the white plastic face of Breadboard [P09] with zero exposed silver pin metal.

22. Photo 22 (stage2_05_bench_overview_boxes_and_boards.jpg) — Step 2.1E: Full benchtop overview showing the component boxes, SanDisk 64GB MicroSD Card [P04] (in blue/black pack), and QMC5883L [P08] (in silver anti-static bag) at top, and the two assembled sub-units (Breadboard + Pico 2 on left, Raspberry Pi 5 + Active Cooler on right) at bottom.

23. Photo 23 (stage2_06_topdown_breadboard_pico2_next_to_pi5.jpg) — Step 2.1F: High-resolution close-up of Breadboard [P09] + Pico 2 [P07] (Rows 1–20) side-by-side next to Raspberry Pi 5 [P01] + Official Active Cooler [P02].

Step 2.2 — Pre-Stage & Plug the 7 Unique Rainbow Jumper Wires ([P10]) Into Pico 2 & Column f

[IMPORTANT]

Unit #2 Zero-Wrong-Turns Mechanical Pre-Check (Before Plugging Anything In):

1. Pre-Stage 2 Separate Groups on the Table (Photo 33): Your ELEGOO box contains 3 different 40-wire ribbons (Male-to-Male, Male-to-Female, Female-to-Female). Peel and lay out two distinct groups flat on the table before wiring:

* Group A — 4 Male-to-Male (Pin-to-Pin) wires (sharp metal pins on both ends): Red, Black, Yellow, Blue (for breadboard-to-breadboard W1–W4).

* Group B — 3 Male-to-Female (Pin-to-Socket) wires (sharp pin on one end, hollow black socket on the other end): Purple, Green, Orange (for breadboard-to-Raspberry Pi 5 W5–W7).

2. How to Push Pins Into Brand-New Stiff Breadboard Clips Without Bending Tips (Photo 35): Never push a jumper wire down while holding the soft colored wire insulation. Always pinch the square black plastic housing 2 mm above the metal pin tip and push straight down vertically (90°). If a pin ever bends, squeeze it flat between smooth flat-nose pliers (Photo 35).

3. Why W1–W4 Go Into Column f (Rows 24–27) (Photo 41): Always plug the lower ends of Red, Black, Yellow, and Blue into Column f (f24, f25, f26, f27 — the innermost column right next to the center ravine). That leaves Columns g, h, i open and gives full clearance for the GY-271 compass module in Column j (j24..j28)!

Check off each wire ([ ]) as you push its metal pin straight into the exact hole listed below:

Check | Wire ID & Color | Wire Type | Plug END 1 Into Breadboard Hole... | Plug END 2 Into... | What This Wire Does

[ ] | W1 — RED | Male-to-Male *(pin both ends)* | Hole j5 *(Right side, Row 5 = Pico 2 Pin 36 3V3_OUT)* | Hole f24 *(Inner-right Col f, Row 24 = GY-271 VCC)* | Sends safe 3.3V power from Pico 2 to Magnetometer.

[ ] | W2 — BLACK | Male-to-Male *(pin both ends)* | Hole j3 *(Right side, Row 3 = Pico 2 Pin 38 GND)* | Hole f25 *(Inner-right Col f, Row 25 = GY-271 GND)* | Connects Ground (0V) to Magnetometer.

[ ] | W4 — YELLOW | Male-to-Male *(pin both ends)* | Hole a7 *(Left side open Col a, Row 7 = Pico 2 Pin 7 GP5 SCL)* | Hole f26 *(Inner-right Col f, Row 26 = GY-271 SCL)* | Carries I2C Magnetic Clock (SCL).

[ ] | W3 — BLUE | Male-to-Male *(pin both ends)* | Hole a6 *(Left side open Col a, Row 6 = Pico 2 Pin 6 GP4 SDA)* | Hole f27 *(Inner-right Col f, Row 27 = GY-271 SDA)* | Carries I2C Magnetic Data (SDA).

[ ] | W5 — PURPLE | Male-to-Female *(pin + socket)* | Pin end -> Hole a11 *(Left side open Col a, Row 11 = Pico 2 Pin 11 GP8 TX)* | Socket end -> Raspberry Pi 5 [P01] Pin 10 (GPIO15 RX) *(5th pin down on the outer-right column of Pi 5)* | Sends TRIG,... wake reason from Pico 2 to Raspberry Pi 5.

[ ] | W6 — ORANGE | Male-to-Female *(pin + socket)* | Pin end -> Hole a4 *(Left side open Col a, Row 4 = Pico 2 Pin 4 GP2 DONE)* | Socket end -> Raspberry Pi 5 [P01] Pin 16 (GPIO23) *(8th pin down on the outer-right column of Pi 5)* | Sends DONE signal from Raspberry Pi 5 back to Pico 2.

[ ] | W7 — GREEN | Male-to-Female *(pin + socket)* | Pin end -> Hole a13 *(Left side open Col a, Row 13 = Pico 2 Pin 13 GND)* | Socket end -> Raspberry Pi 5 [P01] Pin 14 (GND) *(7th pin down on the outer-right column of Pi 5)* | Shared Ground reference between Pico 2 and Raspberry Pi 5.

[TIP]

How to Count the 3 Pins on the Raspberry Pi 5 (8GB) ([P01]) Without Ever Touching the Wrong Column:

1. Look at the 2 long vertical columns of 20 gold pins (40 pins total) along the right edge of the Raspberry Pi 5.

2. CRITICAL: The INNER-LEFT column (right next to the silver heatsink fins) must stay 100% BARE / EMPTY! Look ONLY at the OUTER-RIGHT column of gold pins (the column closest to the outer right green edge of the Pi 5 circuit board).

3. Starting from the very top-right outer corner pin (Pin 2), count straight down that OUTER-RIGHT column:

* 1st outer pin (Pin 2) — leave bare

* 2nd outer pin (Pin 4) — leave bare

* 3rd outer pin (Pin 6) — leave bare

* 4th outer pin (Pin 8) — leave bare

* 5th outer pin (Pin 10) -> Push W5 PURPLE Female socket straight down onto this pin!

* 6th outer pin (Pin 12) — SKIP 1 PIN (leave bare!)

* 7th outer pin (Pin 14) -> Push W7 GREEN Female socket straight down onto this pin!

* 8th outer pin (Pin 16) -> Push W6 ORANGE Female socket straight down onto this pin!

Step 2.3 — Lock the HiLetgo GY-271 (QMC5883L, [P08]) Compass Into Column j (Rows 24–28) Facing RIGHT

1. Take the HiLetgo GY-271 (HW-246 / QMC5883L, [P08]) blue breakout board and its 5-Pin Black Header Strip out of the pink anti-static pouch (Photo 34).

2. Hold the 5-pin header strip so the SHORT GOLD PINS point UP and the LONG GOLD PINS point DOWN (Photo 38).

3. Secure Electrical Contact on the GY-271 Through-Holes (Solder or Safe Zero-Solder Workarounds — NEVER Bend/Twist Pins With Pliers!):

  * WARNING — DO NOT Bend or Twist the Gold Header Pins With Pliers: Cheap 2.54 mm header pins are made of brittle cast brass—bending or twisting them with pliers (even 20°–90°) will snap the pins off! Keep the header pins 100% straight (0°).

  * Best Practice (If a Soldering Iron Is Available): Slide the blue GY-271 board onto the 5 straight short gold pins (chip & white labels facing UP) and touch a small drop of solder to the 4 active pins (VCC, GND, SCL, SDA).

  * If NO Soldering Iron Is Available (3 Safe Zero-Solder Workarounds):

1. Workaround A — Aluminum Foil Conductive Sleeve (Cold-Solder Press-Fit): Fold a narrow 2 mm × 10 mm strip of standard kitchen aluminum foil over the tips of the 4 straight short gold pins (VCC, GND, SCL, SDA — keep individual foil wraps separate so adjacent pins don't touch!) and press the blue GY-271 board straight down over them. The soft aluminum compresses into the 0.18 mm hole gap around each pin for 100% metal-to-metal contact with zero pin-bending stress.

2. Workaround B — Wooden Toothpick Wedge: Slide the blue GY-271 board onto the straight 5-pin header, then gently push the pointed tip of a wooden toothpick into the gap on one side of each of the 4 active holes (VCC, GND, SCL, SDA) so the silver PCB ring is wedged tightly against the straight gold pin, and lock with a dab of Permatex Black Silicone [T11].

3. Workaround C — Direct Jumper-Pin Through-Hole Pinning *(Also works if your 5-pin header broke!)*: Skip the 5-pin header strip completely! Place the blue GY-271 board's 4 active holes (VCC, GND, SCL, SDA) directly over breadboard Column j (Holes j24, j25, j26, j27) and push the bottom metal pins of your 4 jumper wires (W1 Red, W2 Black, W4 Yellow, W3 Blue) straight through the 4 holes of the blue GY-271 board down into the breadboard holes (j24–j27), wedging them snug with a toothpick tip or silicone dab.

4. Slide the 5 holes of the blue GY-271 board (VCC, GND, SCL, SDA, DRDY) down onto the SHORT GOLD PINS with the black square sensor chip and printed white letters facing UP and the blue board hanging out to the RIGHT (Photo 40).

5. Push the 5 LONG GOLD PINS straight down into Column j, Rows 24–28 (Photo 41 & Photo 42) so that the blue GY-271 board hangs out to the right over the +/- power rails (with zero crowding against the wires in Column f):

* Hole j24 (VCC) $\leftrightarrow$ matches RED (f24)
* Hole j25 (GND) $\leftrightarrow$ matches BLACK (f25)
* Hole j26 (SCL) $\leftrightarrow$ matches YELLOW (f26)
* Hole j27 (SDA) $\leftrightarrow$ matches BLUE (f27)
* Hole j28 (DRDY) $\leftrightarrow$ hangs 1 hole lower in Row 28 with no wire needed!

[CAUTION]

Two Critical Real-World Breadboard Rules (Empty-Hole Counting & Hook-Shaped Pin Risk):

1. Do NOT Trust the Printed Row Numbers (1–30) — Count Empty Holes From the Ends Instead (compass_5pin_vs_4wires_1hole_shift_fix_v2.jpg):

* On ELEGOO / zyamy 30-row breadboards, the printed black numbers (1–30) do not align cleanly with the physical holes. Always align by counting empty holes from the green Pico 2 (Row 20) and the bottom edge of the breadboard (Row 30):

* In Column f (4 Wires: Red, Black, Yellow, Blue): Leave 3 empty holes below the green Pico 2 (f21, f22, f23), and leave 3 empty holes at the bottom of Column f (f28, f29, f30).

* In Column j (5-Pin Compass Strip: VCC, GND, SCL, SDA, DRDY): Because the Compass strip has 5 pins (including the unused 5th DRDY pin at the bottom) while there are only 4 wires, you must leave 3 empty holes below the green Pico 2 (j21, j22, j23) and ONLY 2 EMPTY HOLES at the bottom of Column j (j29, j30)! *(If you leave 3 empty bottom holes under the 5-pin strip, the entire Compass shifts 1 hole UP so Red (+3.3V) lands on Compass GND (Pin #2)!)*

2. Watch Out for "Hook-Shaped" Bent Jumper Pins Inside Stiff Holes (stage2_38 & stage2_39):

* If a male jumper pin feels mushy or resists going all the way down into a stiff new breadboard hole, pull it back out and inspect the metal tip (stage2_38_blue_wire_hook_pin_risk.jpg). A pin tip that bends into a J-hook shape either rides on top of the internal spring clip without making electrical contact (open circuit) or curves sideways inside the slot where it can touch/short against an adjacent pin. Always squeeze any hooked or curved jumper pin flat between the smooth jaws of your flat-nose pliers (stage2_39_straightening_hook_pin_with_pliers.jpg) before re-inserting straight down (90°).

* Documented Real-World Steps 2.2 & 2.3 7-Color Jumper Wiring [P10] + GY-271 Compass [P08] + Pi 5 GPIO Golden-Path Photos (16 Chronological Shots, Photos 31–46):

31. Photo 31 (stage2_14_elegoo_jumpers_flatlay.jpg) — Step 2.2A: Flat-lay overview of the ELEGOO 3-ribbon jumper wire box, Breadboard [P09] + Pico 2 [P07], and Raspberry Pi 5 [P01].

32. Photo 32 (stage2_15_unpacked_40wire_jumper_ribbon.jpg) — Step 2.2B: Unpacked 40-wire rainbow jumper ribbon next to Breadboard + Pico 2 and Raspberry Pi 5.

33. Photo 33 (stage2_16_4_male_male_and_3_male_female_wires_selected.jpg) — Step 2.2C: Pre-staging the 2 separate wire groups flat on the table: 4 Male-to-Male (Pin-to-Pin) wires (Red, Black, Yellow, Blue) and 3 Male-to-Female (Pin-to-Socket) wires (Purple, Green, Orange).

34. Photo 34 (stage2_17_unboxed_gy271_compass_and_5pin_header.jpg) — Step 2.2D: Unboxed HiLetgo GY-271 (QMC5883L) [P08] blue compass PCB (VCC, GND, SCL, SDA, DRDY) and its 5-pin male header strip above the labeled ESD pouch.

35. Photo 35 (stage2_18_straightening_jumper_pin_with_pliers.jpg) — Step 2.2E: Using smooth flat-nose pliers (HYCHIKA kit) to keep male jumper pins dead-straight (90°) before vertical insertion.

36. Photo 36 (stage2_19_w1_to_w4_wired_on_breadboard_top_view.jpg) — Step 2.2F: Overhead view of all 7 rainbow wires seated in Column a (a4=Orange, a6=Blue, a7=Yellow, a11=Purple, a13=Green) and Column j (j3=Black, j5=Red) alongside Pico 2.

37. Photo 37 (stage2_20_all_7_wires_on_breadboard_rows_visible.jpg) — Step 2.2G: Angled inspection of the full breadboard wiring with all row numbers visible.

38. Photo 38 (stage2_21_holding_5pin_header_short_up_long_down.jpg) — Step 2.3A: Holding the 5-pin header strip with the SHORT GOLD PINS pointing UP and LONG GOLD PINS pointing DOWN.

39. Photo 39 (stage2_22_pliers_zigzag_bend_for_snug_unsoldered_fit.jpg) — Step 2.3B (CAUTION — Do NOT Bend Brittle Brass Header Pins): Inspecting the 4 active short pins (VCC, GND, SCL, SDA) — solder the 4 pins or use an aluminum-foil sleeve / wooden toothpick wedge (never bend the short brass header pins with pliers, as they snap easily!).

40. Photo 40 (stage2_23_gy271_slid_onto_5pin_header.jpg) — Step 2.3C: GY-271 PCB (VCC, GND, SCL, SDA, DRDY facing up) seated onto the 5-pin header flush against the black plastic collar.

41. Photo 41 (stage2_24_w1_to_w4_in_col_f_and_5pin_strip_in_col_j.jpg) — Step 2.3D: Seating W1–W4 (Red, Black, Yellow, Blue) in Column f (f24..f27) and the 5-pin header in Column j (j24..j28) with 3 open holes (Cols g, h, i) between them.

42. Photo 42 (stage2_25_completed_breadboard_with_gy271_in_col_j_facing_right.jpg) — Step 2.3E: Completed breadboard with the blue GY-271 board seated in Column j (Rows 24–28) hanging out to the right over the +/- power rails!

43. Photo 43 (stage2_26_pi5_gpio_outer_column_verified_side_view.jpg) — Step 2.3F: Verified side view of the Raspberry Pi 5 [P01] 40-pin GPIO header showing all 20 inner-column pins bare and the 3 jumper sockets seated on the Outer-Right column (#1–#4 bare, #5 = Purple, #6 bare, #7 = Green, #8 = Orange).

44. Photo 44 (stage2_27_pi5_gpio_outer_column_verified_angled_view.jpg) — Step 2.3G: Angled verification of the Raspberry Pi 5 [P01] Outer-Right GPIO column (Purple, skip 1, Green, Orange) and Active Cooler [P02].

45. Photo 45 (stage2_28_pi5_gpio_and_cam_disp0_top_view.jpg) — Step 2.3H: Close-up top view of Raspberry Pi 5 [P01] showing the verified Outer-Right GPIO wiring (Purple, skip 1, Green, Orange) and the vertical J4 CAM/DISP 0 camera slot ready for Step 2.4.

46. Photo 46 (stage2_29_completed_stage2_3_full_assembly_golden.jpg) — Step 2.3I: Full golden-path benchtop overview of the completed Stage 2.3 hardware assembly (Breadboard + Pico 2 + GY-271 + 7-wire harness + Raspberry Pi 5 with SanDisk 64GB MicroSD seated underneath and Camera Module 3 box staged on the table).

Step 2.4 — Connect the Official 200mm Orange Camera Ribbon Cable ([P06]) Between Raspberry Pi 5 (J4 CAM/DISP 0) & Camera Module 3 NoIR Wide ([P05])

Make sure the Raspberry Pi 5 (8GB) ([P01]) is unplugged from power while connecting the ribbon cable.

Step 2.4A — Lock the NARROW End (22-Pin Mini, 11.5 mm) Into J4 (CAM/DISP 0) on the Raspberry Pi 5 (Photos 47–50)

1. Identify the Two Ends & Two Sides of the Orange Raspberry Pi 5 Camera Cable (Standard - Mini | 200mm) ([P06], Photos 48–49):

* End 1 — NARROW End (22-Pin Mini, 11.5 mm wide, printed "Mini | 200mm"): Plugs into J4 (CAM/DISP 0) on the Raspberry Pi 5.

* End 2 — WIDE End (15-Pin Standard, 16.0 mm wide, printed "CE FC"): Plugs into the Camera Module 3 NoIR Wide ([P05]).

* Notice that both black stiffeners are on the SAME side (Side A) and both sets of shiny gold metal teeth are on the SAME side (Side B).

2. Locate J4 (CAM/DISP 0) on the Raspberry Pi 5: Look between the Micro-HDMI ports and the silver Gigabit Ethernet port. There are two vertical beige-and-brown slots side-by-side: use the UPPER slot (J4, labeled CAM/DISP 0).

3. Open the Brown Vertical Wedge Latch (1.5 mm ONLY — Do NOT Pull Out or Tilt): Use two fingernails (or the tips of your tweezers) on both the left and right ears of the dark-brown plastic latch on J4 to lift it straight up (90°) by ONLY 1.5 mm until it loosens. *(Never pry at an angle or pull the brown wedge completely out of its thin side tracks.)*

4. Insert the Narrow 22-Pin End & Press Evenly on a Flat Box to Lock (Photo 50):

* Slide the narrow 22-pin end ("Mini | 200mm") straight down into J4 (CAM/DISP 0) with:

  * SHINY GOLD TEETH facing RIGHT (toward the silver Ethernet port)

  * BLACK STIFFENER + WHITE "Mini | 200mm" TEXT facing LEFT (toward the Micro-HDMI ports)

* Support the underside of the Raspberry Pi 5 flat on top of the white 27W USB-C Power Supply box and press straight down (90°) evenly across the top of the brown latch using the flat back/spine of your tweezers until the brown latch sits flush.

Step 2.4B — Swap the Factory White Ribbon on Camera Module 3 NoIR Wide ([P05]) for the WIDE End (15-Pin Standard, 16.0 mm) of Your Orange Ribbon (Photos 51–58)

1. Inspect the Factory White Ribbon Orientation First (Photo 51): Take the Raspberry Pi Camera Module 3 NoIR Wide ([P05]) out of its silver anti-static bag and turn the small square board over onto its back (PCB side). Notice how the factory white Pi 4 ribbon is seated: its colored stiffener tape (Blue) faces UP (on the same side as the brown plastic slide-latch, facing away from the black camera PCB), while its metal contact teeth face DOWN against the black camera PCB (where the 15 internal silver contact pins are soldered).

2. Slide the Camera Latch Out 1.5 mm & Remove the Factory White Ribbon (Photo 52): Use two thumbnails on the left and right ears of the camera's brown horizontal plastic latch to slide it straight out by ONLY 1.5 mm (do not pull it off). Pull the factory white ribbon straight out and set it back in the box.

3. Insert the Wide 15-Pin End ("CE FC") of Your Orange Ribbon & Lock (Photo 53):

* Hold the back of the black camera PCB facing you and orient the wide 15-pin end ("CE FC") of your orange ribbon so that:

  * SHINY GOLD CONTACT TEETH face DOWN against the black camera circuit board (touching the 15 internal solder pins on the board side).

* BLACK STIFFENER + WHITE "CE FC" LETTERS face UP (away from the camera circuit board, on the brown-latch side — matching the blue tape on the factory white ribbon!).

* Push the tip of the wide orange ribbon 2 mm deep inside the slot so that zero shiny gold teeth are visible outside the connector, then push both ears of the brown plastic slide-latch straight back in (1.5 mm) until locked flush (Photo 53).

* Documented Real-World Step 2.4 & Final Stage 2 Verification Golden-Path Photos (12 Chronological Shots, Photos 47–58):

47. Photo 47 (stage2_30a_three_white_boxes_golden.jpg) — Step 2.4A-1: The 3 white boxes for Stage 2.4–2.6: Raspberry Pi 27W USB-C Power Supply, Raspberry Pi 5 Camera Cable (Standard - Mini | 200mm) [P06], and Raspberry Pi Camera Module 3 NoIR Wide [P05].

48. Photo 48 (stage2_32_flat_camera_ribbon_wide_vs_narrow_ends.jpg) — Step 2.4A-2 (Side A — Black Stiffener & Printed Text Side): Left = Wide 15-pin Standard end ("CE FC", for Camera); Right = Narrow 22-pin Mini end ("Mini | 200mm", for Pi 5 J4).

49. Photo 49 (stage2_33_flat_camera_ribbon_gold_teeth_side.jpg) — Step 2.4A-3 (Side B — Shiny Gold Contact Teeth Side): Both sets of gold teeth (15-pin left, 22-pin right) are on this side.

50. Photo 50 (stage2_34_pi5_j4_ribbon_seated_and_locked.jpg) — Step 2.4A-4: Narrow 22-pin Mini end seated 2 mm deep into J4 (CAM/DISP 0) on Raspberry Pi 5 [P01] (gold teeth facing silver Ethernet port, black stiffener facing Micro-HDMI) with the brown vertical latch pressed flush.

51. Photo 51 (stage2_35_camera_module3_factory_white_ribbon.jpg) — Step 2.4B-1: Inspecting the factory white ribbon on the back of Camera Module 3 NoIR Wide [P05] (blue stiffener facing up on latch side, metal teeth facing down against the black camera PCB).

52. Photo 52 (stage2_36_camera_module3_white_ribbon_removed.jpg) — Step 2.4B-2: Sliding the camera's brown horizontal latch out by 1.5 mm and removing the factory white Pi 4 ribbon.

53. Photo 53 (stage2_37_camera_module3_orange_ribbon_seated_and_locked.jpg) — Step 2.4B-3: Completed Step 2.4B — Wide 15-pin end of the orange Pi 5 ribbon seated 2 mm deep inside Camera Module 3 NoIR Wide [P05] (black stiffener + white "CE FC" letters facing up on the brown-latch side, all 15 shiny gold teeth facing down against the black camera PCB with zero teeth exposed outside, and brown latch pushed closed).

54. Photo 54 (stage2_38_blue_wire_hook_pin_risk.jpg) — Step 2.4C-1: Inspecting a bent "J-hook" tip on the blue SDA jumper pin (W3) after meeting resistance in a stiff breadboard hole.

55. Photo 55 (stage2_39_straightening_hook_pin_with_pliers.jpg) — Step 2.4C-2: Straightening the hooked jumper pin flat (90°) with smooth flat-nose pliers so it seats cleanly inside the breadboard spring clip without shorting against neighboring rows.

56. Photo 56 (stage2_40_compass_shifted_down_2_empty_bottom_holes_closeup.jpg) — Step 2.4D-1: Close-up verification of the golden-path hole-counting rule at the bottom of the breadboard: 3 empty bottom holes (f28, f29, f30) under the 4 wires in Column f and 2 empty bottom holes (j29, j30) under the 5-pin GY-271 Compass strip in Column j.

57. Photo 57 (stage2_41_completed_breadboard_topdown_golden.jpg) — Step 2.4D-2: Straight-down overhead verification of the completed golden-path breadboard (Pico 2 in Rows 1–20, 3 empty rows 21–23, W1–W4 in f24–f27 with 3 empty bottom holes, and GY-271 in j24–j28 with 2 empty bottom holes).

58. Photo 58 (stage2_42_complete_benchtop_system_powered_golden.jpg) — Step 2.5/2.6 Golden-Path Overview: Complete Stage 2 benchtop hardware assembly (Breadboard + Pico 2 + GY-271 + 7 rainbow wires + Raspberry Pi 5 + 200mm Orange Camera Ribbon + Camera Module 3 NoIR Wide) powered and ready for live telemetry.

Step 2.5 — Connect Pico 2 ([P07]), Pre-Stage Refactored Code onto SanDisk 64GB MicroSD ([P04] — DONE in Photos 24–30!) & Test QMC5883L (0x0d)

1. Connect Micro-USB to Pico 2 [P07] on Breadboard [P09] & Pre-Stage Refactored pi5_sender_node/ Code onto SanDisk 64GB MicroSD [P04] with Zero macOS Metadata (DONE — Photos 24–30):

* Plug the black Micro-USB data cable (CORD 1) firmly all the way into Pico 2 [P07] on Breadboard [P09] (Photo 24).

* Slide the SanDisk 64GB MicroSD Card [P04] into the SanDisk microSD Adapter (Photos 25–27) and insert it into the right-side SDXC slot of your MacBook Pro ([T02]) (Photo 28).

* On /Volumes/bootfs, our automated staging helper (stage_sd_card_without_macos_metadata() in run_guided_bench_doctor.py) disables macOS Spotlight indexing (mdutil -i off /Volumes/bootfs), writes the refactored pi5_sender_node/ files as pure byte streams (COPYFILE_DISABLE=1, zero ._* or .DS_Store files), enables dtparam=uart0=on & enable_uart=1 in config.txt, configures cloud-init (user-data) to auto-install /home/pi/pi5_sender_node, and safely ejects /Volumes/bootfs.

* Flip Raspberry Pi 5 [P01] upside down (FCC ID: 2ABCB-RPI5) and slide the SanDisk 64GB MicroSD Card [P04] into the underside metal J9 slot with the shiny gold contact teeth facing UP toward the green Pi 5 board (Photos 29–30).

* Documented Real-World Stage 2.5 Pico 2 [P07] USB Cord, SanDisk 64GB MicroSD [P04] Adapter, MacBook Pro Staging & Raspberry Pi 5 [P01] Insertion Photos (7 Chronological Shots, Photos 24–30):

24. Photo 24 (stage2_07_micro_usb_cord_connected_to_breadboard_pico2.jpg) — Step 2.5A: Connecting the black Micro-USB data cord (CORD 1) to Pico 2 [P07] on Breadboard [P09] without holding BOOTSEL (push the Micro-USB plug firmly all the way in past the 2 tiny silver retention barbs!).

25. Photo 25 (stage2_08_sandisk_64gb_microsd_and_adapter.jpg) — Step 2.5B: Holding the black SanDisk microSD Adapter in the left hand and the red/grey SanDisk Ultra Plus 64GB MicroSD Card [P04] in the right hand.

26. Photo 26 (stage2_09_sliding_microsd_into_adapter.jpg) — Step 2.5C: Sliding the SanDisk 64GB MicroSD Card [P04] into the bottom slot of the SanDisk Adapter with both printed labels facing up.

27. Photo 27 (stage2_10_microsd_seated_flush_in_adapter.jpg) — Step 2.5D: Pushing the SanDisk 64GB MicroSD Card [P04] all the way flush into the bottom of the SanDisk Adapter.

28. Photo 28 (stage2_11_sd_adapter_in_macbook_pro_slot.jpg) — Step 2.5E: Inserting the SanDisk Adapter + 64GB MicroSD Card [P04] into the right-side SDXC slot of the MacBook Pro [T02] beside the delete, \, return, and shift keys.

29. Photo 29 (stage2_12_sliding_microsd_into_pi5_underside_slot.jpg) — Step 2.5F: Sliding the updated SanDisk Ultra PLUS 64GB MicroSD Card [P04] into the metal J9 underside slot of Raspberry Pi 5 [P01] (FCC ID: 2ABCB-RPI5), with shiny gold contact teeth facing UP toward the green board and the red-and-grey printed label facing away.

30. Photo 30 (stage2_13_microsd_seated_flush_in_pi5_underside_slot.jpg) — Step 2.5G: SanDisk 64GB MicroSD Card [P04] pushed all the way in and seated flush inside the Raspberry Pi 5 [P01] underside J9 slot — 100% ready for first boot!

2. Upload Gatekeeper Code to Pico 2 ([P07]) & Test the QMC5883L Magnetometer (0x0d):

* Plug the Micro-USB data cable (Photo 24) from the Raspberry Pi Pico 2 ([P07]) into a USB-C port on your MacBook Pro ([T02]) *(do NOT hold BOOTSEL once MicroPython has been flashed via picotool load -v -x)*.

* Go to Terminal on your MacBook Pro, type this exact command from the project folder, and press Enter:

```
mpremote run pi-diode-trailcam/pico_gatekeeper/test_qmc5883l.py
```

3. What you should see:

* Terminal prints Found QMC5883L at 0x0d (decimal [13]) and a steady stream of X / Y / Z / Delta magnetic readings.

* Pick up your small steel screwdriver ([T03]) and wave it within 3 cm of the QMC5883L sensor ([P08]). You will immediately see <-- [MAG TRIGGER] appear on your MacBook Pro screen!

4. Press Ctrl + C on your MacBook Pro keyboard to stop the test, then copy the gatekeeper firmware onto the Pico 2 by typing these exact commands and pressing Enter:

```
mpremote fs cp pi-diode-trailcam/pico_gatekeeper/main.py :main.py
```

```
mpremote reset
```

Note on Sensor Profiles: At boot, main.py automatically detects connected sensors (FULL_TRIPLE when both 0x0d and 0x48 are present, or MAG_RADAR_DUTY when only 0x0d is present). To test automatic single-sensor magnetic wake on the bench before connecting the HLK-LD2450 radar, either plug a jumper wire from Breadboard Hole a19 (GP14) to Hole a18 (GND) at boot or set BENCH_MAG_ONLY = True in pico_gatekeeper/main.py (PROFILE_BENCH_MAG_ONLY).

Step 2.6 — Plug Power & Ethernet into Raspberry Pi 5 (8GB) ([P01]), Copy the Sender Software & Send Your First Live Alert on Your MacBook Pro

1. Plug the official 27W USB-C power supply into a wall outlet and plug its USB-C cord into the USB-C PWR IN port on the Raspberry Pi 5 (8GB) ([P01]). Wait 45 seconds for it to boot.

2. In Terminal on your MacBook Pro, type these two commands one at a time (pressing Enter after each line, and entering your Pi password trailcam when asked) to install the required packages on the Raspberry Pi 5 and copy the sender folder (with COPYFILE_DISABLE=1 so no macOS ._* files are transferred):

```
ssh pi@trailcam-node.local "sudo apt update && sudo apt install -y python3-opencv python3-nacl python3-serial python3-picamera2"
```

```
COPYFILE_DISABLE=1 scp -r pi-diode-trailcam/pi5_sender_node pi@trailcam-node.local:~/pi_sender_node
```

3. Enable the hardware UART serial port on the Raspberry Pi 5 (dtparam=uart0=on, enable_uart=1 with serial login console disabled) by typing this exact command in Terminal and pressing Enter:

```
ssh pi@trailcam-node.local "sudo raspi-config nonint do_serial_hw 0 && sudo raspi-config nonint do_serial_cons 1 && sudo reboot"
```

4. Plug a Cat6 Ethernet cable between the Raspberry Pi 5 (8GB) ([P01]) Gigabit Ethernet port and your MacBook Pro's USB-C Ethernet adapter. Set your MacBook Pro's Ethernet adapter IP address to 10.10.0.100 (subnet mask 255.255.255.0 / /24, allowing UDP port 5514). *(In MacBook Pro Terminal, type networksetup -listallhardwareports and press Enter to see your adapter name, then type e.g. sudo ifconfig en8 10.10.0.100 netmask 255.255.255.0 up and press Enter.)*

5. In Terminal Window #1 on your MacBook Pro, type this single command and press Enter to verify your setup, start both the Collector Receiver and Live Web Dashboard (http://127.0.0.1:8123), and open your browser automatically (or start the two servers manually below; ALLOW_EXAMPLE_KEYS=1 is for isolated bench testing only—before field deployment, generate unique keys with python3 pi5_sender_node/generate_keys.py --node NODE-NORTH-01 --write-node-key ./node_key.json --register-hub ./verified_alerts/authorized_nodes.json):

```
python3 pi-diode-trailcam/run_guided_bench_doctor.py --start-hub --open-browser
```

Or manually in Terminal Tab 1 on your MacBook Pro (Collector Receiver):

```
cd pi-diode-trailcam
```

```
ALLOW_EXAMPLE_KEYS=1 python3 collector_hub/receiver_hub.py --out-dir ./verified_alerts
```

And manually in Terminal Tab 2 on your MacBook Pro (Live Web Dashboard):

```
cd pi-diode-trailcam
python3 collector_hub/dashboard_server.py --alerts-dir ./verified_alerts --port 8123
```

6. Open a new tab in Terminal on your MacBook Pro (press ⌘ Cmd + T), type this exact command, and press Enter to trigger a live camera capture from the Raspberry Pi 5 (8GB) ([P01]) (BENCH_TEST_MODE=1 keeps the Pi 5 powered on after sending so your SSH session does not shut down):

```
ssh pi@trailcam-node.local "cd ~/pi_sender_node && sudo BENCH_TEST_MODE=1 RECEIVER_IP=10.10.0.100 python3 sender_node.py"
```

7. What you should see (✓ STAGE 2 SUCCESS CHECK):

* Open http://127.0.0.1:8123 in your web browser.

* Your live camera photo appears in the dashboard gallery with a green Ed25519 VERIFIED badge!

* Check verified_alerts/alerts_journal.jsonl and confirm "camera_ok": true, and verify that UART telemetry (TRIG,...) and the DONE signal (GPIO23 -> GP2) were recorded.

STAGE 3: Snap Together the 1-Way Optical Fiber Data Diode ([P13]–[P16], [P19])

Once your copper Ethernet bench test works, replace the direct Cat6 cable with the One-Way Optical Fiber Data Diode. Every connection is a tool-free CLICK snap-fit.

[CAUTION]

THREE CRITICAL REAL-WORLD RULES FOR StarTech MCMGBSCSM10 ([P13]) & SC/UPC FIBER (Photos 59–83):

1. 5V ONLY POWER RULE: Both black StarTech Media Converters operate on strictly +5.0V DC (center-positive). Power them using the included 5V wall adapters (Photo 61) or the StarTech USB-to-Type-M 5V barrel cables ([P19]) plugged into a 5V USB port. NEVER connect 12V battery power directly to the Media Converter barrel jack!

2. SLIDE THE FRONT LFP SWITCH DOWN TO OFF BEFORE PLUGGING IN 5V POWER (Photos 79–81):

* Each StarTech MCMGBSCSM10 ships from the factory with its tiny front LFP (Link Fault Pass-Through) slide switch in the ON (UP) position (Photo 79).

* Because Collector Box #2 has its TX port intentionally empty (air-gapped), leaving LFP = ON causes Box #2 to think the fiber link is broken and shut down its RJ45 Ethernet port (leaving only 1 green PWR LED lit!).

* Before plugging in 5V power (or followed immediately by unplugging/re-plugging the rear 5V barrel plug for 3 seconds, since the chip only reads the LFP switch at power-on boot), slide the LFP switch DOWN to OFF on both Box #1 and Box #2 (Photo 80 & Photo 81)!

3. PUSH EVERY BLUE SC/UPC PLUG PAST ITS INTERNAL SPRING UNTIL IT CLICKS (Photos 74–75):

* Every blue SC/UPC fiber plug has a sliding outer rectangular shell and a spring-loaded white ceramic tip inside. If you stop pushing when you first feel the spring, the plug sits 3–4 mm too shallow (Photo 74 bottom) and will not link. Push straight in until the internal metal clip clicks and locks (~1 mm gap inside the Blue Coupler [P15], or ~7 mm of blue plastic remaining outside the StarTech front face).

Step 3.1 — Unbox the Converters, Connect Cat6 Ethernet & Unzip One Single Yellow Fiber Strand (Photos 59–73)

1. Set Up Both Media Converters & Cat6 Ethernet Cables (Photos 59–64):

* Set Media Converter #1 ([P13] — Field / Pi 5 Side) next to the Raspberry Pi 5 ([P01]), and set Media Converter #2 ([P13] — Collector / MacBook Side) across the table toward your MacBook Pro.

* Slide the front LFP switch DOWN to OFF on both Box #1 and Box #2.

* Plug the Black Cat6 Ethernet Cable ([P23]) from the Raspberry Pi 5 Ethernet port into Media Converter #1 RJ45 (Photo 63).

* Plug the Blue Cat6 Ethernet Cable ([P23]) from Media Converter #2 RJ45 into your MacBook Pro's USB-C Gigabit Ethernet adapter ([P24]) (Photo 63 & Photo 64).

2. Prepare the 1×2 Optical Splitter ([P14]) & Blue SC/UPC Coupler ([P15]) (Photos 65–67):

* Unbag the Duttek 1×2 Single-Mode SC/UPC Optical Splitter ([P14]) (Photo 66). Notice its 3 color-coded cords:

* WHITE Cord (1× Input): Receives 100% of Field Box #1's outbound 1310 nm laser from Box #1 TX.

* BLUE Cord (Output Leg 1): Loops 50% of the laser right back into Box #1 RX (local carrier-detect loopback).

* ORANGE Cord (Output Leg 2): Sends the identical 50% laser copy out to the Blue SC/UPC Coupler ([P15]) (Photo 67).

3. Convert the Duplex Yellow Fiber ([P16]) Into a True Single-Strand 1-Way Fiber (Photos 68–73):

* Unbag the 2-meter Yellow Tripp Lite Duplex OS2 Single-Mode Fiber Cable ([P16]) (Photo 68). Because it ships as a 2-strand zip-cord, separate one single yellow strand so there is physically zero return fiber between the two boxes:

1. Slip a small screwdriver tip or tweezers ([T01]) under the center tab of the outer plastic duplex bridge clip at both ends of the yellow cable and pop the clips off (Photo 69 & Photo 70). *(Only remove the outer figure-8 bridge clip—leave the main blue rectangular connector shells on the plugs!)*

2. Locate the white wrap-around adhesive label near one end of the yellow cord (Photo 78) and peel or snip through the paper label between the two yellow jackets (Photo 71 — never cut the yellow jacket!).

3. Gently pull the two yellow fiber strands apart along the thin center zip-cord web from end to end (Photo 72), coil your single yellow strand in a loose loop ($\ge 60\text{ mm}$ diameter, wider than a soda can, Photo 73), and store the 2nd yellow strand as a spare.

Step 3.2 — Click the Fiber Diode Together in Order & Verify All 8 Green LEDs (4 on Box #1 + 4 on Box #2, Photos 74–83)

Pull the clear plastic dust caps off the blue square SC/UPC fiber plugs right before clicking each one in. Match the raised plastic key tab on each blue plug to the keyway slot (facing UP on both StarTech boxes, and aligned with the U-slot on the Blue Coupler [P15]):

1. [Click 1] Mate ORANGE Splitter Leg + YELLOW Fiber Inside the Blue SC/UPC Coupler ([P15], Photo 74):

* Push the ORANGE splitter output plug into one end of the Blue SC/UPC Coupler ([P15]) and one end of your single YELLOW OS2 fiber strand ([P16]) into the opposite end of the Blue Coupler (both raised key tabs lined up with the coupler's U-slot).

* Depth Check: Push both blue plugs firmly toward the center flange until both internal clips click and lock (~1 mm gap to the center flange on both sides).

2. [Click 2] Collector Inbound Fiber (Box #2 RX, Photo 81):

* Plug the far end of the single YELLOW OS2 fiber strand (key tab facing UP) firmly into the RX (Receive — right optical port) on Media Converter #2 (MacBook Side) until it clicks past the internal spring.

* Physical Return-Path Air Gap: Leave the TX (Transmit — left optical port) on Media Converter #2 100% EMPTY / UNPLUGGED (Photo 81)!

3. [Click 3] Field Local Loopback (Box #1 RX, Photo 75):

* Plug the BLUE splitter output leg (key tab facing UP) firmly into the RX (Receive — right optical port) on Media Converter #1 (Pi 5 Side) until it clicks.

4. [Click 4] Power Up Both Boxes & Plug WHITE Splitter Input Into Box #1 TX LAST (Photos 75 & 80–83):

* Plug the 5V DC barrel power cables into both Box #1 and Box #2 (confirming both LFP switches are slid DOWN to OFF; if you slid LFP to OFF while powered, unplug and re-plug the 5V barrel plug for 3 seconds so the chip boots with LFP = OFF, lighting up 3 green LEDs on Box #2 as shown in Photo 80).

* Now push the WHITE splitter input plug (key tab facing UP) firmly into the TX (Transmit — left optical port) on Media Converter #1 (Photo 75)!

* *(Why plug WHITE into Box #1 TX last? In 1000BASE-LX Gigabit fiber, when Box #1 first detects light returning on its RX port, it transmits a short Clause 37 optical auto-negotiation burst (/C1/ -> /C2/ -> IDLE). Plugging WHITE into Box #1 TX after Box #2 RX is already connected—or unplugging and re-plugging WHITE for 2 seconds—forces Box #1 to broadcast a fresh optical handshake simultaneously to both Box #1 RX and Box #2 RX, locking all 4 green LEDs solid ON on both boxes at the exact same instant!)*

5. What you should see (✓ STAGE 3 SUCCESS CHECK — 8/8 GREEN LEDS SOLID ON, Photos 81–83):

* Field Box #1 (Pi 5 Side): All 4 green LEDs (Fiber LNK/ACT, TP LNK/ACT, PWR, 1000) are solid GREEN (Photo 75 & Photo 82).

* Collector Box #2 (MacBook Side): All 4 green LEDs (Fiber LNK/ACT, TP LNK/ACT, PWR, 1000) are solid GREEN (Photo 81 & Photo 83), with TX completely empty!

* Run the manual alert command from Step 2.6 (sudo BENCH_TEST_MODE=1 RECEIVER_IP=10.10.0.100 python3 sender_node.py): a verified JPEG alert arrives on http://127.0.0.1:8123 across a 100% physically unidirectional strand of glass!

* Documented Real-World Stage 3 1-Way Optical Fiber Data Diode Prototype Build Photos (25 Chronological Shots, Photos 59–83):

59. Photo 59 (stage3_01_four_fiber_and_power_packages_staged.jpg) — Step 3.0A: Staging the 4 optical and power packages (StarTech USB-to-Type-M 5V barrel cables [P19], Duttek 1×2 SC/UPC splitter [P14], Tripp Lite 2m OS2 yellow fiber [P16], and Cable Matters Blue SC/UPC couplers [P15]).

60. Photo 60 (stage3_02_two_startech_media_converter_boxes.jpg) — Step 3.0B: Staging the two identical StarTech MCMGBSCSM10 Gigabit Ethernet to Single-Mode SC Fiber Media Converter boxes ([P13], 1310 nm, 10 km).

61. Photo 61 (stage3_03_unboxed_startech_converters_with_5v_power_bricks.jpg) — Step 3.0C: Unboxed StarTech MCMGBSCSM10 metal chassis (Box #1 and Box #2) with their included +5.0V DC center-positive wall adapters.

62. Photo 62 (stage3_04_startech_front_ports_and_5v_dc_jack_closeup.jpg) — Step 3.0D: Close-up inspection of the front SC optical ports (with black rubber dust cap installed), 4 status LEDs, LFP slide switch, RJ45 jack, and rear +5V DC barrel jack.

63. Photo 63 (stage3_05_both_startech_boxes_ethernet_cables_connected.jpg) — Step 3.1A: Connecting the Black Cat6 Ethernet cable ([P23]) between Raspberry Pi 5 [P01] and Field Box #1, and the Blue Cat6 Ethernet cable ([P23]) to Collector Box #2.

64. Photo 64 (stage3_06_blue_ethernet_in_macbook_usbc_adapter.jpg) — Step 3.1B: Plugging the Blue Cat6 Ethernet cable from Collector Box #2 into the USB-C Gigabit Ethernet adapter ([P24]) on the MacBook Pro ([T02]).

65. Photo 65 (stage3_07_duttek_blue_and_orange_plugs_caps_removed.jpg) — Step 3.1C: Pulling the clear plastic dust caps off the Blue (Output Leg 1) and Orange (Output Leg 2) SC/UPC plugs of the Duttek 1×2 splitter ([P14]).

66. Photo 66 (stage3_08_duttek_splitter_all_three_cords_white_blue_orange.jpg) — Step 3.1D: All 3 color-coded SC/UPC cords of the 1×2 Optical Splitter ([P14]) uncapped and identified (White = 1× Input from Box #1 TX; Blue = Output Leg 1 to Box #1 RX; Orange = Output Leg 2 to Blue Coupler [P15]).

67. Photo 67 (stage3_09_blue_sc_coupler_black_dust_caps_removed.jpg) — Step 3.1E: Removing both black protective dust caps from the Blue Cable Matters SC/UPC female-to-female simplex coupler ([P15]).

68. Photo 68 (stage3_10_yellow_duplex_fiber_cable_unbagged.jpg) — Step 3.1F: Unbagging the 2-meter Tripp Lite Duplex Yellow OS2 Single-Mode SC/SC patch cable ([P16]).

69. Photo 69 (stage3_11_prying_open_duplex_clip_on_yellow_fiber.jpg) — Step 3.1G: Gently popping off the outer plastic duplex joining clip holding the two blue SC/UPC plugs together.

70. Photo 70 (stage3_12_duplex_clip_removed_two_yellow_fiber_plugs_separated.jpg) — Step 3.1H: Outer plastic duplex clip removed, separating the two individual blue SC/UPC connector plugs cleanly.

71. Photo 71 (stage3_13b_peeling_white_wrap_label_to_separate_yellow_strands.jpg) — Step 3.1I: Peeling off the white adhesive wrap-around cable label with tweezers ([T01]) so the two yellow fiber strands can unzip cleanly.

72. Photo 72 (stage3_14_unzipping_duplex_yellow_fiber_into_single_strand.jpg) — Step 3.1J: Pulling the two yellow OS2 fiber strands apart along the center zip-cord web to isolate a single unidirectional fiber strand.

73. Photo 73 (stage3_15_single_yellow_fiber_strand_ready_caps_removed.jpg) — Step 3.1K: Single 2-meter yellow OS2 fiber strand coiled loosely ($\ge 60\text{ mm}$ bend diameter) with clear dust caps removed from both blue SC/UPC ends.

74. Photo 74 (stage3_16_blue_coupler_joining_orange_splitter_and_yellow_fiber.jpg) — Step 3.1L: Mating the Orange splitter leg (top) and single Yellow fiber strand (bottom) inside the Blue SC/UPC Coupler [P15] (push both plugs all the way in until both internal clips click!).

75. Photo 75 (stage3_17_pi_startech_box1_white_tx_blue_rx_all_4_green_leds.jpg) — Step 3.1M: Close-up of Field Box #1 (Pi 5 side) with White in TX, Blue loopback in RX, and Black Cat6 in RJ45 — all 4 green LEDs (Fiber LNK/ACT, TP LNK/ACT, PWR, 1000) glowing solid green!

76. Photo 76 (stage3_18_field_side_topdown_pi5_breadboard_box1_splitter_coupler.jpg) — Step 3.1N: Top-down view of the Field Side (Breadboard + Pico 2 + GY-271 + Pi 5 + Camera Module 3 + Box #1 + 1×2 Splitter + Blue Coupler).

77. Photo 77 (stage3_19_field_side_angled_box1_4_green_leds_and_splitter.jpg) — Step 3.1O: Angled view of the Field Side showing all 4 green LEDs glowing on Box #1 alongside the 1×2 Optical Splitter ([P14]).

78. Photo 78 (stage3_13a_locating_white_wrap_label_on_yellow_duplex_cord.jpg) — Step 3.2A: Inspecting the yellow fiber jacket and key-tab orientation before seating the far end into Collector Box #2 RX.

79. Photo 79 (stage3_20_macbook_box2_lfp_on_only_pwr_led_before_switch_off.jpg) — Step 3.2B (Diagnostic Landmark — LFP = ON): Close-up of Collector Box #2 showing why leaving the factory LFP switch in ON (UP) holds the converter in Link Fault Pass-Through lockout (only 1 green PWR LED lit, disabling RJ45).

80. Photo 80 (stage3_21_lfp_switched_off_power_cycled_3_green_leds_ethernet_up.jpg) — Step 3.2C (LFP = OFF + Power-Cycle Reset): After sliding LFP DOWN to OFF and unplugging/re-plugging the 5V DC barrel plug for 3 seconds, TP LNK/ACT and 1000 light up green (3 green LEDs on Box #2).

81. Photo 81 (stage3_22_macbook_startech_box2_all_4_green_leds_rx_verified_closeup.jpg) — Step 3.2D (ALL 4 GREEN LEDS LIT ON BOX #2): Macro close-up of Collector Box #2 after re-plugging White (Box #1 TX) for 2 seconds to sync the 1000BASE-X optical handshake — all 4 green LEDs (Fiber LNK/ACT, TP LNK/ACT, PWR, 1000) solid green, Yellow fiber in RX, and TX 100% empty air-gap!

82. Photo 82 (stage3_23_full_benchtop_angled_both_boxes_all_8_green_leds.jpg) — Step 3.2E: Full angled benchtop verification showing all 4 green LEDs on Field Box #1 (left) and all 4 green LEDs on Collector Box #2 (right) glowing simultaneously (8/8 green LEDs!).

83. Photo 83 (stage3_24_full_benchtop_clean_separated_bundles_golden_overview.jpg) — Step 3.2F (Golden-Path Stage 3 Overview): Cleanly separated top-down overview of the complete Stage 3 system (Collector Box #2 at top, Field Box #1 + Pi 5 + Camera + Breadboard + Pico 2 + GY-271 at bottom-left, and 1×2 Splitter + Blue Coupler + Yellow OS2 Fiber Loop at right).

Step 5: Unpowered Enclosure Dry Fit & Next Steps

Before drilling or wiring the outdoor battery, place the actual 12V 12Ah battery ([P17]) and mounting components inside the QILIPSU enclosure ([P21]) without connecting the battery terminals. Verify board clearances, fiber loop radii (>= 30 mm, wider than a soda can), lid closure, insulation, and service access.

Step 6: Post-Stage 2 Engineering Assessment, Zero-Redesign Vibration Locking & Quantitative Field Risk Matrix

6.1 Why We Keep the Exact COTS Breadboard ([P09]) & 2.54mm Wiring (Zero Custom PCBs, Zero Second Design)

1. Supply-Chain Anonymity & OPSEC: Ordering a custom PCB from a fabrication house (JLCPCB, PCBWay, OSH Park) creates a permanent, uniquely identifiable digital trail (Gerber layout files + account + payment + shipping record) and produces a custom-etched circuit board that forensically links any recovered field unit back to a single batch order. By contrast, ELEGOO 30-row solderless breadboards ([P09]) and standard 2.54 mm jumper ribbons ([P10]) are sold in millions of generic hobbyist kits worldwide—giving the hardware 100% COTS plausible deniability and zero custom fabrication footprint.

2. Cost & Modification Agility: In low-to-medium quantities (1–20 units), custom PCBs do not reduce total cost once connectors, assembly, and shipping are counted (~$15–$35 vs. $1.50 for a COTS breadboard), yet they freeze your pinout in fiberglass. If you ever want to swap a sensor or reassign a GPIO pin in the field, a custom PCB requires cutting traces or waiting weeks for a re-spin, whereas the breadboard lets you move a wire in 5 seconds.

3. Zero Second Build: Switching from the Stage 2 breadboard to a different soldered breakout board for field deployment would force you to build and re-test a second hardware design from scratch. Instead, we keep the exact breadboard assembly you already built and verified in Stage 2 (100% identical hole and pin layout) and mechanically lock its connections in place in 2 minutes using materials already in your kit ([T11]).

6.2 How to Make Your Exact Current Breadboard Build 100% Car & Snowmobile Vibration-Proof (Zero Re-Wiring)

Why do standard single-pin (1x1) Dupont jumpers loosen during vehicle or snowmobile transport?

* 90% of pull-outs happen when the Breadboard and the Raspberry Pi 5 slide or bounce *relative to each other*, pulling on the W5–W7 bridge wires.

* 10% of pull-outs happen when high-frequency engine/track vibration (20–200 Hz) causes isolated 1x1 plastic housings to rock side-to-side and slowly walk upward.

Apply this 3-Step In-Place Mechanical Lock directly to your finished Stage 2 build without unplugging a single wire:

1. Rigid Co-Mounting on the Same Backplate (QILIPSU [P21] Grid Plate): Peel the backing off the 3M double-sided foam tape on the underside of your 30-Row Breadboard [P09] and press it firmly onto the rigid white ABS mounting grid plate right next to the Raspberry Pi 5 [P01] (3 cm apart), then pass a nylon zip-tie around the breadboard waist through the grid holes as a mechanical backup. Because both boards are locked to the same rigid plate, there is zero relative movement between the Breadboard and Pi 5 over rough terrain.

2. Bind Adjacent 1x1 Jumper Shells into Multi-Pin Blocks (3M Tape [T11] or 1x4 Housing): Wrap one tight turn of 3M tape ([T11]) around the 4 adjacent black plastic jumper housings in Column f (Rows 24–27: Red, Black, Yellow, Blue), around the Pico 2 left-column housings (Col a), and around the 3 Pi 5 GPIO sockets (Pins 10, 14, 16). When adjacent 1x1 housings are bound into a unified block, individual pins cannot tilt or pivot, increasing pull-out resistance by 4x. *(Optional COTS upgrade for future units: lift the tiny plastic retention tab on each 1x1 shell with tweezers, slide the metal crimp terminal out, and click all 4 sensor wires into a single standard 1x4 2.54mm Dupont plastic block housing so it snaps in as a single 4-pin plug!)*

3. Peelable Elastomeric Damper Bead (Permatex Black Silicone [T11] — Already in Your Kit): Squeeze a small 2 mm bead of Permatex RTV Black Silicone [T11] along the seam where the black plastic jumper housings meet the face of the breadboard, where the blue GY-271 PCB meets its 5-pin header, and where the 3 female sockets meet the Pi 5 GPIO header. It cures into a flexible, sub-zero-rated (-54°C to +204°C) rubber vibration damper that survives 10G+ snowmobile shocks—yet peels off cleanly in one piece with tweezers if you ever want to service or modify a wire!

6.3 Comprehensive Quantitative System & Technical Risk Assessment (Based on Stage 2 Physical Build Findings)

The table below quantifies every remaining technical risk across the physical mounts, electrical contacts, optical viewport, camera focus pipeline, and cold-weather power budget—showing the Unmitigated Failure Probability, Mitigated Failure Probability, Severity (1–5), Exact Physical Location, and Concrete COTS Mitigation:

| Risk Category & Exact Physical Location / Mount | Lesson Learned Today / Failure Mechanism | Unmitigated Risk (%) | Mitigated Risk (%) | Severity (1–5) | Concrete Zero-Redesign COTS Mitigation

R1 | Breadboard & GPIO Jumper Walk-Out (Col f f24–f27, Col a, Pi 5 Pins 10/14/16) | Individual 1x1 Dupont pins can pivot and back out under 20–200 Hz snowmobile/car vibration or relative board motion. | 65% | < 2% | 5 (Critical) | Co-mount Breadboard [P09] (3M backing + zip-tie) & Pi 5 [P01] on the same QILIPSU grid plate + bind adjacent 1x1 shells with 3M tape + apply a peelable 2 mm Permatex Silicone [T11] bead across shell bases.

R2 | Internal "J-Hook" Bent Jumper Pin Tip (Breadboard Spring Clips, e.g. Blue W3 f27) | Pushing a soft square pin into a stiff new breadboard hole at a slight angle bends the tip into a hook (stage2_38), causing an intermittent open circuit or sideways short to the next row. | 30% | < 1% | 4 (High) | Inspect and flatten any curved pin at 90° with smooth flat-nose pliers (stage2_39) before insertion; verify 0x0d I2C response before applying silicone damper bead.

R3 | Unsoldered GY-271 Compass Through-Holes (Col j Rows 24–28, Pins 1–4) | Loose through-holes (1.0 mm hole vs. 0.64 mm pin) on an unsoldered header lose SDA/SCL contact ([] on i2c.scan()), and bending brittle brass header pins with pliers snaps them. | 80% | < 1% | 5 (Critical) | Solder the 4 active pins (VCC, GND, SCL, SDA) — or if no soldering iron is available, use an aluminum-foil conductive sleeve, a wooden toothpick wedge, or pin the 4 jumper wires directly through the PCB holes into Col j (NEVER bend brittle brass header pins with pliers!).

R4 | CSI Ribbon Micro-Creep in Vertical J4 (CAM/DISP 0) & Camera Module 3 ZIF Latches | 200mm orange ribbon ([P06]) has stiff spring tension; pulling on the ribbon during Zone 1 pod mounting can tug the 1.5 mm brown friction wedge loose. | 25% | < 1% | 5 (Critical) | Seat ribbon 2 mm deep (zero gold teeth visible), lock brown latch flush, and add a 3M tape strain-relief loop anchoring the orange ribbon to the enclosure wall 3 cm behind each connector so zero tension reaches the latch.

R5 | Pico 2 Micro-USB Port (J1) Detent & Cable Leverage (Breadboard Row 1) | The Pico 2 Micro-USB socket has tight metal locking barbs; a stiff USB cord can lever Row 1 of the Pico 2 upward out of the breadboard. | 20% | < 1% | 4 (High) | Push Micro-USB plug 100% flush past the 2 retention barbs while holding Pico 2 down with two fingers; in Stage 5 field deployment, power flows via breadboard rail + zip-tied cable strain relief.

R6 | Camera Autofocus Hunting on Foreground Pine Needles / Snowflakes (Camera Module 3 NoIR Wide [P05]) | By default, IMX708 motorized PDAF focuses on the closest high-contrast twig or falling snowflake (0.2 m), blurring a person or vehicle at 5–15 m on the trail and dropping YOLO recall. | 50% | < 2% | 4 (High) | Lock the VCM lens in sender_node.py to Fixed Hyperfocal Mode (AfMode = 0, LensPosition = 0.5 diopters = 2.0 m focus center), keeping everything from 1.0 m to ∞ razor-sharp with 0 ms focus delay.

R7 | Motion Blur on Walking/Running Targets at Dusk (IMX708 Auto-Exposure Slow Shutter) | In low light under forest canopy, default auto-exposure slows shutter speed to 1/15 s (66 ms), smearing a walking person (1.5 m/s) across 10 cm of pixels. | 45% | < 3% | 3 (Medium) | Cap maximum exposure time in picamera2 controls to ExposureTime <= 8000 (1/125 s day/dusk) or 12000 (1/80 s night IR) and let analog gain (AnalogueGain = 1.0–8.0) compensate so every frame is crisp.

R8 | Internal 940nm IR LED Glare Reflecting Off the 30mm Glass Viewport (Zone 1 Pod [P20]) | If the 2 side 940nm IR LEDs shine onto the inside surface of the 30mm glass disc ([T11]), internal reflection washes out the central IMX708 lens with a white fog halo at night. | 60% | < 2% | 4 (High) | Press the included black neoprene foam baffle ring ([P20]) tightly around the central camera lens barrel so it butts flush against the glass disc, optically isolating the lens from the side IR LEDs.

R9 | SC/UPC Fiber Ferrule Dust Contamination or Micro-Bend Loss (Zone 3 Optical Diode [P13]–[P16]) | A single skin-oil smudge on the 9 µm single-mode glass core or bending the yellow fiber tighter than a 30 mm radius attenuates the 1310 nm laser link. | 20% | < 1% | 4 (High) | Keep white dust caps on until the exact second of clicking ([Click 1–3]), never touch the white ceramic tip, and coil excess fiber in wide loops ($\ge 60\text{ mm}$ diameter, wider than a soda can) inside QILIPSU [P21].

R10 | Sub-Zero Battery Voltage Sag During Pi 5 Cold Boot (-25°C Boreal Winter) | Surface air at -30°C reduces 12V lead-acid/AGM capacity by ~40% and raises internal resistance during the Pi 5 2.5A boot surge. | 35% | < 2% | 4 (High) | Bury Zone 3 Vault [P21] 20–30 cm underground beneath forest moss/snow with 10–20 mm closed-cell foam insulation (holding soil/vault temperature at -2°C to -8°C year-round) + 5V buck converter headroom.

6.4 Real-Time Raspberry Pi 5 YOLO Object Detection, Aspect-Ratio Letterboxing & "No-Vanish" Persistence Tracking (Photos 84–86)

To support real-time human, cell-phone, and camouflaged/fallen-target detection directly on the Raspberry Pi 5 (4× Cortex-A76 @ 2.40 GHz) without requiring an external GPU, sender_node.py and collector_hub/ui/index.html implement a 3-stage detection and persistence pipeline:

1. Physical Raspberry Pi 5 CPU ONNX Runtime Benchmarks (4× Cortex-A76 @ 2.40 GHz):

* yolov8n.onnx (640×640): 167.4 ms/frame (5.97 FPS) — high-detail single-frame burst verification.

* yolov8n_416.onnx (416×416): 66.0 ms/frame (15.16 FPS) — balanced real-time tracking.

* yolov8n_320.onnx (320×320): 39.1 ms/frame (25.58 FPS) — full real-time 25+ FPS streaming (1.18 s for 30 frames).

* yolov8n_256.onnx (256×256): 23.4 ms/frame (42.68 FPS) — ultra-fast 30+ FPS real-time gatekeeping (0.70 s for 30 frames).

2. Aspect-Ratio Letterboxing (letterbox_image — 16.3× Confidence Gain on Horizontal Targets):

* Naive cv2.resize(frame, (640, 640)) squishes 16:9 (1920×1080) or 9:16 (480×848) frames by 1.77×, crushing horizontal or crouched targets down to 0.0038 (0.38%). Preserving 1:1 aspect ratio with (114, 114, 114) letterbox padding increases fallen/floor human confidence by 16.3×.

3. Target-Count Conservation ("No-Vanish" Persistence Zone) + 90° Fallen-Pose Probing (86× Confidence Gain):

* NoVanishPersistenceTracker tracks peak human count inside the camera's field of view and monitors frame borders (border_margin = 0.06). If the standing count drops without a border exit (because a subject crouched, went prone, or blended into floor/forest clutter), the tracker locks a NO-VANISH ZONE around the last interior coordinates and runs probe_rotated_90_roi (cv2.ROTATE_90_COUNTERCLOCKWISE), aligning horizontal fallen subjects with YOLOv8's vertical human prior and boosting confidence 86× (0.0038 -> 0.3273).

4. Documented Boreal Trail Live-Stream Detection Captures (Photos 84–86):

* 84. Photo 84 (stage4_01_boreal_trail_yolo_standing_lock.png) — Frame 100 (4.0s): Real-time multi-person boreal forest trail tracking (PERSON 0.87, PERSON 0.79) with active NO-VANISH ZONE (2/2 CONSERVED IN FOV) conservation overlay.

* 85. Photo 85 (stage4_02_boreal_trail_yolo_crouch_transition.png) — Frame 125 (5.0s): Mid-transition tracking (PERSON 0.86, PERSON 0.54 [LETTERBOX 16.3x]) maintaining NO-VANISH ZONE (2/2 CONSERVED IN FOV) lock as Subject #2 crouches behind snow-covered fern clutter.

* 86. Photo 86 (stage4_03_boreal_trail_yolo_no_vanish_90deg_probe.png) — Frame 150 (6.0s): Continuous conservation tracking (PERSON 0.85, PERSON (90° CCW ROI) 0.72 [86x GAIN]) via probe_rotated_90_roi as Subject #2 goes horizontal/prone on the snowy forest floor.

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* Proceed to Part 2: IKEA-Style Enclosure Build, Camouflage & Tree Installation Manual (Stages 4–6) for step-by-step instructions to build the weatherproof pods, wire the 12V battery safely, camouflage the tree unit, and run the complete system in the field.

* Follow the Field Testing & Validation Plan for full off-grid acceptance checks.

* The 200 m – 20 km All-Weather Troposcatter & Canopy-Bounce Relay (sky-diode-relay Build D) is assembled separately in ~35–45 minutes (zero soldering) and is not required for Stages 1–3 indoor camera/fiber bring-up.

Arctic Trail Cam System — IKEA-Style Manual Part 2: 3-Zone Enclosure Build, Camouflage, Tree Installation & Field Operation (Stages 4–6)

Zero Electronics Experience Required: Before starting this manual, complete Part 1: Indoor Benchtop Assembly & Bring-Up (Stages 1–3) on your kitchen table so you know your camera, Pico 2, Raspberry Pi 5, and optical fiber links work.

* Single electrical pin reference: Wiring & Pinout Schedule

* Parts list & planning BOM: Hardware Configuration & BOM

* Printable PDF booklet (including the True 1:1 Physical-Scale CAD Vector Drilling Template, 1.0 mm = 2.8346 pt): ARCTIC_TRAIL_CAM_BUILD_AND_FIELD_GUIDE.pdf

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Tools ([T04]–[T12]) & 5-Hole Enclosure Drilling Schedule Reference

The complete workshop tool schedule ([T01]–[T12]) and the Exact 5-Hole Enclosure Drilling Table (Holes 1–5) are maintained in HARDWARE_SPECS_AND_BOM.md.

* Plastic Hole-Saw Rule ([T04] + [T05]): Always drill the central 1/4" pilot hole in FORWARD, then flip your cordless drill to REVERSE before the outer circular hole-saw cup touches the ABS/polycarbonate box so the teeth score and shear cleanly without cracking the plastic.

Visual Roadmap: How Your Working Desk Pile Splits Into the 3 Final Units (Stages 3B, 3C, 4 & 5)

Before you drill a single hole in any plastic enclosure, look at the 2-panel real-photo roadmap below. It shows 1) where every working part currently sitting on your desk (Stage 3) goes, 2) which unopened bags to grab next from your Stage 4–5 parts kit, and 3) the two quick bench tests (Step A: 2 m Camera Umbilical + Radar and Step B: 12V -> 5V Power Train) to run on the table before packing the weatherproof boxes.

Physical Unit | What Moves From Your Current Desk Pile (Left Panel) | What Unopened Bags You Add From Your Stage 4–5 Parts Kit (Right Panel)

UNIT #1: Zone 1 Tree Camera & Radar Micro-Pod *(Mounts at 1.8–2.2 m height on pine trunk)* | • Camera Module 3 NoIR Wide [P05] *(unclips from the short 200 mm ribbon on your desk and moves into the tree pod)* | • Arducam CSI-to-HDMI Extension Boards + 2 m Black HDMI Cable [P06] ; • HLK-LD2450 24 GHz Radar + 4-Pin Wire ([P11]) ; • Hammond 1591BSBK Black ABS Box + 30 mm Glass Disc + Black Neoprene Foam Ring + 39 Ω 5W De-Icer ([P20]) ; • Permatex 81158 Black Silicone ([T11]) + 1× Silica Gel Packet

UNIT #2: Zone 3 Underground Base Compute, Battery & Diode Vault *(Buried 20–30 cm in root hummock at base of tree)* | • Raspberry Pi 5 [P01] + Active Cooler [P02] ; • 30-Row Breadboard [P09] (Pico 2 [P07] + 7 Wires [P10] + GY-271 Compass [P08] — stays intact!) ; • Field Media Converter #1 [P13] (LFP = OFF) + Short Black Cat6 ; • 1×2 Optical Splitter [P14] (White -> TX, Blue -> RX, Orange -> Coupler) ; • Blue SC/UPC Coupler [P15] | • QILIPSU IP67 Grey Hinged Enclosure ([P21]) + Closed-Cell Foam Liner ; • 12V 12Ah Battery ([P17]) + Littelfuse 5A Inline Blade Fuse + 6.3 mm Spade Leads ([P18]) ; • MPM3610 5V Buck + 12V->5V 5A Buck + STEMMA Relays ([P12]) ; • StarTech USB-to-Type-M 5V Barrel Cable ([P19]) *(powers Box #1 from the 5V buck instead of the wall outlet!)* ; • 30 m Black Armored Outdoor Fiber ([P16]) *(replaces the 1 m Yellow indoor bench test cord!)*

UNIT #3: Cabin / Collector Receiver Hub *(Stays indoors at your Cabin / Collector)* | • Collector Media Converter #2 [P13] (RX only, TX empty physical air-gap, LFP = OFF) ; • Blue Cat6 Ethernet ([P22]) + USB-C Gigabit Ethernet Adapter ([P24]) -> MacBook Pro [T02] | *Already 100% assembled! In Stage 5/6, simply unplug the 1 m Yellow indoor bench test cord from Box #2 RX and plug in the incoming 30 m Black Armored Outdoor Fiber ([P16]).*

STAGE 4: Build & Camouflage the Zone 1 Tree Micro-Pod (Hammond 1591BSBK [P20])

The Zone 1 Micro-Pod is the tiny, fist-sized black ABS box (112 × 62 × 31 mm, [P20]) that mounts at eye level (1.8 m–2.2 m) inside a vertical pine bark crevice. Look at the reference illustration below (and print the True 1:1 CAD Vector Drilling Template from ARCTIC_TRAIL_CAM_BUILD_AND_FIELD_GUIDE.pdf at 100% / Actual Size) before drilling or gluing.

Parts & Tools Used in Stage 4

* Parts: [P05] (Raspberry Pi Camera Module 3 NoIR Wide or Arducam IMX708 NoIR Camera), [P06] (200 mm 22-to-15-pin Ribbon $9.99 + optional Arducam CSI-to-HDMI Extension Kit w/ 2.0 m Shielded Cable ASIN B06XDNBM63 $18.99 for 2 m split-pod umbilical), [P11] (HLK-LD2450 24 GHz Radar PCB + Harness — $10.94), [P20] (Hammond 1591BSBK Black ABS Box $7.45 [factory IP54 lap-joint; requires Permatex 81158 perimeter lid-channel silicone gasket] or drop-in Hammond 1554EGY factory IP68 gasketed box $13.50 + 30 mm Glass Disc $6.49 + Black Neoprene Foam Ring $7.99 + 39-Ohm 5W Ceramic De-Icer Resistor (3.69 W @ 12.0V) $1.99 + M12 IP68 bottom cable gland $1.99 = $25.91), [P22] (Rust-Oleum Ultra-Flat Earth Brown $8.98 & Deep Forest Green $8.98 spray paint + Gear Aid Camo Form $14.95 = $32.91).

* Tools: [T04] (Cordless drill with Forward / Reverse), [T05] (1-1/8" / 28 mm Bi-Metal Hole Saw Cup + Center Pilot Drill Bit), [T06] (Center awl/nail + 4–22 mm Titanium Cone Step Drill Bit for 18–20 mm viewport hole & 12 mm bottom hole), [T07] (#1 Phillips screwdriver & pliers), [T10] (Scissors/craft knife, ruler, file & 400-grit sandpaper), [T11] (Black waterproof silicone, rubbing alcohol, 3M blue painter's tape & sea sponge), [T12] (Safety glasses & gloves).

Step 4.1 — Drill the Upper Viewport Window (20 mm Cone Step Bit [T06] or 28 mm Hole Saw [T05]) & 12 mm Bottom Cable Hole

1. Put on your Safety Glasses ([T12]), unscrew the 4 corner Phillips screws ([T07]) on the Hammond 1591BSBK box ([P20]), and place the empty lid flat on a scrap piece of wood on your worktable.

2. Mark & Dimple the Single Upper Window Crosshair (Hole 1):

  * Measure 32 mm down from the top edge of the box lid, dead-center between the left and right edges (31 mm from either side edge), or tape the 1:1 paper template (Panel A) directly onto the lid.

  * Press your Center Awl or a Sharp Nail ([T06]) firmly into the center + crosshair to make a 1 mm starter dimple so the drill bit cannot wander on the smooth plastic.

3. Cut the Circular Viewport Window (Two Foolproof Methods — < 2% Risk of Cracking):

  * Method A — Recommended for Raspberry Pi Camera Module 3 NoIR Wide [P05] (20 mm Titanium Stepped Cone Bit [T06] — Zero Snagging Teeth!):

    * Because the Camera Module 3 NoIR Wide (25 × 24 mm PCB) has a compact 11.5 × 11.5 mm square lens barrel (16 mm diagonal) and a 6 mm optical aperture, you do not need a 28 mm hole!

    * Chuck your Titanium Stepped Cone Drill Bit ([T06]) into your drill ([T04]), place the 4 mm tip into your center dimple, and drill smoothly down to the 18 mm or 20 mm step. A stepped cone bit has zero snagging saw teeth—it shaves a perfectly round, burr-free circle in 15 seconds with zero risk of catching or cracking the 2 mm ABS lid, while leaving a wide 5 mm solid plastic ledge all the way around (360°) for the 30 mm Glass Disc ([T11]) to seal against!

  * Method B — For Wider Arducam IMX708 w/ Side IR LEDs (28 mm / 1-1/8" Hole Saw Cup [T05] in REVERSE):

    * Step A (Pilot Hole in FORWARD): Set the drill to FORWARD, place the central 1/4" pilot drill bit into your center dimple, and drill until the center bit pokes through—STOP before the outer circular saw teeth touch the lid!

* Step B (Outer Circle in REVERSE — Zero Cracks!): Flip your drill's direction switch to REVERSE (counter-clockwise). Spin the hole saw in REVERSE at medium speed with gentle downward pressure so the backwards-spinning teeth score and shear a smooth 28 mm circle without grabbing.

* Why Hole Edge Precision Does NOT Affect Waterproofing: The drilled hole edge never seals the water out—the flat inside surface of the lid around the hole forms the seal! Because the 30 mm glass disc ([T11]) overlaps the 20 mm hole by 5 mm on every side, even a slightly off-center hole seals 100% watertight (IP67).

* DO NOT drill a hole for the radar in the lower half! The HLK-LD2450 24 GHz radar ([P11]) sees straight through the solid 2 mm ABS plastic wall with < 0.4 dB loss.

4. Drill the 12 mm (1/2") Bottom Umbilical Hole (Hole 2):

* On the bottom narrow end-wall of the box (62 × 31 mm bottom face), press a center dimple ([T06]) dead-center (31 mm from left/right, 15.5 mm from front/back).

* Using the Titanium Cone Step Drill Bit ([T06]), drill straight down to the 12 mm (1/2") step and thread in the M12 / PG7 IP68 waterproof cable gland (or slot the 12 mm opening into a U-channel if passing a pre-terminated 2 m HDMI umbilical through before silicone potting).

Step 4.2 — Smudge-Free Glass Window Gluing (3M Tape Mask Trick), Foam Baffle & Straight Camera Mounting

1. Apply the 3M Blue Painter's Tape [T11] Center-Mask & Handle Before Touching Silicone (Zero Smudge Risk):

* Wipe both faces of the 30 mm Mineral Glass Disc ([P20]) and the inside of the ABS lid with rubbing alcohol ([T11]) and let dry for 1 minute.

* Cut two 20 mm coin-sized circles of 3M Blue Painter's Tape [T11] and stick one over the center of the front face and one over the center of the back face of the 30 mm glass disc (leaving the outer 5 mm glass rim bare for silicone bonding), with a folded tab of blue tape sticking up in the center like a handle.

2. Bond the 30 mm Glass Disc & Let Cure Overnight (12–24 Hours Recommended):

* Squeeze a continuous 3 mm donut bead of Permatex 81158 Black Silicone ([T11]) onto the flat inside plastic rim around the viewport hole.

* Holding the 30 mm glass disc by its blue tape handle tab, press it firmly into the wet silicone ring until a continuous black rubber gasket forms all the way around (360°). Let it cure (30 min skin-over, 12–24 hrs full waterproof cure), then peel off the inner blue tape circle to reveal 100% factory-clean, smudge-free glass! *(Leave the outer blue tape circle on until after Step 4.4 camo painting!)*

3. Install the Anti-Glare Neoprene Foam Baffle Ring & Mount the Camera Dead-Straight (✓ CRITICAL):

* Press the black closed-cell neoprene foam donut ring (22 mm OD, 12 mm ID, 5–6 mm thick, [P20]) snugly around the 11.5 × 11.5 mm square central lens barrel on the front of your camera board ([P05]). This creates a light-tight optical tunnel that eliminates internal glass glare and reflections.

* Front-Look Alignment & Ribbon Strain-Relief Trick: Temporarily peel back a peek of the outer tape (or look through the glass from the front) while holding the camera board against the inside of the glass with the 200 mm orange ribbon ([P06]) exiting straight down. Looking from the outside front lets you center the lens in the window and align the square PCB edges parallel to the rectangular box walls in 5 seconds.

* Anchor the Ribbon First: Tape the orange camera ribbon flat against the inside back wall 3 cm below the camera with a strip of 3M tape ([T11]) so the springy ribbon cannot twist or tilt the camera board, then secure the 4 corners of the camera PCB with 3M double-sided foam tape + 4 corner dabs of Permatex Black Silicone [T11] so the black foam ring stays gently compressed (1 mm) against the inside of the glass.

4. Mount the 39 Ω, 5W Window De-Icer Resistor (3.69 W @ 12.0V, [P20]):

* Bond the 39 Ω, 5W ceramic de-icer resistor ([P20]) against the lower inside edge of the glass window (outside the foam ring) with thermal tape. Driven from the switched +12.0V rail via STEMMA Relay #2 (GP16), it delivers $P = 12.0^2 / 39\,\Omega = 3.69\text{ W}$ (~3.6 W, 308 mA, operating safely at 74% of its 5.0 W continuous rating during the 4.0 s sub-zero pre-capture pulse).

Step 4.3 — Attach the 4 Front M2 Hex Standoff "Jacks" & Top Focus-Clearance Screws (DO NOT Over-Rotate!), Assemble the 2 m Arducam HDMI Umbilical ([P06]), Wire the Blue HLK-LD2450 Radar ([P11]) & Seal the Pod

[CAUTION]

CRITICAL CAMERA FOCUS CLEARANCE RULE — ATTACH THE 4 FRONT M2 HEX STANDOFF "JACKS" + 4 TOP SCREWS AND DO NOT ROTATE THE SCREWS TOO MUCH (IMG_7394, IMG_7463, IMG_7459, IMG_7465):

* The Camera Module 3 NoIR Wide [P05] uses a motorized voice-coil actuator (VCM) that physically moves the central IMX708 lens barrel forward and backward to focus.

* As proven by the side-profile straightedge check (Stage 4.3B Panel D — IMG_7465 below), the bare black M2 nylon hex standoff ("jack") alone is SHORTER than the protruding IMX708 lens tip!

* If you mount the camera without the 4 front standoffs + top screws—or if you screw the 4 top M2 nylon screws all the way down tight into the jacks—the central IMX708 lens barrel presses hard against the inside of the 30 mm mineral glass window ([P20]), jamming the motorized lens so it CANNOT MOVE OR FOCUS AT ALL!

* The Exact Fix (~1.5–2.0 mm Exposed Thread Gap = ~1.3 mm Lens Focus Clearance):

1. Mount the 4 Black M2 Nylon Hex Standoffs ("Jacks") on the FRONT of the Camera PCB (Panel A — IMG_7394): Open the small clear bag of black M2 nylon hardware included in your Arducam CSI-to-HDMI kit ([P06]). Push the male threads of the 4 black M2 nylon hex standoffs ("jacks") through the 4 corner holes on the front (lens) side of the Camera Module 3 NoIR Wide [P05] and thread the 4 black M2 nylon hex nuts onto the back of the PCB.

2. Thread the 4 Black M2 Nylon Screws Into the TOP of the Jacks — STOP With ~1.5–2.0 mm of Thread Showing (Panels B & C — IMG_7463 / IMG_7459): Using your STREBITO precision screwdriver ([T07]) while holding each hex jack with tweezers ([T01]), thread the 4 black M2 nylon pan-head screws into the top female threads of the 4 hex jacks for ONLY ~2–3 turns—leaving ~1.5–2.0 mm of exposed screw thread visible under each screw head. DO NOT rotate or tighten them flush!

3. Verify With a Straightedge Across the Screw Heads (Panel D — IMG_7465): Hold a flat card or straightedge across the top and bottom screw heads from the side: the 4 partially unthreaded screw heads now stand ~1.3 mm taller than the central IMX708 lens tip. When the camera assembly rests against the inside of the 30 mm glass disc (IMG_7534), the 4 corner screw heads contact the glass perimeter outside the 20 mm viewport hole while the IMX708 lens barrel floats freely ~1.3 mm behind the glass with zero mechanical binding!

1. Assemble the 2 m Arducam CSI-to-HDMI Camera Umbilical Extension ([P06]) & Install the 4 Front Camera Standoff Jacks + Top Focus-Clearance Screws:

* Unclip the 15-pin end of the 200 mm orange Pi 5 camera ribbon from your Camera Module 3 NoIR Wide [P05] and lock it into the Pi-side green Arducam CSI-to-HDMI adapter board (silver/gold contacts facing the green PCB, white/blue backing facing the black latch).

* Attach the 4 black M2 nylon hex standoffs ("jacks") to the front 4 corners of the Camera Module 3 NoIR Wide [P05] (secured with 4 M2 nylon hex nuts on the back), and thread the 4 black M2 nylon screws into the top of the 4 jacks without rotating them too far (leave ~1.5–2.0 mm of exposed thread under each head so the IMX708 lens barrel has ~1.3 mm of clearance behind the glass window).

* Connect one of the short white 15-pin ribbons from the Arducam kit between the Camera Module 3 NoIR Wide [P05] and the Pod-side green Arducam CSI-to-HDMI adapter board, then link both green adapter boards with the 2.0 m black shielded 8K HDMI cable.

2. Wire the Blue HLK-LD2450 (HLHN V1.1) Radar Board ([P11]) with 4 Female-to-Male Jumpers (Red, Black, Green, Blue) to the Pico 2 Breadboard (a1, a2, a3, j1):

* On the blue radar board's 2×4 (8-pin) black header:

* Inner Column (Left of pins): Plug RED (+5V) onto the top pin (5V), skip the 2 middle pins (3.3V and PA9 stay bare), and plug BLACK (GND) onto the bottom-end pin (GND).

* Outer Column (Right of pins): Plug GREEN (RX) onto the top pin (RX) and BLUE (TX) onto the 2nd pin down (TX) (leaving the bottom 2 pins DP and DM empty).

* On the Pico 2 30-Row Breadboard ([P09]):

* Plug GREEN (Radar RX) -> a1 (Row 1, GP0 / UART0 TX)

* Plug BLUE (Radar TX) -> a2 (Row 2, GP1 / UART0 RX)

* Plug BLACK (Radar GND) -> a3 (Row 3, GND)

* Plug RED (Radar +5V) -> j1 (Row 1 right side, VBUS +5V)

3. Cut the 12 mm Bottom Hole Into an Open U-Slot (Serrated Knife [T10]), Drop the Pre-Assembled 2 m HDMI + Green Arducam Board Straight In, Tack-Glue the 4 Raised Standoff Screw Heads with Loctite Super Glue & Smooth the Permatex 81158 RTV with a Metal Teaspoon Bowl (IMG_7097–IMG_7638):

* Saw the 12 mm Bottom Hole Into an Open U-Slot (Panel A — IMG_7097 / IMG_7099): Because the molded black 8K HDMI plug ([P06]) is wider than a closed 12 mm hole, use a serrated knife ([T10], IMG_7097) to saw two short vertical cuts from the open rear rim of the Hammond 1591BSBK bottom wall straight down into the 12 mm hole and pop out the narrow plastic bridge (IMG_7099), converting the closed hole into an open U-slot.

* Drop the Pre-Assembled 2 m HDMI + Green Arducam Adapter Straight In & Strap with 3M ScotchBlue Tape (Panel B — IMG_7100 / IMG_7116): Without unplugging any fragile ribbon or HDMI connector, drop the pre-assembled Camera Module 3 NoIR + White 15-Pin Ribbon + Green Arducam CSI-to-HDMI Board + 2m 8K HDMI Cable directly into the pod so the black HDMI cord seats cleanly into the bottom U-slot (IMG_7100), then strap the green Arducam adapter PCB and white ribbon flat inside the lower bay with 3M ScotchBlue painter's tape ([T11], IMG_7116).

* Tack-Glue the 4 Raised M2 Camera Standoff Screw Heads to the Glass Window Perimeter with Loctite Super Glue (Panel C — IMG_7516 / IMG_7534 / IMG_7535): Never press the camera lens face into wet black RTV silicone (which can smear onto the glass window and jam the IMX708 focus barrel). Instead, apply 1 tiny drop of Loctite Super Glue (Longneck Bottle) (IMG_7516) onto the top of each of the 4 partially unthreaded black M2 nylon screw heads (from Step 4.3B) and press the 4 screw heads squarely against the inside perimeter of the 30 mm mineral glass disc (IMG_7534 / IMG_7535). In 15 seconds, the camera is locked dead-center behind the viewport with ~1.3 mm of free air clearance in front of the IMX708 lens and zero wet black RTV silicone near the lens or glass aperture.

* Tuck the 3M ScotchBlue Protective Barrier with Tweezers ([T01]) & Trowel-Smooth Permatex 81158 Black RTV with a Metal Teaspoon Bowl (Panel D — IMG_7573 / IMG_7635 / IMG_7638): Cover the entire back of the camera PCB, standoffs, and white ribbon with 3M ScotchBlue painter's tape ([T11]), tucking the tape edges neatly around the board perimeter with fine tweezers ([T01], IMG_7573) so zero uncured RTV can touch the electronics. Then squeeze Permatex 81158 Black RTV Silicone ([T11], IMG_7635) around the outer perimeter and bottom HDMI U-slot and use the rounded back of a metal teaspoon bowl (IMG_7638) like a miniature mason's trowel to pack and smooth the black RTV cleanly into the joints while keeping all 4 corner lid screw posts 100% unobstructed.

* Place the HLK-LD2450 Radar ([P11]), Add Anti-Fog Desiccant & Close the 4-Screw Lid: Seat the blue HLK-LD2450 radar in the lower half of the pod aimed forward through the unpainted front wall, drop one small silica-gel desiccant packet inside (-30°C anti-condensation rule), and fasten the rear lid with all 4 corner Phillips machine screws ([T07]).

Step 4.4 — Paint the Zone 1 Pod in Solid Matte Forest Green ([P22])

1. Confirm the 3M blue painter's tape circle ([T11]) completely covers the outside of the glass window so zero paint touches the glass.

2. Lightly scuff the outside of the black plastic box with 400-grit sandpaper ([T10]) for 60 seconds and wipe with rubbing alcohol ([T11]).

3. Spray 2–3 light, uniform coats of Rust-Oleum Specialty Ultra-Flat Deep Forest Green ([P22], RAL 6003 / 6005) from 25 cm away and let dry 20 minutes between coats. The uniform solid matte forest green finish gives the unit a clean, non-threatening civilian forestry/environmental sensor appearance that blends naturally against pine foliage and bark while its non-metallic mineral pigment remains 100% radar-transparent (24 GHz).

4. Peel the blue painter's tape off the 20 mm glass window once dry.

Step 4.5 — Real-World Stage 4 Zone 1 Pod Verification: 3M Tape Shielding, Silicone Potting, Live Tripod Test, 4-Screw Closure & Optical Window Clarity

* Internal 3M ScotchBlue Protective Barrier + Cured Permatex 81158 Black RTV Silicone (Panel A): Applying 3M ScotchBlue painter's tape ([T11]) over the back of the Camera Module 3 NoIR Wide [P05], short CSI ribbon, and Arducam CSI-to-HDMI adapter PCB ([P06]) before potting isolates all delicate electronics from direct RTV contact while the cured Permatex 81158 black silicone locks the perimeter and forms a heavy-duty waterproof strain-relief boot around the 2 m thin black HDMI umbilical exit. All 4 corner screw posts remain 100% clear of silicone.

* Flush 4-Screw Rear Lid Closure (Panels B & C): Because the 4 corner screw bosses are unobstructed, the Hammond 1591BSBK rear cover seats with a 0.0 mm flush tongue-and-groove seam and fastens cleanly with all 4 corner Phillips machine screws ([T07]).

* Optical Window Clarity & Cure-Haze Physics (Panel D): Any faint RTV silicone off-gassing residue is confined to the outer perimeter rim of the 20 mm aperture outside the active optical cone. Directly over the central 6 mm optical entrance pupil of the Camera Module 3 NoIR Wide (IMX708) (11.5 × 11.5 mm square lens barrel), the 30 mm mineral glass disc is crystal-clear—verified by the fact that even the sub-millimeter white "Camera Module 3 NoIR Wide" PCB silkscreen inside the pod is crisp and legible through the glass (Panel D). Because the lens sits ~1.5 mm behind the glass locked at hyperfocal infinity (AfMode = 0, LensPosition = 0.5), peripheral rim haze outside the 6 mm cone has zero measurable impact on image contrast or YOLOv8 detection confidence.

STAGE 5: Pack, Insulate, Wire & Commission the Zone 3 Underground Base Vault (QILIPSU IP67 [P21])

The Zone 3 Base Vault is buried 20–30 cm underground in a free-draining root hummock (*mätäs*) at the base of the pine tree beneath 5–10 cm of insulating forest soil, moss, and winter snowpack, and lined with 10–20 mm closed-cell neoprene/XPS thermal insulation foam. This provides 100% visual & thermal concealment (zero above-ground box outline) AND geothermal freeze protection (keeping the battery bay at -2°C to -8°C even when surface air drops to -30°C to -40°C).

Look at the exact physical layout in the diagram below—it matches your physical QILIPSU enclosure (285 × 195 × 130 mm, [P21]) with the closed-cell thermal foam liner and the ExpertPower 12V 12Ah battery (151 × 98 × 95 mm, [P17]) sitting wall-to-wall across the bottom third.

Step 5.1 — Drill the 3 Gland/Vent Holes, Line with Thermal Foam, Mount the Grid Backplate & Strap the Battery

[CAUTION]

KEEP BATTERY TERMINALS UNPLUGGED UNTIL STEP 5.4! Cover both silver metal tabs (Red + and Black -) on top of the 12V 12Ah battery ([P17]) with a piece of blue painter's tape ([T11]) while placing components inside the box.

1. Drill the 3 Holes in the Empty QILIPSU Box ([P21]) Before Installing Electronics:

* Holes 3 & 4 (2× Bottom M20 Cable Glands — 20 mm / 3/4"): On the bottom wall (285 × 130 mm), mark and dimple two crosshairs with your center awl ([T06]): one at 75 mm from the left edge (65 mm from front) and one at 210 mm from the left edge (65 mm from front). Using your Cordless Drill ([T04]) and 3/4" (19–20 mm) Hole Saw Cup ([T05]) in REVERSE (or the 20 mm step of your Cone Step Bit [T06]), cut the two circular openings and thread in the M20 IP68 cable glands.

* Hole 5 (1× Right-Side M12 Breather Vent — 12 mm / 1/2"): On the right side wall (45 mm down from top, 65 mm from front), dimple the crosshair and drill a 12 mm (1/2") hole using the 12 mm step on your Titanium Cone Step Drill Bit ([T06]), then thread in the M12 ePTFE breather vent.

2. Line the Vault Interior & Door with Closed-Cell Thermal Foam Insulation (10–20 mm Neoprene / XPS):

* Press self-adhesive 10 mm closed-cell neoprene or XPS foam sheets against the inside of the hinged door and around the lower battery bay walls (leaving the M12 ePTFE breather vent opening unobstructed).

3. Screw the white ABS grid mounting plate into the back of the QILIPSU IP67 enclosure ([P21]) using the 4 corner Phillips screws ([T07]).

4. Place the ExpertPower 12V 12Ah AGM battery ([P17]) horizontally across the foam-lined bottom third of the enclosure and cinch it tightly to the grid slots with the heavy-duty Velcro strap ([T12]).

Step 5.2 — Crimp the Littelfuse Inline Fuse, Wire the 12V-to-5V 5A Buck & StarTech USB2TYPEM Barrel Cable, and Verify +5.0V DC with the Multimeter ([T09])

Using your Wire Stripper & Crimping Pliers ([T08], IMG_6652), 3M ScotchBlue Tape ([T11], IMG_6677), and 2.5 mm Flathead / #1 Phillips Screwdriver ([T07]), with the 5A tan blade fuse ([P18]) REMOVED from its fuse holder, strip ~10 mm of insulation off the wire ends (Panel B Top-Left — IMG_6652), tightly finger-twist the bare copper strands together (IMG_6664), and wrap all 3 wire splices in multi-layer 3M ScotchBlue tape (Panel B Bottom-Left — IMG_6677) to complete the fused 12V -> 5V USB-C Y-harness (Panel B Right — IMG_6709) before packing the components into the white QILIPSU enclosure:

Step | Wire From... | Wire To... | Why / Orientation Check

[Pwr 1] | Battery Red (+) Spade Clip | Input wire of Littelfuse Holder ([P18]) *(within 10 cm of battery + tab)* | Protects the entire box if any wire ever shorts.

[Pwr 2] | Output wire of Littelfuse Holder ([P18]) | +12V IN (+) of the 12V-to-5V 5A Buck Converter ([P12]) *(or via MPM3610 / STEMMA High-Side Relay [P12])* | Feeds fused +12V from the 12V 12Ah battery into the step-down buck stage.

[Pwr 3] | Battery Black (-) Spade Clip | Common GND (-) of the 12V-to-5V 5A Buck Converter (Pico GND, Radar GND, Compass GND) | Every single board shares the same Black (-) Ground reference.

[Pwr 4] | +5.0V DC USB-C Output of the 12V-to-5V 5A Buck Converter | Raspberry Pi 5 ([P01]) USB-C PWR IN port *(after multimeter check below!)* | Supplies regulated +5.0V DC (5A) directly from the 12V 12Ah battery.

[Pwr 5] | Raspberry Pi 5 USB-A Port #1 | Pico 2 ([P07]) Micro-USB cable | Powers the Pico 2 + GY-271 compass + HLK-LD2450 radar (j1 VBUS +5V).

[Pwr 6] | Raspberry Pi 5 USB-A Port #2 (Blue USB 3.0) | StarTech USB2TYPEM USB-A-to-5V-Barrel Cable ([P19]) -> StarTech Media Converter #1 ([P13]) 5V DC jack | Zero wall bricks in the field box! The Raspberry Pi 5's USB-A port powers StarTech Media Converter #1 directly via the StarTech USB2TYPEM (3ft/90cm) USB-to-Type-M barrel cable (IMG_6811 / IMG_7727).

[Pwr 7] | *(Optional Eco/Sub-Zero Duty-Cycling)* Pico 2 GP15 / GP6 / GP16 | STEMMA Relay #1 ([P12]), Pololu 2810 Radar Switch ([P12]), STEMMA Relay #2 (39 Ω, 5W De-Icer [P20]) | Optional hardware power-gating switches for multi-month deep-sleep duty-cycling and sub-zero window de-icing.

Also confirm your sensor & handshake wires on the Pico 2 ([P07]):

* Magnetometer (GY-271 QMC5883L [P08] in Col j Rows 24–28, 2 empty bottom holes j29–j30) & W1–W4 (Col f Rows 24–27, 3 empty bottom holes f28–f30): W1 Red (j5 -> f24 = VCC); W2 Black (j3 -> f25 = GND); W4 Yellow (a7 -> f26 = SCL); W3 Blue (a6 -> f27 = SDA); j28 = DRDY (5th pin, bare).

* Radar (HLK-LD2450 [P11]): Green (a1 / GP0) -> Radar RX; Blue (a2 / GP1) -> Radar TX; Black (a3 / GND) -> Radar GND; Red (j1 / +5V VBUS) -> Radar 5V.

* Pi 5 Outer-Right GPIO Handshake (W5–W7): W5 Purple (a11 -> Pi 5 Outer-Right #5 / Pin 10 GPIO15 RX); W7 Green (a13 -> Pi 5 Outer-Right #7 / Pin 14 GND); W6 Orange (a4 -> Pi 5 Outer-Right #8 / Pin 16 GPIO23 DONE).

[IMPORTANT]

DO NOT SKIP THE 30-SECOND MULTIMETER +5.0V DC CHECK ([T09])! Before plugging the USB-C power plug into the Raspberry Pi 5 ([P01]), verify the voltage on the bench with your Digital Multimeter ([T09]):

1. Leave the Pi 5 USB-C power cable UNPLUGGED from the Pi 5.

2. Push the 5A tan blade fuse into the Littelfuse holder ([P18]) and slide the red and black spade clips onto the 12V 12Ah battery ([P17]) (Red to Red +, Black to Black -).

3. Turn your Digital Multimeter ([T09]) dial to DC Volts (V= / 20V):

* Check 1 (Battery Rail): Touch Red probe to Fuse output and Black probe to Battery Black (-). Confirm +12.4V to +13.0V DC.

* Check 2 (Regulated +5.0V Buck Output): Confirm the buck output reads strictly +4.95V to +5.15V DC (never 12V!).

4. Unplug the Black (-) battery clip, plug the 12V-to-5V 5A USB-C cable into the Pi 5, plug the Pico 2 USB-A cable and the StarTech USB2TYPEM USB-A-to-5V-Barrel cable ([P19]) into the Pi 5's two USB-A ports (Panel D below), and plug the 5V barrel end into StarTech Media Converter #1 ([P13]).

Step 5.3 — Seat & Strap the Battery, Breadboard, Pi 5, Radar (3M Strain-Relief) & Optical Fiber Lid Spool Inside the White QILIPSU Vault ([P21])

[IMPORTANT]

Architectural Design Decision — Keep the Exact Stage 2 COTS 30-Row Breadboard ([P09]) & 2.54 mm Wiring (Zero Custom PCBs, Zero Second Design):

1. Supply-Chain Anonymity & OPSEC: Custom PCBs require uploading Gerber files to a fabrication house (JLCPCB/PCBWay), creating a permanent digital blueprint and uniquely traceable board batch. Using 100% generic COTS ELEGOO breadboards ([P09]) and standard 2.54 mm jumper ribbons ([P10]) guarantees zero custom manufacturing trail and total COTS plausible deniability.

2. Cost & Field Modification Agility: Custom PCBs save $0 in prototype/small-batch quantities while freezing pin assignments in fiberglass. Keeping the breadboard allows 5-second field pin modifications and eliminates building or testing a second hardware design.

3. 3-Step In-Place Vibration Lock (Reduces Snowmobile/Car Transport Risk from 65% to < 2% Without Unplugging a Single Wire):

* Lock 1 (Rigid Co-Mounting on the Same Backplate): Peel the backing off the 3M adhesive foam tape underneath your finished Stage 2 30-Row Breadboard [P09] (Pico 2 [P07] in Rows 1–20, W1–W4 in Col f Rows 24–27 with 3 empty bottom holes, and soldered / foil-wedged GY-271 [P08] in Col j Rows 24–28 with 2 empty bottom holes) and stick it directly onto the white QILIPSU grid plate 3 cm beside the Raspberry Pi 5 [P01], adding a nylon zip-tie around the breadboard waist through the grid slots. Because both boards are fixed to the same rigid plate, zero relative movement ever tugs on W5–W7!

* Lock 2 (3M Tape Multi-Pin Block Binding): Wrap one tight turn of 3M tape ([T11]) around the 4 adjacent black jumper shells in Col f (Rows 24–27), around the left-column Pico 2 shells (Col a), and around the 3 Pi 5 Outer-Right GPIO sockets (Pins 10, 14, 16). Binding single 1x1 shells into a solid block stops individual pins from pivoting (4x pull-out resistance).

* Lock 3 (Peelable Permatex Black Silicone [T11] Damper Bead): Squeeze a 2 mm bead of Permatex RTV Black Silicone [T11] along the seam where the black jumper shells meet the breadboard face, where the blue GY-271 PCB meets its 5-pin header collar, and where the 3 female sockets meet the Pi 5 GPIO header. It cures into a sub-zero (-54°C) rubber vibration damper that survives 10G+ snowmobile shocks yet peels off cleanly in one piece with tweezers if you ever modify a wire!

Now pack and secure the bench-tested assembly inside the foam-lined white QILIPSU IP67 enclosure (IMG_7723–IMG_7804):

1. Bottom Floor — Seat the 12V 12Ah Battery ([P17]) & StarTech Media Converter #1 ([P13]):

  * Place the heavy 12V 12Ah battery ([P17]) and Field Media Converter #1 (StarTech MCMGBSCSM10 [P13]) onto the black closed-cell foam floor (Stage 5.3A Panel B below). Verify Media Converter #1's front LFP DIP switch is slid DOWN to OFF, connect the short black Cat6 patch cable from the Pi 5 RJ45 port to Media Converter #1 RJ45, and connect the StarTech USB2TYPEM ([P19]) cable from the Pi 5 USB-A port to Media Converter #1 5V DC.

2. Wrap the HLK-LD2450 Radar 4-Wire Header in 3M ScotchBlue Tape ([P11] / [T11]):

  * Wrap a tight collar of 3M ScotchBlue tape ([T11]) around the 4 female jumper sockets (Red, Black, Green, Blue) and the radar board's 2×4 pin header (Stage 5.3A Panel C below) so the sockets can never pull loose or touch adjacent components.

3. Middle Layer — Cross-Strap the 30-Row Breadboard ([P09]), Raspberry Pi 5 ([P01]), and 12V 12Ah Battery:

  * Anchor the Pico 2 + GY-271 Breadboard ([P09]), Raspberry Pi 5 ([P01]) with its Active Cooler ([P02]) and 200 mm orange ribbon + Arducam Pi-side HDMI adapter ([P06]), and the 12V-to-5V 5A USB-C Buck Converter ([P12]) inside the white QILIPSU enclosure using 3M ScotchBlue retention straps (Stage 5.3A Panel D and Stage 5.3B Panels A–B).

4. Hinged Lid — Strain-Relieve the 1×2 Optical Splitter ([P14]) & Blue SC/UPC Coupler ([P15]):

  * Plug the Splitter's White Input strand -> Media Converter #1 TX (left optical port) and Blue Output 1 strand (50%) -> Media Converter #1 RX (right optical port).

  * Plug the Splitter's Orange Output 2 strand (50%) -> Blue SC/UPC Coupler ([P15]) -> Outbound Yellow / 3 mm Outdoor Fiber Cable ([P16]), pushing both blue SC/UPC plugs firmly past the internal spring (~1 mm gap at the center flange).

  * Coil the excess 0.25 mm bare fiber and 900 µm buffer in wide, gentle loops (>= 30 mm radius) and tape them flat inside the hinged white enclosure lid (Stage 5.3B Panel D).

Step 5.4 — Boot-Order & Power-Glitch Hardening for Buried Underground Operation (RP1 GPIO 32 ETH_RST_N Zero-Inrush Handshake + Zero-Config ff02::1 Dual-Stack)

[IMPORTANT]

ZERO-TOUCH UNDERGROUND HARDENING (WORKS REGARDLESS OF WHICH BOX BOOTS FIRST OR LOSES POWER):

* Why Unhardened 1000BASE-X Media Converters Can Miss Link If Surface Power Glitches: When StarTech Media Converter #1 ([P13]) inside the buried White Vault boots, its IEEE 802.3z Clause 37 (1000BASE-X) state machine transmits /C/ optical configuration ordered sets for only ~0.5 seconds before locking with its own Blue 50% splitter loopback leg (RX). If Collector Media Converter #2 (MacBook / Starlink / Cabin end) boots *later* or suffers an intermittent surface power loss while the buried vault stays running, Box #2 wakes up waiting for /C/ codes and its Top-Left (1000M Fiber Link) LED would remain dark without re-negotiation.

* 3-Layer Autonomous Hardening Built Into sender_node.py & receiver_hub.py (Zero Manual Unplugging Ever Needed Underground):

1. Layer 1 — Zero-Inrush Gigabit Ethernet PHY Reset via RP1 GPIO 32 (ETH_RST_N): Instead of hard-switching the Pi 5 +5V USB VBUS rail (GPIO 42, which would cause a 470 µF capacitive inrush spike into Field Box #1 on a cold-soaked battery), renegotiate_optical_diode_link() in sender_node.py pulses the Raspberry Pi 5's logic-level RP1 GPIO 32 (ETH_RST_N, BCM54213PE Gigabit Ethernet PHY reset) low for 0.65 seconds (pinctrl set 32 op dl) and high (pinctrl set 32 op dh) with 0.00 A +5V power surge. When Field Media Converter #1 inside the buried vault has its front LFP (Link Fault Pass-Through) switch slid UP to ON *(while Surface Collector Box #2 keeps LFP = OFF DOWN)*, dropping eth0 via GPIO 32 causes Field Box #1 to briefly squelch its 1310 nm laser and—upon eth0 recovery 0.65s later—transmit a fresh 0.5-second 1000BASE-X /C/ optical configuration burst down the yellow fiber so Surface Box #2 automatically relocks its Top-Left Green LED (4 + 4 = 8 Green LEDs) with zero 5V rail dip!

2. Layer 2 — Zero-Config Dual-Stack IPv6 Link-Local Multicast (ff02::1%eth0 -> ff02::1%en10) + Static IPv4: Whenever either box loses power, standard OS network managers drop temporary IPv4 addresses and flush ARP tables while waiting for DHCP (which can never reply across a one-way optical fiber). transmit_over_diode() transmits every signed alert burst simultaneously over RFC 4291 IPv6 All-Nodes Link-Local Multicast (ff02::1, deterministic Ethernet MAC 33:33:00:00:00:01) and static IPv4 (10.10.0.100:5514), while receiver_hub.py continuously re-joins ff02::1 every 2 seconds across all active interfaces—requiring zero DHCP, zero ARP, zero IPv4 address, and zero sudo privileges.

3. Layer 3 — Duty-Cycled Domain B Cold-Boot + Bench Quick-Reset Tip: In duty-cycled field mode (STEMMA / Pololu high-side switch [P12]), Field Box #1 cold-boots from 0.00V on every sensor wake event. During initial bench setup, you can also trigger an instant manual handshake at any time by connecting the Black (-) battery spade clip LAST (or running pinctrl set 32 op dl && sleep 1 && pinctrl set 32 op dh on the Pi 5).

Step 5.5 — Autonomous Field Commissioning: Provision Unique Node Keys & Install diode-node.service

Before closing the QILIPSU vault for unattended field operation, transition the Raspberry Pi 5 [P01] from manual bench mode to autonomous cold-boot service execution (ALLOW_EXAMPLE_KEYS=0, unique Ed25519 node identity, hardware Wi-Fi/BT disabled, and read-only OverlayFS enabled):

1. Generate the Unique Ed25519 Keypair & Node Identity on Raspberry Pi 5 [P01]:

```
cd ~/pi-diode-trailcam/pi5_sender_node
python3 generate_keys.py CAMERA-NORTH-01
```

* This writes node_key.json (containing both "node_id": "CAMERA-NORTH-01" and "private_key_hex") and prints the public-key registry snippet.

2. Register the Node's Public Key on Your Cabin Collector Hub ([T02]):

* Copy the printed CAMERA-NORTH-01 public key entry into pi-diode-trailcam/collector_hub/authorized_nodes.json on your Collector laptop/hub. *(Once authorized_nodes.json is present on the Collector Hub, receiver_hub.py automatically clears the demo NODE-NORTH-01 example key and trusts only the provisioned nodes in authorized_nodes.json.)*

3. Run the One-Command Production Fast-Boot & Service Installer on Raspberry Pi 5 [P01]:

```
sudo bash install_pi5_fastboot.sh CAMERA-NORTH-01
```

* What install_pi5_fastboot.sh configures automatically:

1. Copies /opt/trailcam/sender_node.py, installs /etc/trailcam/node_key.json (chmod 600), and writes /etc/trailcam/node.env (NODE_ID=CAMERA-NORTH-01, ALLOW_EXAMPLE_KEYS=0).

2. Configures /boot/firmware/config.txt with enable_uart=1, dtoverlay=disable-wifi, and dtoverlay=disable-bt (0.00 W RF at hardware boot).

3. Enables /etc/systemd/system/diode-node.service (EnvironmentFile=-/etc/trailcam/node.env) so every time Pico 2 closes STEMMA Relay #1, Pi 5 boots directly into sender_node.py, signs the alert as CAMERA-NORTH-01, blasts the 3× optical carousel, asserts GPIO23 (DONE), and halts cleanly.

4. Enables read-only OverlayFS (raspi-config nonint do_overlayfs 0) so sudden power cuts can never corrupt the MicroSD card.

4. Autonomous Cold-Boot & Cold-Soak Chamber Verification (✓ STAGE 5 SUCCESS CHECK):

* Reboot/power-cycle once with no SSH session attached (connecting the Black (-) battery spade terminal LAST so all 8 green LEDs—4 on Field Box #1 inside the White Vault + 4 on Collector Box #2—lock solid green). Wave your magnetic screwdriver ([T03]) over the sensor: Relay #1 clicks ON, Pi 5 + Media Converter #1 boot in < 7 seconds, the Collector Hub dashboard (http://127.0.0.1:8123) receives a verified alert signed by CAMERA-NORTH-01, GPIO23 (DONE) goes HIGH, and Relay #1 clicks OFF (0.00 W Pi standby)!

* Optional Pre-Deployment Cold-Soak Test in Bottom Freezer Compartment on Frozen Blue Igloo Ice Bricks (Fast Freeze Mode): Line the bottom pull-out freezer compartment with pre-frozen blue Igloo ice bricks (IMG_7825), place the latched White QILIPSU Vault and Black Zone 1 Pod directly on top of the frozen blue bricks (IMG_7829) running off the internal 12V 12Ah battery (with StarTech Box #1 powered by the USB2TYPEM cable from the Pi 5), and press Fast Freeze (snowflake icon red LED lit + 6/6 coldness bars active, IMG_7833). In cold-soak mode (run_freezer_soak.py), the Pi 5 captures a 1280×720 IMX708 frame, LD2450 radar distance, Pico 2 RP2350 internal temperature, SoC temperature (vcgencmd measure_temp), and regulated buck input voltage (EXT5V_V = 5.15V) every 10 seconds to /var/log/trailcam_cold_soak.jsonl so you can verify zero voltage sag and crystal-clear optics under rapid sub-zero chilling!

STAGE 6: Tree Installation, Collector Setup & Operating the System in the Field

Step 6.1 — Strap Zone 1 to the Pine Trunk (or Stand on Tripod for Rapid Trail Validation) & Bury + Insulate Zone 3 Underground

1. Pick the Right Tree (3–5 meters off the trail edge):

* Choose a mature pine or spruce tree (>25 cm trunk diameter) growing on a slightly raised, free-draining root hummock (*mätäs*) with deep vertical bark grooves and low branches that shade the trunk from direct sky view.

2. Mount the Zone 1 Micro-Pod ([P20]) at 1.8 m–2.2 m Height (25° Oblique Angle) — or 0.6 m–0.9 m in Dense Off-Trail Forest:

* Tuck the camouflaged Hammond 1591BSBK pod into a vertical bark crevice at eye level (1.8 m–2.2 m).

* Aiming Rule (25° Oblique): Point the single circular glass window 25° obliquely down the trail or driveway (not 90° straight across the road). This gives the camera and radar a 45 m view down the approach path so walking intruders or vehicles stay in the frame for several seconds.

* Dense / Thick Off-Trail Spruce Forest Rule (0.6 m–0.9 m "Under-Canopy Trunk Tunnel"): In thick spruce or conifer stands where drooping green branches at eye level (1.2 m–4.5 m) block optical sightlines beyond 3–5 m, non-optical sensors still wake the node through the trees (SM-24 geophone [P11-SEIS] senses footsteps 25–60 m away through the interlocking root mat with 0 dB foliage loss, [P11-AIR] 100–600 Hz acoustics diffract around trunks at 35–80 m, and the QMC5883L magnetometer [P08] sees steel gear/ATVs with 0.00 dB wood attenuation). Lowering the Zone 1 pod to 0.6 m–0.9 m (knee-to-hip height) underneath the lowest green boughs—where shaded lower branches naturally self-prune between 0.3 m and 1.1 m above the moss—opens a 15–35 m clear horizontal "trunk tunnel" for the 24 GHz radar and hyperfocal-locked camera (AfMode=0, LensPosition=0.5, which never autofocus-hunts on foreground twigs) to capture hikers' legs, wildlife, or ATV wheels moving between the bare trunks.

* Secure the pod with a matte brown bark strap or two stainless wood screws through the rear mounting tabs, and tuck a piece of natural pine bark / lichen over the top seam.

3. Wrap the Zone 2 Trunk Umbilical in Gear Aid Camo Form ([P22]):

* Press the 2 m umbilical cable straight down a natural vertical groove in the tree bark to the ground, and wrap it with self-clinging matte Gear Aid Camo Form fabric tape ([P22]) so no black wire is visible.

4. Bury & Insulate the Zone 3 Base Vault ([P21]) 20–30 cm Underground (Concealment + Freeze Protection):

* Dig a Shallow 20–30 cm Pit in a Free-Draining Root Hummock (*Mätäs*): Never bury the box in a low puddle or swamp hollow where spring snowmelt collects. Dig between the raised roots on the high side of the tree and spread 5 cm of coarse gravel, crushed rock, or dry pine cones across the bottom of the pit as a free-draining base bed.

* Wrap & Insulate with Closed-Cell Foam (10–20 mm XPS / Neoprene): In addition to the interior 10 mm closed-cell foam liner installed in Step 5.1, wrap a 10–20 mm closed-cell extruded polystyrene (XPS) or neoprene foam sleeve around the exterior sides and top of the latched QILIPSU IP67 vault before setting it into the pit.

* Backfill with 5–10 cm of Soil & Living Forest Moss (Sphagnum / Pleurozium): Lay the lifted carpet of living forest moss, pine duff, and soil (5–10 cm thick) back over the top of the insulated vault so 100% of the box is underground and invisible from the trail.

* Why Underground Burial + Foam Insulation Doubles Cold-Weather Battery Autonomy (WITH vs. WITHOUT Comparison): Even when Arctic air plunges to -35°C to -40°C (-45°C under snow), geothermal heat from the forest soil combined with the R-value of closed-cell foam, moss, and winter snowpack keeps the buried battery compartment between -2°C and -8°C—preventing electrolyte freezing, eliminating any thermal hot-spot visible to FLIR cameras, and preserving full 1.5 to 2.5+ year battery life!

Component / Configuration | Reliable Cold-Weather Limit (Outside Air) | Internal Vault / Battery Temp | Usable Battery Capacity & Runtime (12Ah AGM / 24Ah LiFePO4) | Why It Stops / Survival Mechanism

Zone 1 Tree Micro-Pod ([P20] at 2.0 m in open air) | -35°C to -40°C (-31°F to -40°F) | Equal to outside air (+12°C at lens during 3.69W de-ice pulse) | *Powered from Zone 3 via 2 m umbilical* | Contains zero batteries or liquid electrolytes—only solid-state IMX708 CMOS, LD2450 MMIC radar, and the 39 Ω, 5W ceramic window de-icer resistor ([P12-HEAT]).

Zone 3 Base Vault WITHOUT Burial & WITHOUT Foam (Above-ground in open air, standard 12Ah AGM [P17]) | -15°C to -20°C (+5°F to -4°F) *(Brownout / freeze risk below -22°C)* | Equalizes to outside air (-15°C to -20°C) within 3–5 hours | 45%–55% capacity (~140–190 days on 12Ah AGM) | Below -22°C, uninsulated AGM internal resistance (ESR) triples, causing voltage sag (<10.5V brownout) during the Pi 5's 2.5A cold-boot spike; partially discharged AGM electrolyte can freeze below -22°C.

Zone 3 Base Vault WITHOUT Burial + WITH 10–20 mm Foam Liner + Supercap Buffer ([P21-FROST], exposed rock site) | -25°C to -28°C (-13°F to -18°F) | Buffers overnight dips by +4°C to +6°C | 50%–65% capacity (~170–220 days AGM; 1.0+ yr LiFePO4) | For bare rock/concrete sites where digging is impossible: 2× 2.7V 50F series supercapacitors absorb the Pi 5 2.5A boot surge so high battery ESR at -28°C never triggers a brownout.

Zone 3 Base Vault WITH Underground Burial (20–30 cm) + 10–20 mm Closed-Cell Foam + Moss/Snow (Recommended) | -35°C to -40°C (-31°F to -40°F) *(Down to -45°C with 30+ cm snowpack)* | -2°C to -8°C (+28°F to +18°F) all winter | 85%–92% capacity (270–365+ days 12Ah AGM; 1.2–2.5+ years 15Ah–24Ah LiFePO4) | Geothermal heat rising from unfrozen subsoil (+2°C to +4°C at 1 m depth) trapped beneath closed-cell XPS/neoprene foam (R-4), moss (R-3.5), and snow (R-1/inch) keeps the battery near freezing even in -40°C blizzards.

Extreme Exposed Arctic Bedrock (-40°C to -45°C Open Air, Zero Soil) | -40°C to -45°C (-40°F to -49°F) | Equal to outside air (-40°C to -45°C) | 75%–85% capacity (1.5–2.5+ years) | Requires swapping [P17] to Lithium Titanate (LTO) or Li-SOCl2 primary pack ([P17-ARCTIC]) + supercap boot buffer ([P21-FROST]).

Step 6.2 — Connect Your Collector Unit (Choose Your Scenario: A, B1, B2, or B3)

Look at the three illustrations below to choose the exact Collector setup that matches your property:

* If You Have a Cabin Within 30 m to 10 km (Scenario A — Direct Moss-Buried Fiber):

1. Unroll the 3 mm black steel-tube armored single-mode OS2 fiber cable ([P16]) from the tree's Zone 3 Base Vault to your cabin, pressing it 2 cm under the soft forest moss with a garden trowel (no trench digging needed).

2. Inside the cabin, confirm the front LFP switch on Media Converter #2 ([P13]) is slid DOWN to OFF, push the incoming blue SC/UPC fiber connector firmly past the spring into the RX port of Media Converter #2 (leave TX capped as an empty physical air-gap!), power Media Converter #2 with its 5V adapter ([P19]), and connect Cat6 Ethernet to your cabin computer (10.10.0.100) so all 4 green LEDs light solid ON.

* If There Is NO Cabin Nearby (Scenario B1 — 200 m Buried Dead-Drop Vault):

1. Run 200 m of moss-buried armored fiber ([P16]) from the trail tree deep into the woods to a second buried QILIPSU Collector Vault (Variant 3B) containing a 12V battery, 5V buck, Media Converter #2 (RX only), and a headless Collector Pi storing verified JPEGs to a 512 GB NVMe SSD. Even if an intruder finds and smashes the camera at the trail, the signed photos are already safe 200 m away under the moss.

* If You Need Alerts Sent 200 m – 20 km Over a Forested Hill With NO Line of Sight (Scenario B3 — 200 m Lateral Fiber + Build D 2.4 GHz Troposcatter & Canopy-Bounce Bridge):

1. How Scenario B3 (sky-diode-relay Build D) Defeats Jamming & Interference (6-Layer Defense-in-Depth):

* Layer 1 — 200 m Lateral Fiber Separation + L-com HG2414P 14 dBi Flat-Panel Spatial Nulling: Because the camera at the trail (Zone 1–Zone 3) uses 100% passive glass fiber ([P16]), a handheld RF jammer carried down the trail has zero effect on the camera or its 200 m fiber link to the buried woods vault. By routing the 200 m fiber laterally into the forest and mounting the weatherproof square L-com HG2414P 14 dBi Flat-Panel Patch Antenna (203 × 203 × 38 mm) tilted $+2^\circ$ into the sky/canopy scatter volume toward the farmhouse (plus the matching HG2414P 14 dBi flat-panel antenna at the farmhouse aimed at the woods vault's sky scatter volume, NOT the trail), any jammer on the trail sits in the >20 dB off-axis/rear null of both flat-panel antennas and is attenuated by dense trees and terrain!

* Layer 2 — Out-of-ISM Guard-Band Hopping (2380.373 MHz, 2487.818 MHz, 2459.496 MHz) & Mini-Circuits VBF-2435+ Bandpass Filtering: The ADALM-PLUTO SDR + Skyworks SE2576L-EK1 (1 W / +30 dBm) PA hops across 3 cryptographic frequencies derived from SHA256(shared_secret || time_slot) outside standard Wi-Fi (2412–2462 MHz), while the gold Mini-Circuits VBF-2435+ (2370–2500 MHz) coaxial bandpass filter ahead of the Skyworks SKY65405-21EK1 (1.2 dB NF) LNA provides >40 dB stopband rejection against LTE/5G/VHF/UHF/GPS blockers.

* Layer 3 — VK7MO Coherent Lock-In Integration (+55.8 dB Processing Gain) & FFT Spectral Excision: Demodulates signals 22.0 dB BELOW the thermal noise floor (+14.1 dB to +62.3 dB SNR margin from 200 m to 20 km in clear sky, 24h summer sun, rain, fog, and blizzard snow) while automatically zeroing narrowband/swept jammers in the FFT domain.

* Layer 4 — Zero-Data-Loss 512 GB NVMe Vault Storage & Ed25519 Anti-Spoofing: Even against a high-power wideband RF jammer, the signed JPEG is already permanently stored on the 512 GB NVMe SSD inside the buried 200 m woods vault (<2 ms after capture) and re-transmits automatically via persistent store-and-forward (5s to 300s exponential backoff) once jamming clears. And because the farmhouse verifies the camera's Ed25519 cryptographic signature before accepting any alert, an attacker can never inject fake photos or spoofed alerts.

2. For complete zero-solder assembly (~35–45 min) of the Build D 200 m – 20 km Troposcatter & Canopy-Bounce hardware, see sky-diode-relay/hardware/SKY_RELAY_HARDWARE_SPEC_AND_BOM.md. To provision production Ed25519 keys (pi5_sender_node/generate_keys.py) and launch the live Collector Hub & Dashboard (collector_hub/receiver_hub.py + dashboard_server.py), follow the Software Stack & CLI Reference (pi-diode-trailcam/README.md).

3. Once installed, execute the 20-walk acceptance test, night 940 nm zero-glare verification, and 48-hour (192-event) unattended soak test in the Field Testing & Validation Guide (FIELD_TESTING_AND_VALIDATION_GUIDE.md).

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10. International Air-Shipping & Battery-Free Transit Guide (Shipping Pre-Assembled Units)

Because the pre-assembled Zone 1 Micro-Pod ([P20]), Zone 3 Base Compute Vault ([P21]), Cabin Receiver ([P13]), and Optics/Cables ([P14]–[P16]) contain zero lithium or lead-acid batteries when shipped dry, you can legally airmail or courier the entire pre-assembled, pre-tested system anywhere in the world as 100% standard non-hazardous air cargo (No IATA Dangerous Goods).

10.1 What to Leave Out of the Air Parcel (3 Strict Exclusions)

1. Lead-Acid / LiFePO4 12V 12Ah Main Battery ([P17]): Leave out (saves 7.72 lbs / 3.50 kg of shipping weight and avoids UN2800 / UN3480 battery rules). Keep the Littelfuse 5A Inline Fuse Holder [P18] and 6.3mm (F2) Red/Black Slide-On Spade Terminal Leads pre-crimped inside the QILIPSU box so the recipient can plug in a local 12V 12Ah battery in 30 seconds without tools.

2. Optional ML2020 Lithium Coin Cell on Raspberry Pi 5 [P01]: Unplug the tiny 2-pin J5 BAT RTC coin cell if fitted (UN3091). The Pico 2 [P07] hardware timer timestamps every wake event automatically without a coin cell.

3. Aerosol Spray Cans ([P22]) & Uncured Liquid Silicone Tubes ([T11]): Paint the enclosures and cure the 30mm glass viewport (24 hours) before packing. Once cured, dry paint and cured silicone rubber are 100% inert solids.

10.2 Nesting Trick ("Russian Doll" Packing in a Single 14" × 10" × 8" Box — 8.4 lbs / 3.81 kg)

Because the lower third of the QILIPSU Base Vault ([P21], 285 × 195 × 130 mm) is normally occupied by the 12V 12Ah battery, that entire bottom compartment is empty during shipping:

1. Wrap the pre-assembled Hammond 1591BSBK Camera/Radar Pod ([P20]) and the Cabin Receiver Media Converter ([P13]) in anti-static bubble wrap and nest both inside the empty lower battery bay of the QILIPSU Base Vault, strapped down with the nylon battery cinch strap ([T12]) so nothing shifts in transit.

2. Place a piece of bubble wrap inside the QILIPSU lid so the Raspberry Pi 5 [P01], Active Cooler [P02], and Pico 2 [P07] cannot vibrate loose, and close the QILIPSU latches.

3. Pack the closed QILIPSU box alongside the 30 m Armored Fiber Spool ([P16]) and umbilical/Ethernet cables inside a standard 14" × 10" × 8" (36 × 25 × 20 cm) corrugated box (Total packed weight: ~8.4 lbs / 3.81 kg).

Air Carrier Service (US -> Nordic / EU) | Typical Rate (8.4 lbs / 3.81 kg) | Transit Time | Customs / Tracking Notes

UPS Worldwide Expedited *(via Pirate Ship)* | ~$62 – $72 | 3 – 5 Days | Door-to-door air tracking + electronic customs pre-clearance

USPS Priority Mail International *(Commercial Rate)* | ~$68 – $74 | 6 – 10 Days | Handed off to national postal service (*Posti* in Finland)

DHL Express Worldwide *(via Pirate Ship)* | ~$82 – $93 | 2 – 3 Days | Fastest express air courier with SMS recipient customs link

10.3 Customs Declaration (CN23) & 5-Minute Local Battery Hookup at Destination

* Harmonized Tariff (HS) Codes for Smooth Customs Clearance:

  * HS 8471.50 — *Single-board microcomputer & digital camera sensor module (Raspberry Pi 5 + IMX708, no battery)* (0% EU/Finland duty tariff)

  * HS 8517.62 — *Ethernet-to-optical-fiber media converter modules & passive glass fiber cable (non-RF)* (0% EU/Finland duty tariff)

  * HS 8538.10 — *Empty polycarbonate weatherproof electrical junction enclosure (QILIPSU + Hammond, no battery)*

  * Important Note on Box Exterior: Tape a clear notice reading: "CONTAINS NO BATTERIES — PASSIVE ELECTRONICS & OPTICS ONLY (NO LITHIUM / NO LEAD-ACID)".

* 5-Minute Local 12V 12Ah Battery Hookup Upon Arrival (~€26.90 in Finland/EU):

  * The recipient buys a standard 12V 12Ah (151 × 98 × 95 mm, F2 / 6.3mm spade terminals) AGM or LiFePO4 battery locally (e.g., Motonet Product #90-0014 *Yuasa NPW45-12 / MTX Energy 12V 12Ah AGM* or Biltema Product #80-412 *12V 12Ah AGM*, ~€26.90).

  * Open the QILIPSU Base Vault, remove the nested Hammond Camera Pod and Cabin Receiver, strap the local 12V 12Ah battery into the bottom bay, push the pre-crimped Black (-) and Red (+) 6.3mm spade connectors onto the battery tabs (Red to +, Black to -), and insert the 5A blade fuse ([P18]). The system is immediately live!

Electrical connections

This is the maintained pin schedule for modular bench integration. Pin names describe the firmware interfaces; they do not certify the generated carrier layout. Check orientation against the physical board before connecting power. All Pi/Pico signal connections below use 3.3 V logic and a common ground.

Raspberry Pi Pico 2 ([P07]), QMC5883L ([P08]) & Raspberry Pi 5 (8GB) ([P01]) Bench Harness

(Note: Use your primary Raspberry Pi 5 (8GB) [P01] board—identified by its bottom-right Ethernet port, two narrow 22-pin CAM/DISP slots, and two CPU cooler mounting holes. While the 40-pin GPIO header pinout on the secondary Raspberry Pi 4 Model B [P01-ALT] is identical for Pins 10/14/16, only Raspberry Pi 5 (8GB) [P01] fits the Active Cooler [P02], Hailo-8L [P03], and 22-pin Camera Ribbon [P06].)

Signal | Source | Destination

Sensor power (W1 Red) | Raspberry Pi Pico 2 [P07] physical 36, 3V3_OUT (Right Row 5, Hole j5) | QMC5883L / GY-271 [P08] VCC (Row 24: Wire in Hole f24 -> Sensor Pin 1 in Hole j24)

Sensor ground (W2 Black) | Raspberry Pi Pico 2 [P07] physical 38, GND (Right Row 3, Hole j3) | QMC5883L / GY-271 [P08] GND (Row 25: Wire in Hole f25 -> Sensor Pin 2 in Hole j25)

I2C SDA (W3 Blue) | Raspberry Pi Pico 2 [P07] physical 6, GP4 (Left Row 6, Hole a6) | QMC5883L / GY-271 [P08] SDA (Row 27: Wire in Hole f27 -> Sensor Pin 4 in Hole j27; Pin 5 DRDY in Hole j28 left bare)

I2C SCL (W4 Yellow) | Raspberry Pi Pico 2 [P07] physical 7, GP5 (Left Row 7, Hole a7) | QMC5883L / GY-271 [P08] SCL (Row 26: Wire in Hole f26 -> Sensor Pin 3 in Hole j26)

Telemetry TX (W5 Purple) | Raspberry Pi Pico 2 [P07] physical 11, GP8 (Left Row 11, Hole a11) | Raspberry Pi 5 (8GB) [P01] physical 10, GPIO15 RX (5th down on right)

DONE (W6 Orange) | Raspberry Pi 5 (8GB) [P01] physical 16, GPIO23 (8th down on right) | Raspberry Pi Pico 2 [P07] physical 4, GP2 (Left Row 4, Hole a4)

Shared ground (W7 Green) | Raspberry Pi Pico 2 [P07] physical 13, GND (Left Row 13, Hole a13) | Raspberry Pi 5 (8GB) [P01] physical 14, GND (7th down on right)

Mechanical Verification & Field Vibration-Locking Rules (Stage 2 -> Stage 5)

1. Bottom Empty-Hole Counting Landmark (3 Empty Holes in Col f vs. 2 Empty Holes in Col j):

- The 4 breadboard jumper wires (W1 Red, W2 Black, W4 Yellow, W3 Blue) occupy Rows 24, 25, 26, 27 in Column f, leaving 3 empty holes (f28, f29, f30) at the bottom of Column f (and 3 empty holes f21–f23 between Pico 2 Row 20 and Row 24).

- Because the GY-271 / QMC5883L [P08] module uses a 5-pin header (VCC, GND, SCL, SDA in Rows 24–27, plus unused 5th pin DRDY in Row 28), Column j has 3 empty holes (j21–j23) between Pico 2 and VCC, and 2 empty holes (j29, j30) at the very bottom of Column j (Row 30). Always verify 3 empty bottom holes in Col f (f28–f30) and 2 empty bottom holes in Col j (j29–j30) so the compass is never shifted down by 1 row.

2. Unsoldered Header Press-Fit & 90° Straight-Pin Check:

- Before pressing the GY-271 [P08] into Column j (Rows 24–28), grip each of the 4 active short gold pins (VCC, GND, SCL, SDA) with flat-nose pliers ([T08]) and give a 15°–20° axial twist so the diagonal corners of the 0.64 mm square post bite into the 0.90 mm plated through-hole wall (100% gas-tight spring press-fit without soldering).

- Inspect every jumper pin before insertion and straighten any bent "J-hook" tips with pliers so each pin enters its breadboard socket at a true 90° perpendicular angle.

3. Design Decision — Keep the COTS Breadboard ([P09]) + 3-Step In-Place Vibration Lock (Zero Custom PCBs / Zero Second Build):

- Rather than fabricating a custom PCB (which creates third-party manufacturing records, compromises supply-chain OPSEC/anonymity, saves $0 at prototype scale, and prevents quick pin changes) or tearing down the tested Stage 2 build for a second harness, the exact Stage 2 Breadboard [P09] is locked for car/snowmobile transport (<2% failure risk) in 10 minutes during Stage 5 using three reversible steps:

1. Rigid Co-Mounting on the QILIPSU [P21] Grid Plate: Stick the Breadboard [P09] foam adhesive backing (plus one safety zip-tie) onto the white QILIPSU mounting grid plate 3 cm beside Raspberry Pi 5 [P01] so zero differential wire tug can occur.

2. 3M Multi-Pin Block Tape Binding: Wrap a strip of 3M tape ([T11]) across the plastic housings of W1–W4 (Col f), W3–W7 (Col a), and W5–W7 on the Pi 5 GPIO header (Pins 10, 14, 16) so individual pins (~25 g pull-out force) become a unified multi-pin block (>500 g pull-out force).

3. Peelable 2 mm Permatex Black Silicone [T11] Damper Bead: Run a thin 2 mm bead of Permatex Black Silicone across the plastic base where the jumper shells meet the breadboard face and across the top of the GY-271 5-pin header (>1,500 g vibration retention, while remaining peelable in 5 seconds with tweezers [T01] if a future modification is ever desired).

Power the Raspberry Pi 5 (8GB) [P01] (27W USB-C) and Raspberry Pi Pico 2 [P07] (Micro-USB) from their appropriate bench USB supplies. Keep the 12 V battery ([P17]) disconnected during this bench exercise. See the bench procedure for flashing and diagnostic steps.

Additional firmware interfaces & complete split-pod field harness (Stages 4–5)

Function | Pico 2 [P07] GPIO | Exact Switching Module & Pin-Level Field Connection

Domain B main power (Pi 5 + Media Converter #1) | GP15 (Pin 20) | Drives SIG on Adafruit STEMMA High-Side Relay #1 (PID 4409) ([P12]) (VIN = 3V3_OUT, GND = GND). Relay COM connects to fused +12.0V battery ([P17]/[P18]); Relay NO feeds the 12V-to-5V 5A (25W) Buck ([P12]) powering Raspberry Pi 5 [P01] and StarTech Media Converter #1 [P13].

Radar +5.0V power switch (180 mA load) | GP6 (Pin 9) | Drives CTRL/ON on Pololu 2810 High-Side MOSFET Switch (or STEMMA Relay, [P12]). Switch VIN connects to Always-On +5.0V (Adafruit MPM3610 output); Switch VOUT feeds HLK-LD2450 [P11] Pin 1 (5V) in the Zone 1 Pod. (GP6 never sources the 180 mA radar current directly.)

Radar UART0 TX / RX (256000 baud) | GP0 / GP1 (Pins 1 / 2) | GP0 (UART0 TX) -> HLK-LD2450 [P11] Pin 3 (RX); GP1 (UART0 RX) $\leftarrow$ HLK-LD2450 [P11] Pin 2 (TX); Pin 4 (GND) -> Common GND via the 2.0 m Zone 2 umbilical.

Window de-icer heater switch (3.69 W @ 12.0V) | GP16 (Pin 21) *(bench alias GP22)* | Drives SIG on Adafruit STEMMA High-Side Relay #2 (PID 4409) ([P12]). Relay COM connects to fused +12.0V ([P17]/[P18]); Relay NO feeds the 39 Ω, 5W Ceramic Window De-Icer Resistor ([P20]) returned to Common GND ($P = 12.0^2 / 39\,\Omega = 3.69\text{ W} \approx 3.6\text{ W}$, $I = 308\text{ mA}$, operating at 74% of its 5.0 W continuous rating during the 4.0 s sub-zero boot pulse).

Seismic ADC & alert (optional V1.1) | GP4 SDA / GP5 SCL / GP14 Alert | ADS1115 [P11-SEIS] on shared I2C0 bus (0x48); ALERT/RDY wired to GP14 (Pin 19).

[P12-BELT] Non-RF Rotorcraft Belt & 120 ms Bushing Net (100% Non-Chinese) | GP4/GP5 (I2C0) + GP18–GP20 (PIO0 PDM) + GP21 (Pin 27) | • Modality 2 (60 GHz Blade Micro-Doppler): SparkFun / Acconeer XM125 (1568-24540-ND, Sweden/USA) on I2C0 (0x52). ; • Modality 3 (LWIR Thermal Tripwire): Adafruit / Melexis MLX90640 (1528-4407-ND, Belgium/USA) on I2C0 (0x33). ; • Modality 1 (4× PDM Acoustic Array): Infineon IM69D130 (DE) / STMicro MP34DT01-M (EU, 1528-4346-ND) on GP18 (PDM_CLK), GP19 (PDM_DATA0), GP20 (PDM_DATA1). ; • 120 ms Dielectric Net Release: GP21 (Pin 27) drives SIG on Adafruit STEMMA Relay (PID 4409, 1528-4409-ND, USA/JP) switching fused +12.0V to Guardian Electric LT6X12-I-12D / Southco R4-EM (USA) solenoid latch holding the 2.5 × 2.5 m Dyneema® (UHMWPE) bushing curtain.

Primary / bench DONE handshake | GP17 (Pin 22) / GP2 (Pin 4) | Raspberry Pi 5 (8GB) [P01] GPIO23 (Pin 16) asserts 3.3 V HIGH when optical transmission completes; Pico 2 debounces for 75 ms (3 samples) before cutting GP15.

Gatekeeper telemetry (UART1, 115200 baud) | GP8 TX / GP9 RX (Pins 11 / 12) | Pico 2 GP8 (Pin 11) -> Raspberry Pi 5 [P01] GPIO15 (Pin 10, UART RX).

2.0 m Split-Pod Camera Bus (Zone 1 $\leftrightarrow$ Zone 3) | Pi 5 CAM/DISP 0 | Indoor bench (200 mm): Raspberry Pi Camera Module 3 NoIR Wide (IMX708) [P05] connects directly to Pi 5 J4 CAM/DISP 0 via the 200 mm 22-pin to 15-pin FPC ribbon ([P06]). ; Separated 2.0 m Tree Pod (Stage 4–5): Pi 5 J4 CAM/DISP 0 (200 mm 22-to-15-pin ribbon) -> Arducam CSI-to-HDMI Base Adapter (ASIN B06XDNBM63, [P06]) -> 2.0 m Round Shielded HDMI Cable through M20/M12 glands -> Arducam CSI-to-HDMI Pod Adapter inside Zone 1 -> Camera Module 3 NoIR Wide [P05] (plus auxiliary AUX1/AUX2 lines for the 39 Ω 5W heater).

Converter supply and optical topology

StarTech MCMGBSCSM10 ([P13]) requires 5 V DC, center positive. Do not connect it to a switched 12 V output. Its supplied adapter is rated 5 V, 2 A (or use the StarTech 5V USB-to-Barrel Cable [P19]). Confirm the connector and polarity on the actual unit. These details and the standard 10 km / 0–50°C ratings come from the manufacturer datasheet; they do not establish performance in the proposed splitter arrangement.

The proposed optical routing is:

1. Raspberry Pi 5 (8GB) [P01] Gigabit Ethernet to Field Media Converter #1 ([P13]) RJ45.

2. Field optical TX to the input of the 1x2 50/50 single-mode splitter ([P14]).

3. One splitter output to field optical RX (local carrier-detect loopback); the other to the outdoor fiber ([P16]) via coupler ([P15]) and collector optical RX.

4. Collector optical TX unconnected/capped; collector RJ45 to the hub Ethernet adapter or local switch.

Prove link negotiation and delivery on the short bench configuration before extending the route. The hub application listens on UDP 5514; the example hub address is 10.10.0.100/24. A direct Cat6 bench connection remains bidirectional hardware.

Optional sky relay interface

The sky-diode-relay Build D (200 m – 20 km All-Weather 2.4 GHz Troposcatter & Canopy-Bounce Relay) uses a standalone zero-solder SMA coaxial and +5.0V DC screw-terminal harness (ADALM-PLUTO SDR + Skyworks SE2576L-EK1 1W PA + SKY65405-21EK1 LNA + Mini-Circuits VBF-2435+ filters + TDK-Lambda LS50-5 5V PSUs; ~35–45 min assembly) documented in SKY_RELAY_HARDWARE_SPEC_AND_BOM.md.